

PUEBLO POTTERY MAKING

A STUDY AT THE VILLAGE OF SAN ILDEFONSO



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A STUDY AT THE VILLAGE OF SAN ILDEFONSO ***

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NUMBER TWO

PUEBLO POTTERY MAKING



PLATE 1
Courtesy Wesley Bradfield
MARIA MARTINEZ

The most skillful potter of San Ildefonso.

DEPARTMENT OF ARCHAEOLOGY
PHILLIPS ACADEMY · ANDOVER · MASSACHUSETTS

PUEBLO POTTERY MAKING

A STUDY AT THE VILLAGE OF SAN ILDEFONSO

BY
CARL E. GUTHE



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INTRODUCTION

The present paper is a careful study by Dr. Guthe of pottery making at San Ildefonso, a typical Pueblo Indian town on the Rio Grande, north of Santa Fe, New Mexico. The field-work was undertaken in 1921 as part of an archaeological survey of the Southwest, that has been carried on for a number of years by the Department of Archaeology of Phillips Academy. From prehistoric archaeology to modern pottery making may seem a far cry, but in the Southwest the past merges almost imperceptibly into the present, and the Pueblos of today live in almost exactly the same way, and practise almost exactly the same arts, as did their ancestors of a thousand years ago. In the Southwest, therefore, the archaeologist has the invaluable opportunity of observing, and of studying at first hand, the life whose earlier remains he unearths from the ancient ruins. When one considers what such a privilege would mean to the excavator in, for example, the mounds of the Mississippi Valley, or the Neolithic village-sites of Europe, it becomes obvious that the Southwestern archaeologist should devote a not inconsiderable part of his time to the study of that industrious, kindly, hospitable, and thoroughly charming folk, the Pueblo Indians.

Living a sedentary, agricultural life in an arid country it was inevitable that the Pueblos should early have developed into expert and prolific potters; and so pottery, in the form of sherds scattered about their former dwellings, and of vessels piously interred with their dead, is the most striking, the most abundant, and the most readily accessible form of evidence to be dealt with by the Southwestern archaeologist. The value of pottery to the student of the past cannot be more happily expressed than in the words of the historian Myres: "When with the soft clay which has, so to say, no natural shape or utility at all, the human hand, guided by imagination, but otherwise unaided, creates a new form, gourd-like, or flask-like, or stone-bowl-like, but not itself either gourd, or skin, or stone, then invention has begun, and an art is born which demands on each occasion of its exercise a fresh effort of imagination to devise, and of intellect to give effect to, a literally new thing. It is a fortunate accident that the material in question, once fixed in the given form by exposure to fire, is by that very process made so brittle that its prospect of utility is short; consequently the demand for replacement is persistent. The only group of industries which can compare with potmaking in intellectual importance is that of the textile fabrics; basketry and weaving. But whereas basket-work and all forms of matting and cloth are perishable and will burn, broken pottery is almost indestructible, just because, once broken, it is so useless. It follows that evidence so permanent, so copious, and so plastic, that is to say so infinitely sensitive a register of the changes of the artist's mood, as the potsherds on an ancient site, is among the most valuable that we can ever have, for tracing the dawn of culture."

Myres wrote of arid Egypt, and in many ways conditions in the likewise arid Southwest, where a primitive people were also building up for themselves through agriculture a new type of civilization, closely parallel those of the Nile Valley in predynastic times. And the analogy, naturally enough, holds good in the matter of archaeological methods. The rise and spread of the predynastic cultures of Egypt are being traced out in large part by studies of

ceramic types and of their stratigraphic relation one to another. The same methods are applicable, and indeed are now beginning to be applied, to the problems of the Southwest.

To understand the trend of recent archaeological work in the Pueblo field it is necessary to take a bird's-eye view, so to speak, of the region. Over a vast area in New Mexico, Arizona, Utah, Colorado, and the state of Chihuahua in Old Mexico, are found the remains of the prehistoric Pueblos. Their villages, now in ruins, were built in all sorts of places, on the tops of mesas, in open plains, in narrow canyons, on the ledges of great cliffs, and in the shelter of caves. They range in size from one-or two-room houses of the roughest construction, to great communal buildings of five hundred or even a thousand rooms, compactly built of excellent masonry and terraced to a height of three, four, or five stories. The artifacts, too, found in them vary bewilderingly, both in degree of perfection, and in type. All Southwestern houses, however, and all Southwestern artifacts, have a certain family resemblance that allies them to each other and makes it evident that they are all the product of a single culture, a culture distinct, for example, from that of Central Mexico or that of the Mississippi Valley. To reconstruct the history of that culture, to trace its origin, to follow its growth, and to explain how and why it developed in the peculiar way it did, are the tasks which confront the Southwestern archaeologist.

As in any scientific problem, the first step must be to collect and classify the data; and although Pueblo sites are numbered in thousands, and an overwhelming amount of merely descriptive work must still be done, the outlines of a classification have been achieved. We know, for example, what sort of ruins are found in the San Juan drainage of northern New Mexico; what kinds of pottery occur along the Gila river in southern Arizona. But what relationship, genetically and in time, there existed between, say, Pueblo Bonito and Casa Grande, we do not know. The time element, or in other words the historical sequence of our material, remains in large part to be determined.

To set up an historical outline we must first determine the relative ages of the different ruins, and then estimate the size and distribution of the Pueblo tribes from the earliest times to the present. At the top, so to speak, of our series the problem is simple enough—we are acquainted with the present location of the tribes, and the various Spanish accounts tell us where they have been living during the past three hundred and fifty years. But for the prehistoric period (and everything in the Southwest prior to 1540 is prehistoric) we must rely almost wholly on such evidence as may be turned up by the archaeologist's shovel, for of native written records there are none, nor can native legendary testimony be safely depended upon. This, of course, throws a heavy burden upon the archaeologist, a burden which is made heavier by the fact that stratigraphy of remains is so rarely found in the Pueblo country.

Stratigraphy, in other words the superposition of the more recent upon the more ancient, has been the Open Sesame to all the reconstructive sciences. The very framework of geology, for example, has been built up from stratigraphic observations. In archaeology, too, stratigraphy has revealed the sequence of the Stone Ages; made clear the development of the early Mediterranean cultures, and the rise of predynastic civilization in Egypt. Therefore, as has just been said, the general lack of stratigraphic conditions in the Southwest renders the task of the student a particularly hard one. For some reason, not yet clearly understood, the Pueblos ancient and modern were very prone to shift from one

dwelling place to another, and a site once abandoned was seldom reoccupied. Although their houses were of the most permanent construction, and their agricultural life should have tended to render them solidly sedentary, they moved about to a surprising extent. The result of this is that one seldom finds a ruin which was lived in for more than a few decades or, at most, centuries; and few have so far been discovered to contain superimposed remains illustrating any long period of development. Where such evidence is so rare, what can be found naturally becomes of the greatest importance; hence the recent diligent search for, and excavation of, such sites as show signs of long occupancy.

The choice of Pecos for investigation by Phillips Academy was due to the above considerations. The ruin was a large one, was occupied at the time of the Discovery and was not abandoned until 1838. A surface examination also showed that it must have been tenanted for a long time prior to the Conquest because its mounds were scattered over with potsherds not only of recent date, but also of several distinct prehistoric types, each one well enough known to students, but whose relative ages were entirely a matter of conjecture. It was hoped, therefore, that excavation might disclose some definite cases of superposition, and that several prehistoric periods might thereby be arranged in their proper chronological order.^[1]

The results have been more than satisfactory. Pecos proved to have been built on the edge of a sharp-sided mesa, a fact not suspected before digging began, because the rubbish from the town had heaped up to such an extent against the original cliff as completely to mask its steepness. The first inhabitants naturally threw their refuse over the edge of the mesa, their descendants added to the accumulation, and the process continued down the centuries until there grew up a midden of enormous extent and, for the Southwest, of unusually great depth. It is stratified as neatly as a layer-cake.

When the exploratory trenches revealed the size and probable importance of the Pecos rubbish heap, all other projects were postponed, and two full field-seasons were devoted to the meticulous dissection of large areas of the deepest deposit. At frequent intervals stratigraphic tests were made, in which all the specimens from each successive stratum were kept separate and shipped to the Museum for study. It was found that many changes in culture had taken place during the long occupancy of Pecos; in the stone and bone implements, in the pipes, and in burial customs. But the most abundant, the most easily gathered, and the most readily interpreted evidence of cultural change was offered by the thousands of pottery fragments that filled the mound from subsoil to surface.

We have been able to recognize about twenty distinct wares, to arrange them into eight chronological groups, and to determine the exact sequence of these groups. This information, derived from the stratigraphic study of the pottery in the mounds, has been of the greatest value. Its application has been both local and non-local. In the excavations that we have since carried on at Pecos it has enabled us to date relatively to each other the various kivas, cemeteries, and small refuse mounds that occurred on the mesa top, and also to unravel much more confidently than we would otherwise have been able to do, the extraordinarily complex jumble of ruined, torn-down, stone-robbled, rebuilt, abandoned, and reoccupied rooms that we encountered when we attacked the pueblo itself. As helpful as has been the knowledge of the sequence of the pottery types in working out the details of local archaeology, its usefulness in that regard is small as compared with the flood of light which

has been thrown on much larger and more vital problems. It has just been stated that some twenty types of pottery were identified; the majority of these are not peculiar to Pecos; many of them occur throughout large areas in the Rio Grande drainage; and so we are now able, in most cases by a hasty examination of the surface sherds, to assign to its proper place in the chronological series any ruin at which our types are present. Thus mere reconnaissance (a cheap and rapid undertaking) now serves to make clear the major outlines of Rio Grande archaeology. But the usefulness of the stratigraphic studies at Pecos does not end with the territory in which the Pecos types of pottery are found, for Pecos, because of its size and because of its situation on the main route between the Pueblo country to the West and the buffalo ranges to the East, was an important trade centre.

From its mounds we have taken potsherds from almost all parts of the Southwest, shells from the Pacific, spindle-whorls from Central Mexico, as well as pottery and stone objects from the Mississippi drainage. The importance of such finds is evident; every sherd from an outside culture found in a datable stratum at Pecos helps to fit into our general chronological scheme the culture from which it came; as conversely, does every Pecos or even Rio Grande sherd that turns up beyond the limits of the Rio Grande. Only a start has been made, but enough has already come to light at Pecos and at such stratified sites as have been excavated by other institutions, to provide us with a surprisingly full knowledge of the rise and growth of the Pueblo civilization. Ten years ago it would have been hard to believe how much could be accomplished by the stratigraphic work carried on by Nelson in the Galisteo Basin, Hodge at Hawikuh, Judd at Pueblo Bonito, Morris at Aztec, Guernsey in the Kayenta country, and the writer and his associates at Pecos.

The work, as I have said, has just begun, but the prospects are bright. Success will depend, as in any such endeavor, upon intelligent excavation, careful collection of data, and accurate observation of specimens, but the investigator cannot hope to derive the best results from his labors if he does not hold to a very broad view of his field. He must familiarize himself not only with the material of the locality he is working in but must also know as much as possible about that of other regions, for it is of the last importance that trade-objects be recognized wherever they occur. Of all trade-objects, vessels and potsherds are likely to be the commonest and most easily recognizable.

When the members of the Pecos expedition realized how vitally important was to be a close knowledge of the pottery, not only of that particular site, but also of the Rio Grande in general, and, indeed, of the entire Southwest, they devoted a large part of their time, both in the field and at the Museum, to the study of ceramics. A difficulty was at once encountered in our ignorance of the technique of Pueblo pottery making. We were constantly obliged to compare the paints and slips of different wares, to decide whether observed variations were technically fundamental, or were merely accidents of clay-mixing or firing. No full published accounts existed, nor had any of us had the opportunity for more than casual observation of potters at work. Dr. Guthe, accordingly, spent the month of August, 1921, at San Ildefonso in making detailed studies. His results are published in the hope that they may be of use to Southwestern archaeologists and ethnologists, and also to students of the primitive pottery of other areas who may be in need of material for comparative purposes.

In order that the reader may envisage the long history that lies behind the work of the modern potters, I give a summary of our present knowledge of the origin and development

of pottery in the Southwest.

To go back to the very beginning, it is not yet certain whether pottery making was introduced into the region from Mexico, or whether it was a local invention. There is evidence in favor of each theory. As will presently be shown, there can be traced in the Southwest an almost unbroken development from the earliest and crudest forms, through intermediate stages, to the perfected styles of later times; nor can Mexican influence be seen in the wares themselves at any period from beginning to end. Hence it might appear that the art was of purely local origin and growth. On the other hand, Mexico was the breeding ground for all the higher elements of native American culture; corn-growing, the cultivation of cotton, the use of stone masonry, the true loom, etc., and it is practically certain that all these traits worked into the Southwest from Mexico. It would thus seem reasonable to suppose that the concept of pottery came from the same source, particularly as pottery does not appear in our area until some time after the introduction of corn, and the mere presence of corn proves contact of one sort or another with Mexico, where presumably pottery had been in use for a considerable period. My present feeling is, however, that the importation consisted of no more than the bare idea that vessels could be made of clay, and that the subsequent development of the art was entirely unaffected by outside influences. This question, so important for its bearing on general problems of dissemination *versus* independent origin of culture traits, can only be settled by work in the archaeologically still unknown field of northern Mexico.

The earliest inhabitants of the Southwest of whom we have certain knowledge were the Basket Makers, so called by their discoverer because basketry rather than pottery was buried with their dead. They lived in southeastern Utah and northern Arizona, were an agricultural or semi-agricultural people who built no permanent houses, and who had no true pottery. In some of the caves that they occupied, however, there have been found a few fragments of crude, unfired clay dishes whose thick, bungling sides bear the imprints of the fingers of their makers. Save for occasional hesitating lines of incised marks, or straggling daubs of black paint, these vessels are undecorated. They bear no slip, nor, as was said above, were any of them subjected to the burning which would have converted them into true pottery. In the course of time, though whether that time should be counted in tens, or hundreds, or even thousands of years, we have at present no way of telling, the Basket Makers began to build houses of stone slabs, to group their hitherto scattered dwellings together into small communities, and, most important of all from the point of view of our present inquiry, to make hard, thin-walled, thoroughly fired vessels, in other words real pottery. The culture stage characterized by these

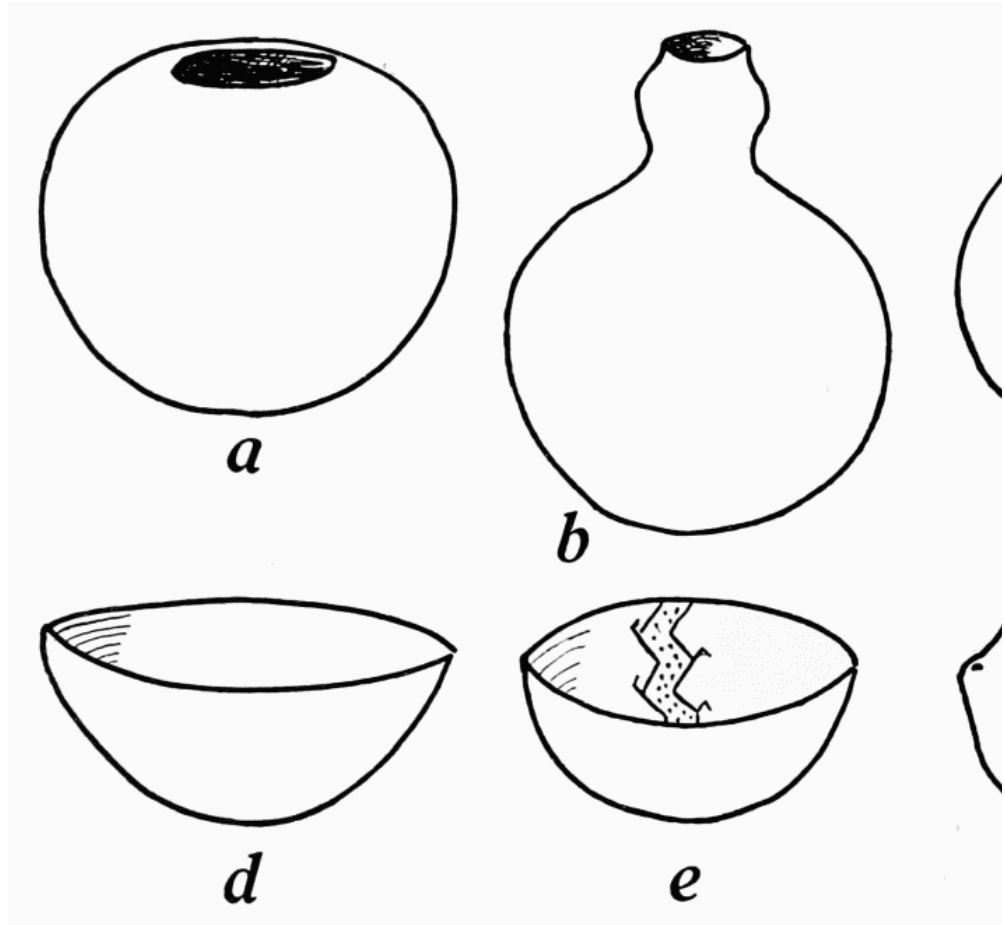


FIG. 1. Outlines of Post-Basket Maker vessels. (Courtesy S. J. Guernsey.)

remains and further distinguishable from the classic Basket Maker on the basis of house structure, sandal form, and other traits, was first identified by S. J. Guernsey, and named by him “post-Basket Maker”.

Post-Basket Maker pottery is simple in form and in decoration. It is normally light gray in color. The vessel forms most commonly found are bowls (fig. 1, d, e), globular vessels with restricted orifices (fig. 1, a), and water jars, the latter often having a swollen neck (fig. 1, b). The body of the ware in cross-section has a markedly granular appearance, due to the large size of the bits of tempering material included in it. A small percentage of the bowls bear a thin white slip on the interior

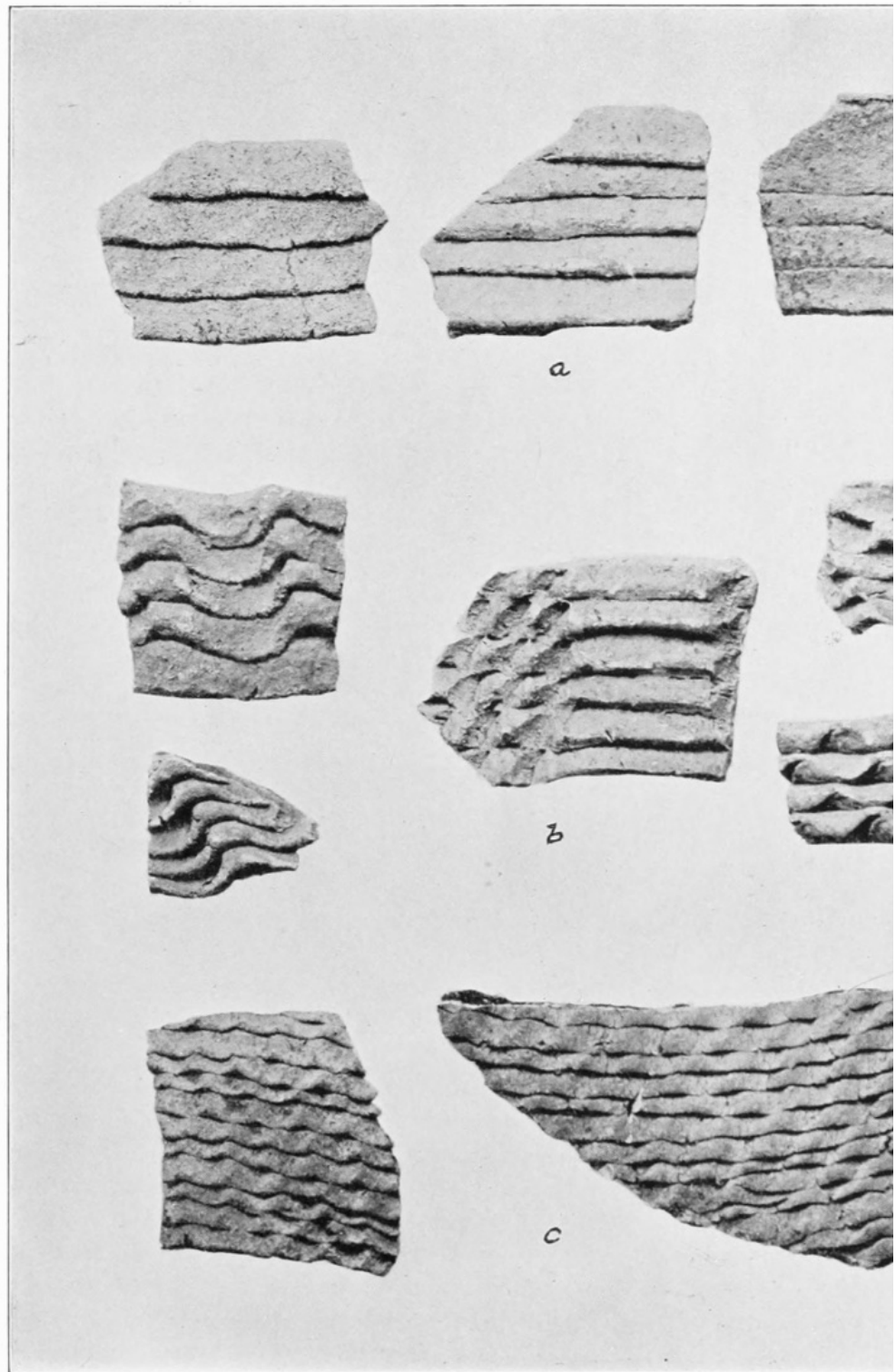


PLATE 2
ANCIENT CORRUGATED WARES
a. Pre-Pueblo. b. Early Pueblo. c. Classic Pueblo period.

and are ornamented with irregularly drawn designs in black paint (fig. 1, e). Many of the patterns suggest basketry prototypes.

All in all, post-Basket Maker pottery, judging from the few specimens so far recovered, is surprisingly well made; so well as to raise the suspicion that an earlier stage still awaits discovery. Hints of such a stage have, indeed, been found in the form of heavy clay dishes, tempered with shredded juniper bark, unbaked but distinctly superior in size and workmanship to the Basket Maker product. These specimens have turned up in post-Basket Maker sites, but in caves stratigraphically so confused that it is still uncertain whether they antedated, or were contemporaneous with, the more highly-perfected post-Basket Maker wares. They are too crude, however, to be considered the direct forerunners of post-Basket Maker pottery. If the development took place in the San Juan, one step at least is still lacking. These questions will be answered by further work of the sort now being carried on by Guernsey in the Kayenta country and by Morris in the Canyon del Muerto.

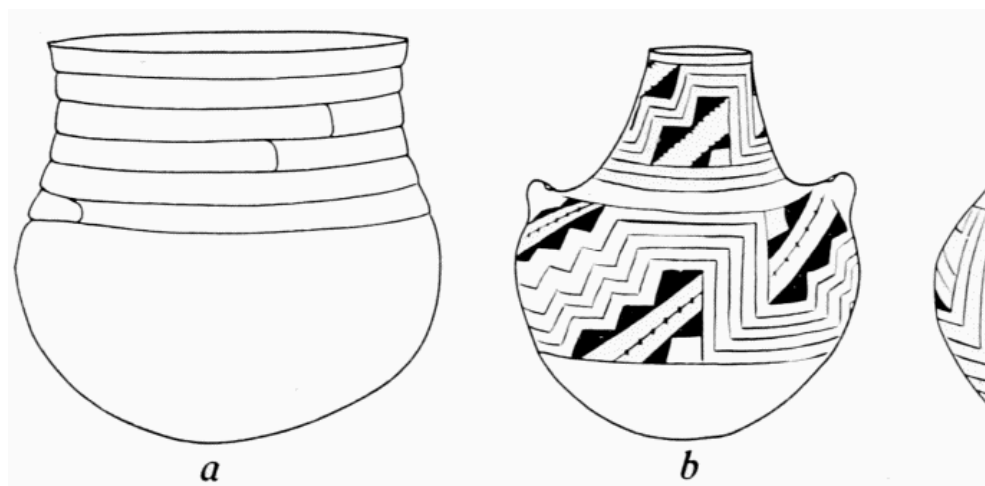


FIG. 2. Pre-Pueblo vessels.

From post-Basket Maker times down to the present day, however, the development of Southwestern pottery can already be outlined with considerable confidence. The next stage is usually called the pre-Pueblo. It is marked by further improvements in house-building, by the introduction of cranial deformation, and by certain innovations in ceramics. It had been the custom with the post-Basket Makers (as among many primitive people) to build up their vessels by adding to the growing walls successive rings of clay. When the process was complete and the sides had reached the requisite height, all trace of the rings was obliterated by rubbing down the surface. In the interiors of some post-Basket Maker pieces, however, with orifices so small that a smoothing tool could not be introduced, there were left the telltale junction-lines between the structural rings. In pre-Pueblo times it became the fashion purposely to leave unsmoothed the last few rings at the necks of certain small cooking vessels (fig. 2, a; pl. 2, a). This was the beginning of the elaborated coiled or corrugated technique, later so widely used.

Pre-Pueblo black-on-white ware takes the form of bowls, pitchers, ladles, small jars, and ollas. Its paste is hard, is less heavily tempered than post-Basket Maker paste, and the surfaces of the pieces are better smoothed. The decoration is characterized by boldness and

dash, the brushwork being correspondingly hasty. The lines are of irregular width, nor was care exercised that the junctions of strokes should be accurate. As is shown by the accompanying drawings (fig. 2, b, c), the commonest elements are the stepped figure and the triangle, both often emphasized by sets of widely spaced thin lines which follow the outer edges of the basic patterns.

The transition between the pre-Pueblo and the Pueblo was evidently a gradual one. There was no violent break, either in culture or in physical type; and so it is hard to say just where the dividing line should be drawn. The introduction of indented corrugated pottery, however, forms a convenient milepost on the road of ceramic progress. The change was brought about as follows: the plain, broad rings of pre-Pueblo ware were replaced by a continuous thin fillet of clay applied spirally; the junctions between the successive laps of the fillet were left unobliterated, not only at the neck, but over the entire vessel; and the fillet itself was also notched or pinched or otherwise indented to produce various ornamental effects (pl. 2, b, c). Thus was made the well-known coiled or corrugated cooking-ware so characteristic of all the archaic true Pueblo ruins, a style introduced at about the same time as the development of the above-ground rectangular living-room, the clustered method of building, and the use of horizontally coursed masonry, all elements that are commonly recognized as typical of true Pueblo culture. The earliest varieties of Pueblo pottery are imperfectly known, for little work has so far been done in the small and often nearly obliterated mounds that mark the house-sites of the first pueblo-dwellers. It is certain, however, that in the San Juan and Little Colorado drainages there was a strong growth in corrugated ware. This technique was experimented on and played with. The variety and even virtuosity of the results are extraordinary: waved coiling, pinched coiling, incised coiling, and all manner of indenting were practised (pl. 2, b). The black-on-white ware evidently did not advance so rapidly, the vessels, to judge from the sherds, were not particularly symmetrical as to shape or notable as to finish. The decoration ran to hachure in wavy lines and to poorly drawn spiral figures.

As I have just said, little is known of the early Pueblo sites. The period which they represent, however, was undoubtedly a long one. The date of its beginning is entirely problematical, nor can we even say with any assurance when it ended, but the probabilities are that as early as the opening of the Christian era the Pueblos had begun to gather together into larger settlements, to develop the closely-knit community life, and the many-storied terraced style of architecture which typify the golden age of their existence. Between that time and 1000 or 1100 A.D. the more or less uniform early culture split up into distinct and often highly specialized local sub-cultures, each of which followed its own line of growth in house-building, pottery-making, and the minor arts. It was during this general phase of Southwestern history that there came into being the great pueblos and cliff-dwellings that housed entire villages: Pueblo Bonito and the other Chaco Canyon towns; the cliffhouses of the Mesa Verde and of the Kayenta country; the enormous adobe constructions such as Casa Grande on the Gila and Casas Grandes in Chihuahua.



PLATE 3

OLD MODERN DECORATED WARE

a. Acoma. b. Zuñi. c. Sia. d. Hopi. e. San Ildefonso. f. Tesuque. g. Santo Domingo. h. Cochiti.

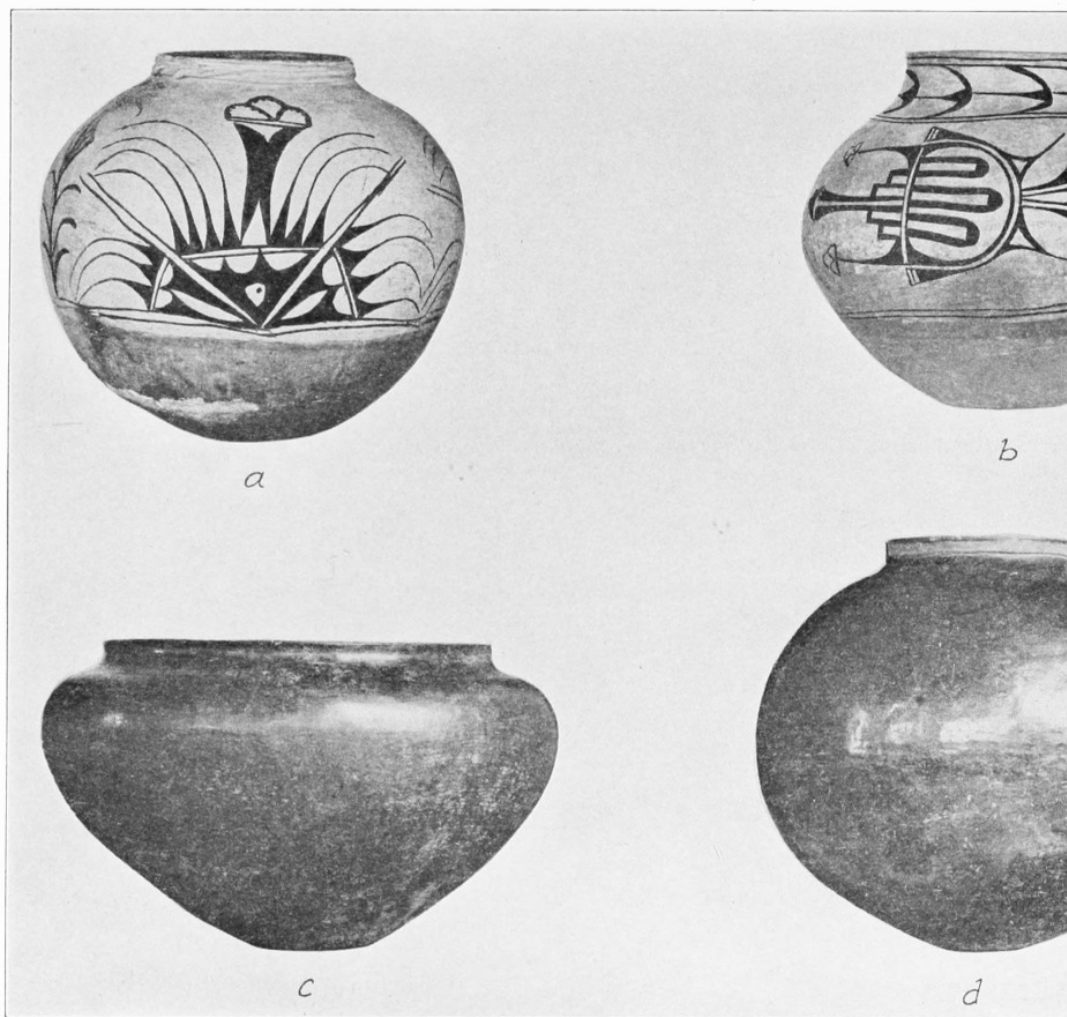


PLATE 4

OLD PIECES FROM SAN ILDEFONSO

a, b. Black-on-buff. c, d. Polished black.

To describe the many different wares of the classic Pueblo period would entail far more space than is available in this introduction. Brief accounts of the more important styles are included in the first paper of the present series,^[2] and a handbook of Southwestern pottery will, it is hoped, appear as a subsequent number. A single

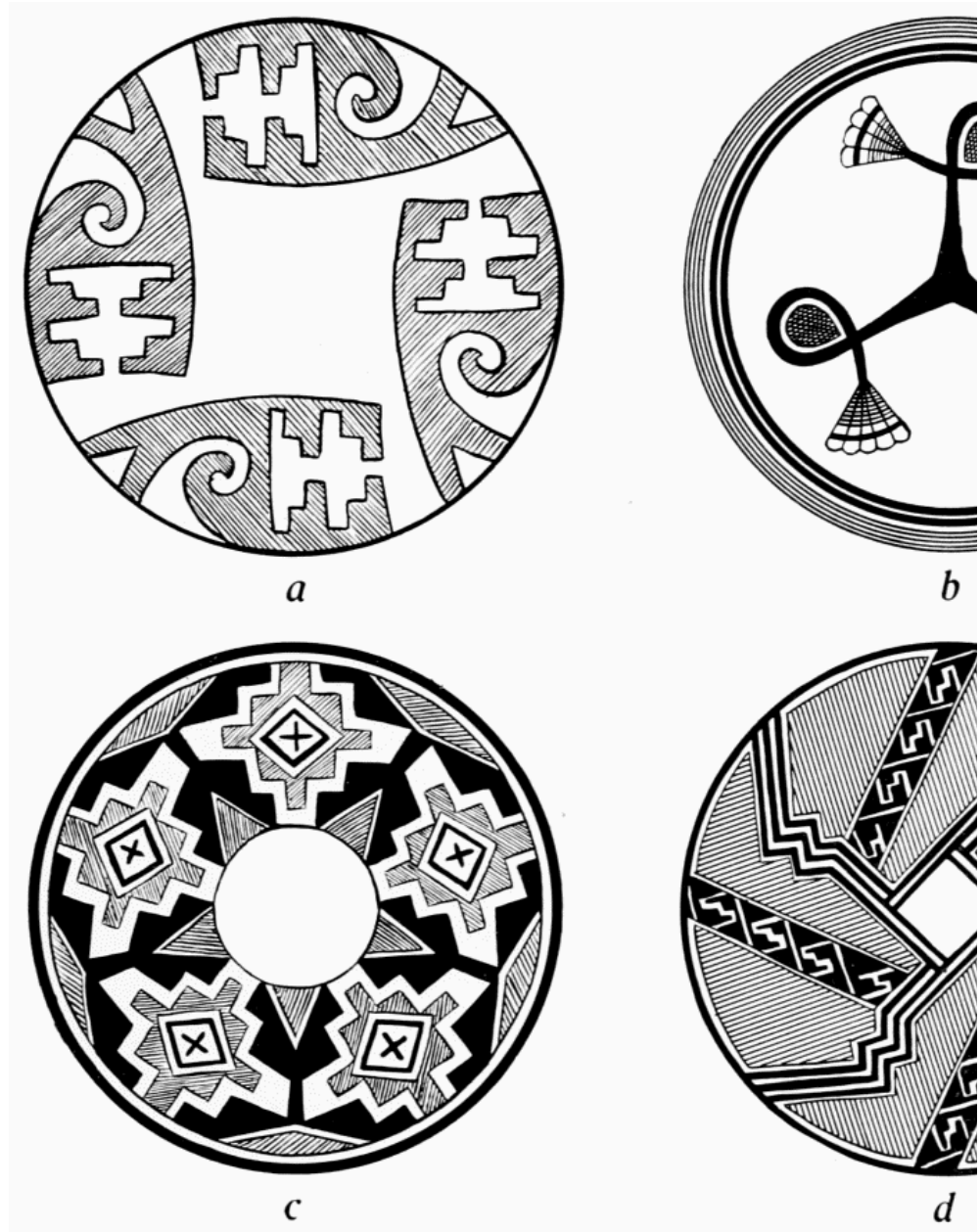


FIG. 3. Black-on-white bowls of classic Pueblo period. a. Chaco Canyon. b. Mimbres Valley. c. Upper Gila River. d. Mesa Verde.

illustration ([fig. 3](#)) must suffice to show the bewildering variety and the high artistic perfection which were reached at this time.

It seems ordained that periods of high achievement shall be followed by periods of slackening and even atrophy. So it was with the Pueblos. The years between about 1000 A.D. and the Spanish invasion of 1540 were evidently a time of tribulation. The formerly prosperous communities of the North and the South were abandoned, the population shrank, the arts degenerated, and when Coronado entered the Southwest he found the Pueblos occupying a mere fraction of their former range. What brought about this decline is not surely known, droughts, intertribal wars, pestilences, inbreeding, may all have been in part

responsible; but its principal cause was probably the ever-increasing attacks of nomadic enemies. At all events Pueblo culture of the sixteenth century was not what it had been in the past, and pottery making suffered with the other arts. By that time the beautiful and elaborate, but difficult, corrugated technique had been given up, and cooking vessels had become mere unornamented pots; black-on-white ware with its handsome, intricate geometric decoration had long been extinct, replaced by various local styles, few of which equalled it in technical excellence or artistic perfection. As a matter of fact the ceramic products of the historic period are very little known. In old times pottery formed an essential accompaniment of the dead, hence at nearly every prehistoric ruin may be found a burial place richly stocked with mortuary vessels. With the arrival of the priests, however, missions and Catholic cemeteries were established, all interments took place under the supervision of the Padres, and no more bowls or ollas went into the graves.

Pottery, of course, is short-lived, few vessels, other than reverently guarded ceremonial jars, can be expected to escape breakage for more than a generation or two, and so we have practically no material to illustrate the changes which took place between the Conquest and the present day. Much will eventually be learned in regard to seventeenth and eighteenth century Pueblo pottery, but the work will have to be done largely on the basis of sherds from the rubbish heaps of the older towns, and such broken pieces as may be found under the fallen walls of abandoned dwellings. We can, however, supplement this research by working backwards, so to speak, from the modern wares. If we can acquire a thorough knowledge of them it will be much easier to understand the fragmentary material we shall have to deal with in studying the early historic period. We should accordingly collect all possible data as to the wares turned out in recent years at the various pottery-making Pueblos.

The term pottery-making Pueblos is used advisedly, for, queerly enough, no pottery is manufactured at a number of the towns. If this condition had come about within the last decade it would be easy to account for, as tin oil cans for holding water, enamelled pots for cooking, and china dishes for serving food have lately been pushing native pottery out of use. But no pottery, other than rough cooking ware, has been made for a long time at Taos, Picuris, San Felipe, Jemez, and Sandia; while the art is practically or wholly extinct at Laguna, Isleta, and Santa Ana. Potters occasionally marry into these towns and continue to exercise their craft, but they seem to have no disciples among their daughters or among the local tribeswomen. At all the other Pueblos, however, potting is still carried on, but in most cases with a decreasing output and a lamentable degeneration in technique. The vessels, being mostly made to sell to tourists or curio dealers, are carelessly fashioned, crudely decorated, and insufficiently fired. Good contemporary pieces

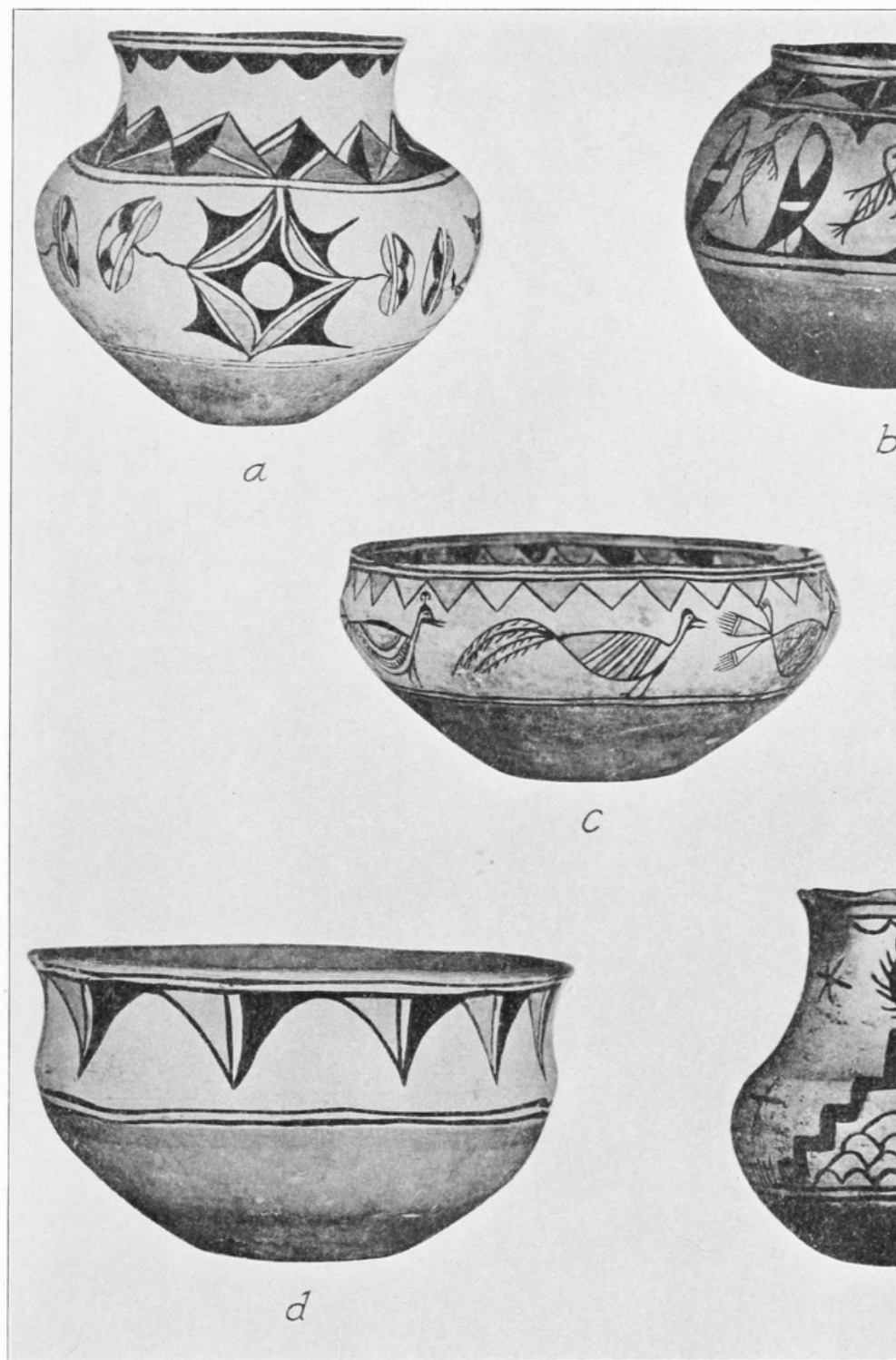


PLATE 5
 OLD MODERN SAN ILDEFONSO WARES
 a, c, d. Polychrome. b, e. Black-on-red.



Co

PLATE 6

are becoming rare, and the older vessels, which are naturally the best material for study, are now almost gone from the villages. Some pieces, however, are still left, and many more are in the hands of Mexicans and Americans living in the Southwest. These should be got into the custody of museums as rapidly as possible, for pottery is fragile stuff and every year sees the breakage of many unreplacable old vessels.

Excellent work in collecting and preserving the older pieces is being done by certain residents of Santa Fe, who have organized the "Southwestern Pottery Fund". In spite of limited resources they have got together a most remarkable collection from the Eastern pueblos, and are constantly adding to it as specimens come into the market.

A plate showing a typical example of old modern decorated ware from each of the pottery-making pueblos is here given, in order that the reader may appreciate the striking differences which obtain (pl. 3); San Ildefonso vessels of this period are illustrated elsewhere (pls. 4, 5).

In closing this introduction something should be said of the older San Ildefonso wares, and of the remarkable renaissance which has led to the high artistic achievements of the present-day potters.

Of the pottery made at San Ildefonso during the early historic period we have as yet no definite knowledge, but from the fact that the sherds at the ruined pueblo of Cuyamongé, a Tewa village only a few miles from San Ildefonso which was abandoned in 1698, are closely similar to the seventeenth century wares of Pecos, we may conclude that the San Ildefonso product of that time also bore resemblance to that of Pecos. The Pecos wares in question have as yet been but briefly described.^[3] The following are the leading varieties: rough black cooking ware; polished red; polished black; and decorated ware of two varieties, black-on-buff, and black-and-red-on-buff. The ornamentation is in heavy black lines, for the most part sloppily drawn, but the designs are bold, free, and effective. Although these styles all differ to some extent from the ceramics of late nineteenth century San Ildefonso, the artistic and technical relationship of the two groups is evident at a glance. The principal differences lie in the much greater amount of polished red found at Pecos, and in the absence from Pecos of a black-on-red decorated ware made at San Ildefonso.

What we know of nineteenth century San Ildefonso pottery is derived from pieces found in the pueblo when collecting began about 1870; and from vessels secured between that date and 1900. The oldest examples we have were thus probably not made much before 1850, and the great bulk of the specimens now in museums and in private hands were presumably turned out during the last third of the century. This material (aside from cooking pots, of which the older collections contain practically no examples) is divisible into five types, two plain and three decorated. The plain wares are polished black and polished red; the decorated are black-on-buff, black-and-red-on-buff (usually called polychrome), and black-on-red.

The polished black and polished red are, as Dr. Guthe brings out in greater detail below (p. 74), merely variants of the same ware—the red being produced by a preliminary burning in a more or less open fire—the black by smothering the fire with pulverized manure at a

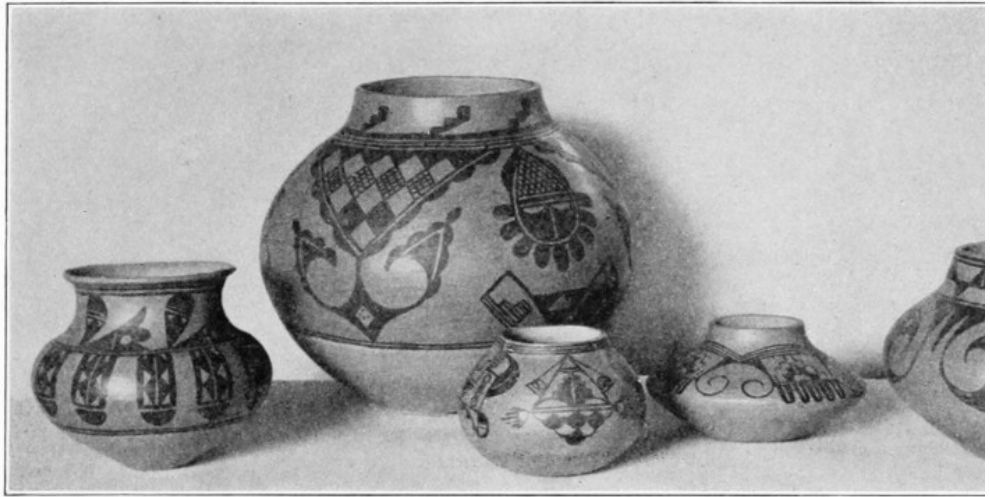
late stage in the burning. Vessels of both sorts were evidently made at San Ildefonso during the period under discussion, the red rarely, the black fairly commonly. It is usually supposed that the making of these wares, particularly the polished black, was confined to the nearby town of Santa Clara, and vessels of the latter type are ordinarily referred to as “Santa Clara black”. This is a natural enough error as the Santa Clara turn out nothing but the polished wares, and until the last ten years they led all the other pueblos in the excellence of their manufacture. The Indians of San Ildefonso, however, say that they have always made polished black pottery, and while its production may have partly or wholly lapsed during the 90’s and early 1900’s, there is little doubt that many of the older specimens, such as the ones here illustrated (pl. 4), are actually from San Ildefonso. The majority of the examples of old black are large storage jars with full, round bodies and relatively small orifices; there are also capacious bowl-like vessels (called by the Mexicans *cajetes*), which were used for the mixing of dough. The old pieces are less meticulously finished than are those of today, but their deep, velvety black lustre is of even greater beauty than the extremely high polish of the modern pieces.

The black-on-buff decorated ware was evidently the commonest variety of nineteenth century pottery. It took the form of prayer-meal bowls (small open vessels with terraced sides); medium-sized pots (pl. 3, e); and large storage jars (pl. 4). The ware, like all San Ildefonso work, is thick and heavy, giving a “woodeny” sound when tapped. The surface color is a creamy buff and the decoration is in deep black pigment. The bottoms of all the pieces, and the interiors of the necks of the jars, are painted red. The general nature of the elaborate designs is best brought out by the illustrations.^[4]

Black-and-red-on-buff or polychrome was apparently not an abundant style in old times. Its occurrence at Cuyamongé and at Pecos, however, seem to indicate that it was always made at San Ildefonso. There are also in museums and private collections pieces which appear to be as old as any of the examples of non-polychrome. The style was seemingly limited to bowls and small jars (pl. 5). The ware itself is the same as the foregoing, but various areas in the decorations (which themselves seem to differ from those of the black-on-buff) are filled in with a purplish red paint of very characteristic shade.

The red-on-black, like the polychrome, was not common. As was said above, it is absent from Pecos and Cuyamongé, therefore it is apparently of fairly recent origin in the Rio Grande. Its introduction or invention may, indeed, date from as recent a time as 1850. It differs from the buff wares in having a dark red slip, much more highly polished than are the buff slips. The decoration is in deep black. No large pieces, as far as I know, were made, most specimens being small globular jars without necks. One of these and a vase-like vessel are illustrated (pl. 5).

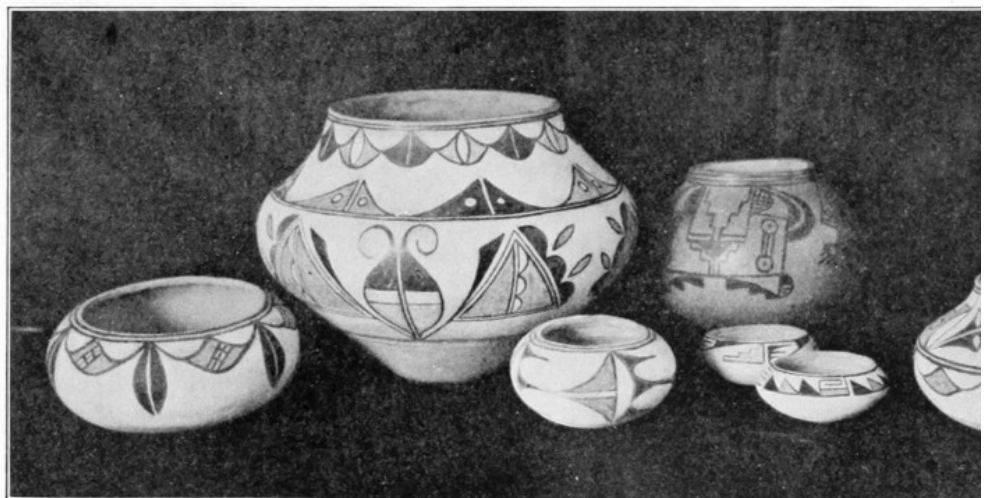
Such was the pottery of San Ildefonso during the second half of the nineteenth century. Toward the end of that period, however, a marked degeneration set in.



a



b



c

c

Courtesy K. M. Chapman

PRESENT-DAY SAN ILDEFONSO POTTERY

a, b. Black-on-red by Antonita Roybal. c. Polychrome, by Maria Martinez
—one black-on-red piece by Maximiliana Martinez.

This was due to the substitution for native pottery of American utensils for cooking, food-serving, and containing water; and to the growing tendency to produce pottery for sale to curio stores and tourists rather than for home use. The black-on-buff ware and the polished black were given up entirely, the potters concentrating on the more showy black-on-red and particularly on the polychrome. In the case of the black-on-red the high gloss of the older pieces was no longer produced, and the paint, being neither carefully mixed nor sufficiently aged (see p. 26) developed on firing a dull, bluish cast. The polychrome also fell off in technical perfection and in artistic excellence. The old, dull, purplish reds were replaced by brighter colors, the designs became flamboyant, and many ugly non-Indian vessel shapes were introduced (fig. 4). This state of affairs went from bad to worse until, in 1907, the potter's art at San Ildefonso was in a thoroughly bad way. A few women, however,

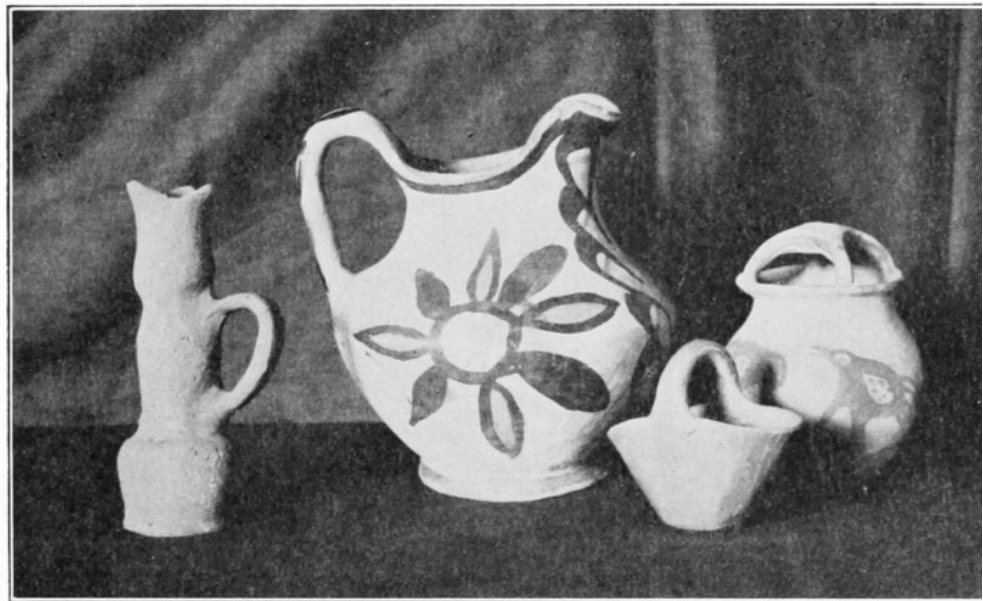


FIG. 4. Bad examples of modern pottery—un-Indian shapes and slipshod decoration.

retained something of the old craftsmanship and were ready to profit by the opportunity which was about to present itself.

In 1907 Dr. E. L. Hewett of the School of American Research began a series of excavations in the ancient ruins of the Pajarito Plateau. The diggers were all Tewa Indians from San Ildefonso. They proved to be excellent shovelmen, who took a keen interest in everything they found. They helped us identify many specimens which would otherwise have been puzzling, and their comments on the pottery, and especially on the designs, were most illuminating. The women of the Pueblo, when visiting camp, often held animated

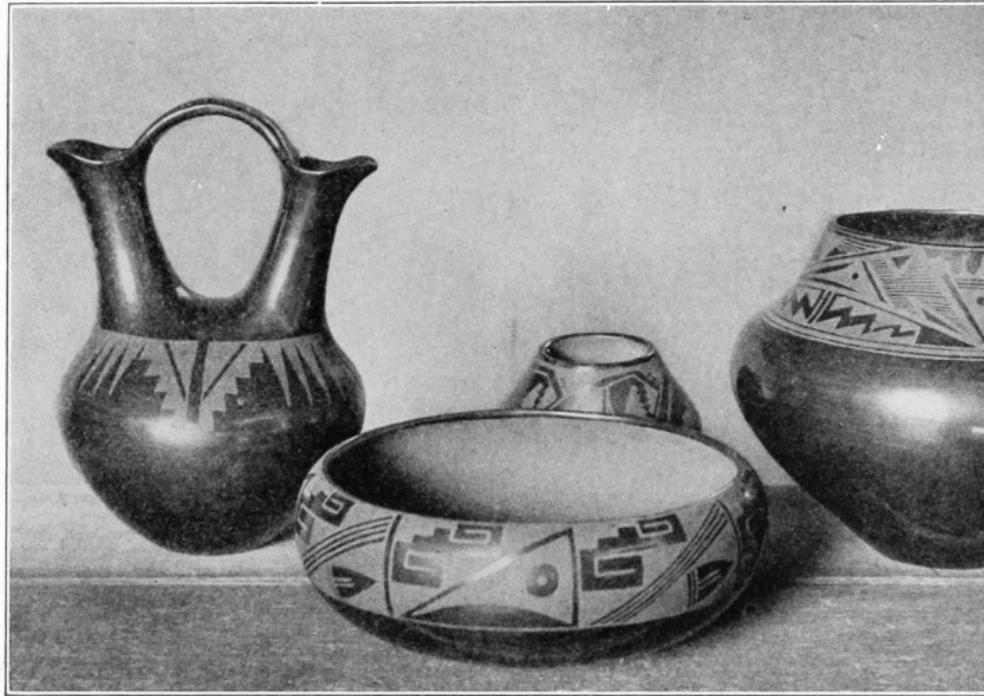
discussions as to the vessels from the ruins, and it was suggested to some who were known to be good potters, that they attempt to revive their art, and try to emulate the excellence of the ancient wares. While the response was not immediate, there was observable, during the next few years, a distinct improvement in the pottery of San Ildefonso. Realizing the importance of this, the authorities of the Museum of New Mexico and the School of American Research threw themselves heartily into the task of stimulating the industry. They urged the women to do better and better work, and in particular induced them to return to the sound canons of native art. Some old pieces remained in the pueblo, many others were in the Museum, of still others photographs were obtained. These were all brought to the attention of the potters.

The undertaking was not an easy one, however, for it was difficult to get most of the women to go to the trouble of making good pieces when the tourists, who were still the principal purchasers, were equally or even better pleased with imitations of china water-pitchers, ill-made rain gods, and candlesticks. The problem thus resolved itself into one of supplying a market. The Museum bought many good pieces, and Mr. Chapman, who from the beginning had been a leading spirit in the attempt at rehabilitating the art, himself purchased large amounts of pottery, never refusing a creditable piece, never accepting a bad one.

Progress was slow, but eventually certain women, becoming interested in their work, made real progress both technically and artistically. Their products began to sell more freely and at better prices than did those of others. Antonita Roybal, Ramona Gonzales, Maximiliana Martinez, and Maria Martinez all turned out fine vessels, the two latter being greatly aided by their husbands, who developed into skillful decorators. Maria especially shone. By 1915 she had far surpassed all the others, her pots were in great demand, and at the present time she has a ready market, at prices which ten years ago would have seemed fantastic, for everything she can find time to make. Her income is probably not less than \$2,000.00 a year, and following her example, many other women are now doing fine work and are earning substantial amounts. The beneficial effect of this on the pueblo has naturally been great. New houses have been built, new farm machinery, better food, and warmer clothing have been bought, and, best of all, there has been acquired a wholesome feeling of independence and of accomplishment, the value of which cannot be gauged in dollars and cents.

From the point of view of ceramics the development has been most interesting. Maria began with the manufacture of polychrome ware, as that was the style most commonly being made at the time. The shapes were improved, the finish of the surfaces was given greater attention, and the decorations were applied with a surer, more delicate touch (pl. 6; pl. 7, c). Black-on-red ware was taken in hand, the oldtime polish was restored, the vegetable paint properly prepared and aged (pl. 7, a, b). Polished black was also reintroduced; pieces of this sort, with their simple, graceful shapes and high black gloss sold so well that they soon became an important product (pl. 8, b). In 1921, as Dr. Guthe records (p. 24), Maria discovered how to apply to polished vessels dull designs which give the appearance of being etched (pl. 8, a). This method was in its infancy when Dr. Guthe made his study, but it has since proved so remarkably successful that it bids fair to drive out completely the making of plain polished ware. Late in the past autumn there appeared at the

Indian Fair in Santa Fe, some pieces of a warm pinkish color, an entirely new departure in San Ildefonso pottery making, and one which promises much of interest. Developments and changes in design have kept pace with the improvements in



a



b

c

PLATE 8
Courtesy K. M. Chapman

technique, yet the development has been purely Indian, and the basic processes of today are essentially the same as they were a hundred or a thousand years ago.

A somewhat similar revival has taken place at the Hopi towns in Arizona. In 1897 Dr. J. W. Fewkes of the Bureau of Ethnology was excavating the ruin of Sikyatki near First Mesa. Nampeo, a woman of Hano, the wife of one of the workmen, observed the beautiful old bowls that came from the graves, skillfully copied their designs and then searched out beds of clay that could be burned to the exact shades of the ancient pieces. Her work was a direct imitation of a lost style rather than a development, such as Maria's, of a current one; but the effect has been no less extraordinary. Nampeo's bowls and ollas were so attractive that they sold well; they were copied by other women for sale and for their own use, and thus the potter's art of the Hopi was not only greatly stimulated but also radically changed.

In addition to their economic importance, the cases of San Ildefonso and Hopi are of great interest as anthropological phenomena. The stimulus at both places was of course partly commercial, but it could scarcely have acted without an inherent capacity for rapid artistic and technical progress among the Indians themselves, nor could it possibly have brought about such great results without the inspiring example of exceptionally gifted leaders. It is, therefore, quite conceivable that other stimuli, as the discovery of new clays of surpassing excellence, or of fine new pigments, may have proved equally potent in ancient times, and that some of the striking mutations in pottery making, which have so puzzled archaeologists, may have been originally started by such discoveries, and been brought to fruition by the genius of prehistoric Marias and Nampeos.

A. V. KIDDER
Director Pecos Expedition

PUEBLO POTTERY MAKING

BY CARL E. GUTHE

The Pueblo of San Ildefonso is a Tewa village of about one hundred people, situated on the east bank of the Rio Grande some twenty miles northeast of Santa Fe, New Mexico. San Ildefonso was chosen for the present study because its women have always been skillful potters, and under the wise and friendly encouragement of the authorities of the School of American Research at Santa Fe have of recent years been steadily improving in their work without, however, sacrificing any of their native methods. Furthermore, they make more varieties of pottery than are turned out at any other one village. Last, and in some ways most important, the men of San Ildefonso have for a long time been accustomed to work in the various excavations carried on by the School, cordial relations have been established,

the confidence of the Indians has been thoroughly won, and I did not encounter that reticence toward strangers, so characteristic of the Rio Grande Pueblos, because I was known to be a friend of their friends.

To the Pueblo woman pottery making is simply one of the mechanical household tasks, just as dishwashing is among us. Nearly every woman at San Ildefonso is a potter, good, bad, or indifferent. In two cases, at least, women have developed into real artists, and are relieved of other household duties in order to devote their time to pottery making. Each potter of today watched her mother make innumerable pots while she was growing up, and now with a family of her own, she makes pottery just as did her mother. Many steps and small details are carried out for no other reason than that the mother used to do likewise. Obviously every potter has her own technique, which differs slightly from that of others. The daughters in one family work in more nearly the same manner than outsiders, because they all have had the same teacher. In this report the attempt has been made not only to record the essential steps in the making of pottery, but also to note individual variations as an index of the amount of latitude a given process may permit.

Because religion and ceremony play so important a role in the everyday life of the Pueblo Indians, it is certain that there must be involved in the making of pottery and particularly in its decoration, a mass of esoteric beliefs and practises. The Pueblos, however, are so loath to refer in any way to the mystical side of their existence, and, if it is even mentioned, at once become so suspicious, that it seemed best to steer clear of all allusion to such matters. The present report, therefore, confines itself to a description and discussion of the purely technical side of the potter's art.

The writer is indebted to the staff of the School of American Research, especially to Mr. Kenneth M. Chapman, for the facilities given for and the interest shown in the work; also to the Governor of San Ildefonso, Juan Gonzales, for his cordial coöperation. Of the eight informants used, Maria, wife of Julian Martinez, and Antonita, wife of Juan Cruz Roybal, should specially be mentioned because of their constant patience, and their willingness to answer the many questions which to them must often have seemed absurd. Thanks are also due to the American Museum of Natural History, New York, for the photographs of pottery making at Zuñi (pls. 28-31) which are introduced in this paper for comparative purposes.



a



b

PLATE 9
WINNOWER CLAY

- a. Hand method. As the material is sifted through the uplifted hands, the wind blows the fine, pure clay upon the shawl at the left, while the heavier impurities drop back on the pile.
- b. Basket method. The clay is repeatedly tossed upward from the basket, until all the fine particles have been blown out and have fallen upon the shawl.

RAW MATERIALS; COLLECTION, AND PREPARATION

INGREDIENTS

Red Clay^[5]

The region most often visited for obtaining this clay is directly south of San Ildefonso, about a quarter of a mile from the village itself, in the arroyos of the low hills.^[6] The deposit lies directly under a sandstone ledge and spreads over a considerable area. The clay is in the form of soft earth, easily scooped up with the hand. In gathering it the Indians first scrape off the top half-inch or so, which contains impurities such as small pebbles and twigs. When an area from two to three feet in diameter has been cleared in this way, the clay itself is scooped into a loose pile in the centre of the space, and transferred by the cupped hands into a woman's shawl or a gunny-sack, and so transported to the house on the back.^[7]

One method of cleaning the clay may be used either at the beds or at home. The material is winnowed like wheat to remove small pebbles and fine gravel, either of which will ruin a vessel. The larger impurities are of course picked out by hand. The cloth containing the clay is placed in front of the woman and a second cloth or shawl is spread on the ground to leeward of her. A gusty wind requires several shiftings of the shawl, to the woman's evident annoyance. A double handful is lifted and allowed to sift through the fingers; the fine particles and dust are blown upon the shawl by the wind, while the heavier pieces fall again upon the pile of uncleaned clay (pl. 9, a). The height to which the hands are lifted, varying from three to five feet, the speed of the movements, and the rapidity with which the material is allowed to sift through the fingers, all depend upon the force of the wind. Sometimes, instead of lifting the clay in the hands, it is allowed to flow over the side of a shallow basket tilted toward the woman at the level of her shoulder; the coarser particles fall straight down in front of her on the pile of uncleaned clay. Another variation is to toss the clay into the air from a shallow basket (pl. 9, b). The movement is repeated quickly and rhythmically; the wind as before blowing the fine stuff on the shawl, while the coarse particles fall back into the basket. After this tossing has been repeated a dozen or more times, the residue, which consists mostly of coarse impurities, is thrown away. The process occupies approximately half an hour, depending somewhat upon the amount of clay handled. The cleaned clay, which is now ready for mixing, is by no means entirely dust; it contains no lumps, but flaky particles fully three-sixteenths of an inch across are often found in it.

If at the time of gathering the wind is not strong enough for winnowing, the uncleaned clay is brought to the house. Here it may be kept for a windy day, or may at once be sifted through an ordinary kitchen sieve of medium large mesh.

The clay, when cleaned, may be stored in a variety of receptacles (pottery vessels, sacks, or boxes), or, if it is the intention of the woman to use it within a few days, it is simply left in the shawl. During the fall the Indians gather great quantities of clay, and pile it on the floor in the corner of a room for use during the winter when the clay beds are frozen.

The preparation of the clay for use consists of two processes, mixing and kneading. The mixing, which as a rule immediately precedes the kneading, consists of the addition of temper (see page 21). This is done while both ingredients are dry. Different varieties of clay are not mixed together. The work is done on a piece of canvas, an opened cement-sack, or the inner side of a skin (usually that of a goat or dog).^[8] Either the clay or the temper may be put on the mixing surface first; the other ingredient is then added, and the whole is sifted through the fingers until the mass is uniform in color. The Indians have no definite idea of the necessary proportions. They judge simply by the color of the resulting mixture. The proportions used by three different informants seemed to be about one-third temper and two-thirds clay. Before mixing, the clay is reddish brown; the addition of temper lightens the color several shades.

The scrapings from partly finished vessels (see page 54) and the ground-up fragments of pots which have cracked badly in the course of sun-drying are used a second time. This re-used clay, since it already has temper in it, does not need to be mixed over again. It is mingled with newly mixed clay just before kneading, or is kneaded by itself, as occasion demands.

The kneading is usually done on the same canvas or skin that was used for the mixing. If a large amount of clay is to be treated at one time, a quantity of water is poured into a hollow formed in the centre of the pile. Later, as the worked clay or paste approaches the proper consistency, water is sprinkled over it with one hand, just as clothes are sprinkled, at first copiously, later more sparingly. At the beginning, the entire mass is kneaded until the water has been thoroughly absorbed. When the paste is wet, it becomes of course considerably darker. It is then divided into masses which can easily be handled, about the size of two large loaves of white bread. The woman works the paste to a uniform consistency in exactly the same way that dough is kneaded (pl. 12, a), and piles the masses, now ready for use in moulding, on a board or canvas, covering then with a piece of cloth to keep them damp until they come under the potter's hands. The consistency is that of putty, just dry enough to crack if pinched. The mixing and kneading can be completed in half an hour.^[9]



a



b

PLATE 10

- a. Indian digging material for tempering pottery from an outcrop near the village.
- b. Guaco, or Rocky Mountain Bee Plant, from the juice of which is made black paint for decorating pottery.

White Clay

White clay is used interchangeably with red, but the two are never mixed. It is obtained in the bad-lands to the northeast of San Ildefonso, from the side of an arroyo at some distance from the village. The Indians dig it, bring it home, and prepare it in exactly the same manner as the red clay. The color, when dry, is greyish white; when wet, brownish grey.

Temper

The Indians of San Ildefonso obtain their tempering material from outcrops along the Santa Fe-Española road, in the vicinity of the landmark known as Camel Rock. Near the crest of the first high hill crossed by the road north of Camel Rock, just west of the road, in the eastern bank of the cut of the old abandoned road, is a small cave made by the digging of this temper (pl. 10, a). There is another outcrop in the ridge of which Camel Rock is a part. This mineral is a very light grey in color. When freshly exposed to the air it is soft and crumbly—fragments as large as the fist can be crushed in the hand—but after being exposed for some time it hardens. When chopped out with an axe or hatchet it takes the form of small fragments and fine powder. There are veins and nodules of a hard yellow rock in the stratum, which are chopped out and thrown away. The source of the temper is at such a distance from the pueblo that trips to it are made only at long intervals when a large quantity is secured in cement-bags, and carried home in a wagon. Two cement-bags may be filled with from fifteen to twenty minutes' work at the quarry.

Some of the Indians, while still at the quarry, pound the temper with an axe or hatchet, or a stone, to break the larger pieces and remove the coarser impurities. Others wait to clean it until reaching the house. In the latter case the temper is spread on a cloth or skin; the skin being preferred as offering a smoother, more resistant surface. It is then broken up by trampling it with moccasined feet. Formerly the Indians ground the temper on their metates. The final step in the preparation is to pass the material through a sieve to remove the smaller impurities such as sand and crushed rock. Before the sieve was known, the Indians spread the temper in a very thin layer on a board or skin, and went over it carefully and painstakingly with their fingers, feeling for any slight irregularities or rough places. The temper when cleaned is a fine powder, which looks and feels like dry cement. It is stored in old cement-bags ready to be mixed with the clay.

The purpose of the addition of tempering material is to counteract the tendency of the pure clay to crack during the shrinkage that takes place in sun-drying and in firing. If, however, there is too much temper, the clay becomes “short” and loses its adhesiveness. The Indians understand this fully, for if vessels crack badly during sun-drying, they are ground up, and more temper is added to the paste before it is used again; if, on the other

hand, a pot will not retain its shape while being moulded, it is pulled down, more dry clay added to the paste, and the mass kneaded.^[10]

Cooking-Vessel, or Apache, Clay

This is very different from the two other kinds of clays used. It is light brown in color, with a large quantity of fine mica flakes in it^[11]; it also has a distinctive odor. When wet it smells cleaner and fresher than the red or white clay, like a newly plowed field after a shower. Some of the San Ildefonso Indians obtain this clay from Las Truchas, a mountain district to the northeast of the village^[12]; others get it from Santa Fe Canyon. It is dug either with the hand or with a small stick, and is brought to the pueblo in the form of small lumps. While still in this condition, it is stored in vessels or bags.

The preparation for use is very different from that of the other clays. The lumps are put in a receptacle, such as an enameled washbasin, and sufficient water is added to saturate them. If too much water has been mixed in, the mass is put in the sun to dry out for a while. After the clay has been thoroughly softened by the water, the next step is to remove the pebbles. A small handful is carefully and slowly squeezed and kneaded in the left hand, while the thumb and forefinger of the right are used to pick out the numerous small pebbles as they are encountered. After most of these have been removed, the clay is put on a canvas or skin, and a small part of it at a time is flattened out with the heel of the right hand, pressing away from the body, and against the cloth or skin so as to produce a very thin layer, in which the smaller pebbles are readily felt and from which they can easily be eliminated. Finally when this has been done with all the clay, the whole mass is kneaded together, just as the red and white clays are kneaded. When a uniform consistency has been attained—that of putty—the mass is ready for use. The removal of pebbles and the kneading consume from one-half to three-quarters of an hour. The pebbles which have been removed are placed in a small lard-pail containing water, and when the clay adhering to them has dissolved, they are thrown away and the clayey water is used in the process of moulding the paste.

The most important difference between this clay and the others is that no temper of any kind is mixed with it; the clay, after being cleaned, is used in the vessels just as it came out of the claypit. The fine mica flakes probably act as temper.

SLIPS AND PAINTS

Native Slip^[13]

This is a white, flaky, fairly soft mineral, which is used in solution to give a white outer coating to the bodies of vessels. It is found at some distance from the pueblo, beyond the bad-lands which are visible to the northeast. When the Indians reach the place they pick up small sticks to use as digging tools, but if the deposit happens to be soft enough, the material is merely scooped out with the hand. It is carried home in shawls or bags, and placed in the sun to dry thoroughly; then stored in pottery vessels, usually ollas. It receives no further treatment at this time. When preparing it for use, the Indians simply place it in water as it is, in small lumps. The container is either a small pottery vessel or an enamelled

basin. A sufficient amount is mixed with the water to give it the color of milk, but not to thicken it. Occasionally the consistency of thin cream is obtained. Undissolved lumps remain in the solution.

Santo Domingo Slip

This substance, known to the Indians of San Ildefonso either as “Santo Domingo white” or “Cochiti white”, is usually obtained from the Santo Domingo people; it is dug, according to some informants, in the same manner as the native slip. In color and general appearance it is exactly like the latter, but its surface feels more soapy. Santo Domingo slip has largely replaced the native product, because when applied it does not need to be polished with a stone, as does the native slip. It is prepared for use in exactly the same manner as the native slip.

Red Slip

This is a rather loose red earth, which is dug either with the hands or, if conditions demand it, with a small stick. The San Ildefonso Indians obtain it near Santa Fe, but not all of the women know the exact location of the beds. One informant said it was found in Santa Fe Canyon, east of the town, a short distance below the Apache clay beds, at a spot where the convicts from the Penitentiary get material for their bricks. There are three different colors of clay at this place, red, yellow, and white. The red is the clay under discussion. The white is used to color the women’s moccasins.^[14]

This slip is handled in much the same manner as the other clays. It is brought home in the usual receptacles—either shawls or bags—and placed in the sun to dry thoroughly. It is then stored, usually in ollas or other earthenware vessels, without further treatment.

There are two ways of preparing red slip for use, according to the kind of vessel for which it is intended. For polished black ware, it is simply mixed to a thin solution with water. For decorated red ware, the process is slightly complicated. At some previous time, equal parts of temper and native slip have been mixed in water and allowed to dry in cakes. When desired for use on a vessel, the cakes are broken and redissolved in water. A sufficient amount is put in to give the water an opaque, milky color, but not to thicken it. To this mixture is added a thin solution of the red slip. There is, apparently, no definite rule in regard to the amount of the red solution to be added. The woman simply puts in enough to give to the slip, when applied to a pot, the proper shade of red. In some cases this mixing of the red and white is done once for all at the beginning of the application of the red slip. In other cases the woman has a bowl of each solution beside her, and from time to time, as she works, adds some of the red to the white. In each bowl lumps of the undissolved substances still remain. Occasionally the slip is stirred to the consistency of cream, but nearly always it has that of water.

Orange-Red Slip

This substance is a yellow clayey earth, in texture somewhat like the two white slips. It occurs in the “Valle” to the west, beyond the first Jemez range, near Ojo Caliente. It is dug

with a stick, in the same manner as the native white, and is carried home in shawls and bags. Before being stored it is put out in the sun to dry thoroughly, then placed in ollas and kept until needed. Like the other slips, it is prepared for use by being mixed with water. A saturated solution is made, but the consistency remains that of water.

This material, which in solution is a brilliant yellow, is used for two purposes—as a slip to color the bases of bowls and ollas, and as a paint to supply the red elements of polychrome designs. After being fired it assumes an orange-red or burnt-sienna color.^[15]

Black Ware Paint

This is a paint used for making matte designs on polished black ware, a new departure in decorative technique first used by Maria and Julian Martinez of San Ildefonso, in June, 1921. The substance is a hard yellow stone, said to occur in the “Valle”, west of the Jemez range, near Ojo Caliente, in the same district as the orange-red paint.

The first step in preparing the paint for use is to scrape the stone with a knife. The resulting powder is mixed with water, and there is then added about one-fourth as much dissolved “guaco” (see below) as there is paint. It is said that the purpose of the guaco is to make the paint “stick” to the polished surface. This paint, when ready for use, is kept in a small earthenware or china dish. The consistency of the mixture, like the other paints, is that of water.

Black or Guaco Paint

This is the only vegetable paint used by the San Ildefonso Indians for the decoration of their pottery. It is obtained from a very common weed, known as the Rocky Mountain Bee Plant, or Guaco,^[16] which grows in the moister flats of this district in great profusion. It flowers late in July and early August, and the seedpods open toward the first of September. The Indians say it grows at a given place only in alternate years. In favorable spots the plants attain a height of five or six feet (pl. 10, b).

The plant is gathered during April and May, when the shoots are only six to ten inches high, and therefore tender. Great quantities of it are carried home in shawls and bags. It is placed with water in Apache-clay cooking-pots, and is boiled over an open fire, never on a stove. As guaco has a bitter taste (“hot like pepper”, the women describe it) when not cooked sufficiently, it is necessary to boil it until this unpleasant flavor has disappeared. Over a hot fire half a day is long enough, but often a whole day is required. Then the liquid is drained into wide-mouthed earthen vessels and placed in the sun to harden into a solid, black, rather sticky, mass. Some women reboil the liquid until it thickens, before placing it in the sun. Metal containers are not used, because they are said to spoil the color. The informants stated that hardened guaco was not eaten. When it has become completely solidified it is set away for future use in painting. It will keep indefinitely.

The residue in the pot, after the liquid has been poured off consists of tender stems and leaves. This is eaten by the Indians, and is not unlike spinach.

The hardened guaco juice is stored at least a year before it is used in painting; the longer it is kept the better black it produces. If it is used during the first year, the resulting color on

the vessel is a streaky blue-black.

Guaco is prepared for use in painting in the same manner as the other pigments. A chunk of the hardened mass is broken off and dissolved in water. One informant makes a solution which, although black, has the consistency of water. The others use a syrupy solution almost like thin molasses. The paint has a distinctive, not unpleasant, vegetable odor, and is rather sticky. Any small, wide-mouthed Indian bowl, not too shallow, is used as a paint-cup. In this bowl there is always a stirring-stick, usually a splinter from a board.

FUEL

Manure

Both cow-manure and horse-manure are used for fuel in burning pottery, but sheep-dung, when obtainable, is preferred, because it is thought to make a hotter fire.

The cow-manure is collected from the corrals, and while still soft is patted into large circular cakes, sixteen to twenty inches in diameter. These, after being hardened in the sun, are stored in a dry room until needed. Such circular cakes, with the hand-prints showing, are usually employed for making the top of the oven.

The horse-manure is obtained from the corrals in the early spring, where during the winter it has been trampled down by the horses into a compact layer a foot or more thick. When the corrals are cleaned during April this deposit is cut with an axe into chunks roughly two feet square. After being removed, the squares are split with an axe into slabs several inches thick, which are then stood up against the bases of house or barn walls to dry in the sun, and later stored indoors until needed. Disused bread-ovens sometimes serve as temporary storage places. Further splitting of the slabs may be done at the time of firing. When used in the ovens the slabs are from one to two inches thick.

Sometimes the slabs of manure are placed in the sun for a day or two before being burned, but often they are taken directly from the store-room to the fire, where they are supplemented by dried manure collected from the pastures.

During the burning of polished black ware (see p. 74) the oven is smothered with loose manure, which is brought in galvanized washtubs directly from the stables and corrals, either at the time of burning, or on the preceding day.^[17]

Kindling

For starting the fire in the oven no wood but cedar is ever used. This is cut into pieces from six to eighteen inches long and is split into fine kindling at the time of burning.

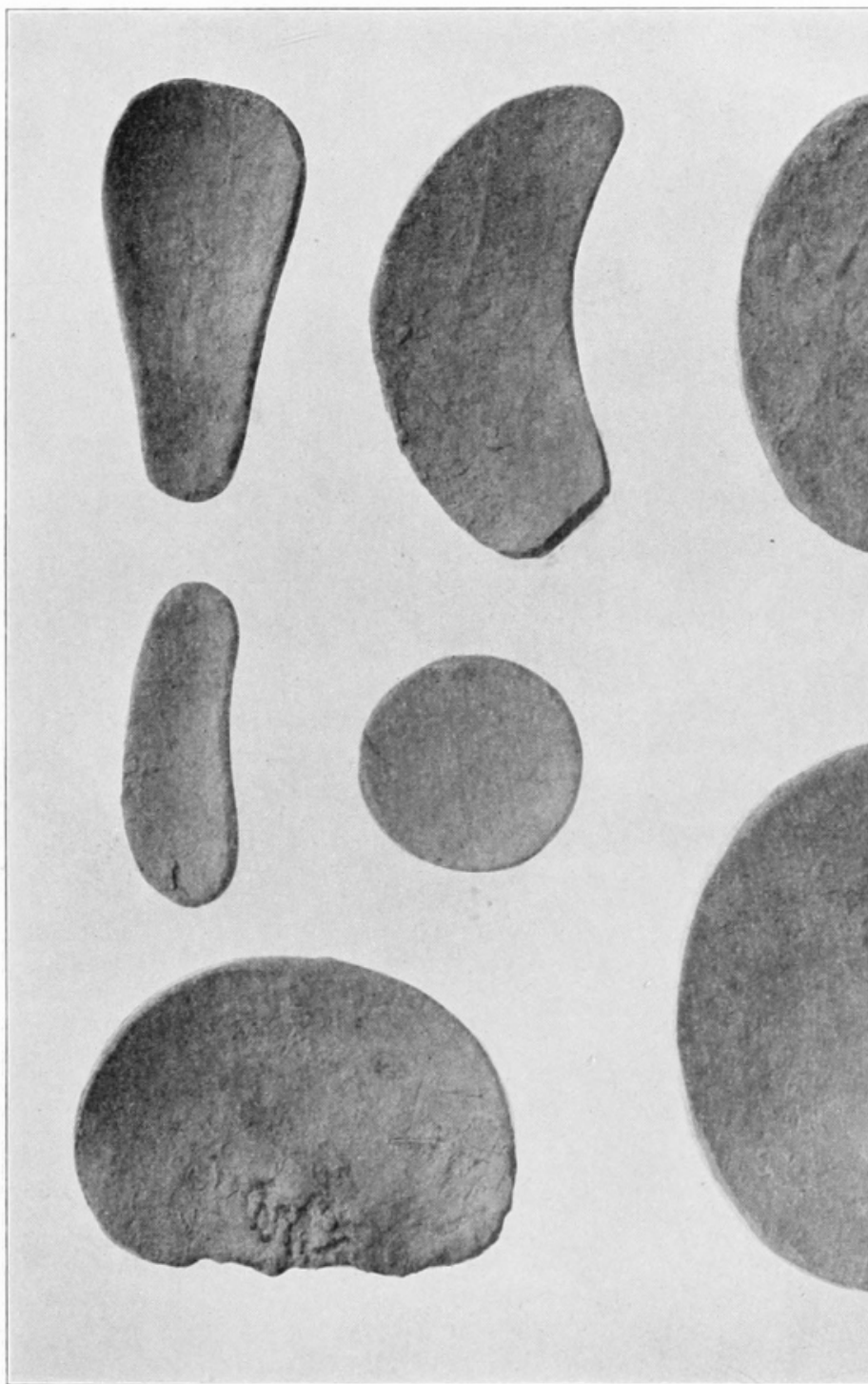


PLATE 11

Gourd spoons, or kajepes, used to shape and smooth the walls of vessels.

PARAPHERNALIA

PRIMARY PARAPHERNALIA

Earthenware Moulds or “Pukis”

These moulds are usually the bases of broken bowls or ollas. Occasionally pieces of pottery are made especially for moulds; in such cases, although they are fired, they are neither finished carefully nor decorated.

When a mould is to be used, a small amount of temper or wood ashes is placed in it, so that the vessel being moulded will not stick to it.^[18]

Gourd Moulding Spoons or “Kajepes”

These spoon-like implements (pl. 11) are made from pieces of gourd-rind, usually from broken rattles or dippers. They vary greatly in size, ranging in diameter from a little over one inch to as much as four, or four and a half, inches. There are also many different shapes: round, several forms of ovals, and a few which have one concave edge. The various shapes are used for the different parts of the vessels. As a general rule the larger kajepes are used on the larger vessels. Each woman has from four to a dozen of these implements, which are distinctly individual, for one potter shows awkwardness in handling the spoons of another. It is said that potsherds were formerly used for the same purpose as these gourd spoons.^[19]

Scrapers

Two kinds of scrapers are used at the present time. One is the top of a baking-powder can, which is popular because its shape makes it possible to apply it to any part of a vessel. The other scraper is an ordinary kitchen case-knife. The scraping of the vessels is done after they have been dried in the sun. It is said that potsherds, with edges sharpened on a piece of sandstone, were formerly used.

Polishing Stones

Very smooth, fine-grained pebbles of various colors are used to put the final finish on certain kinds of pottery. They vary considerably in size, from three-quarters of an inch long to three, or even three and a half, inches. It is impossible to classify them by shape, except that the larger ones are usually more flat than spherical. Most of the stones appear to be waterworn pebbles, without any acute angles or sharp edges. Usually two surfaces of the stone are highly polished, most frequently at the ends, where they come in contact with the vessel. Some specimens show signs of great wear.

Each potter has several polishing stones, the number varying from seven to sixteen; they are distinctly the personal property of the potter, and apparently have a semi-sacred significance. For the most part they are heirlooms, handed down from mother to daughter, but additional stones are picked up from ruins which the potters have visited. One informant has four stones which her mother gave her, and three which she found at the ruined pueblo of Puyé. Another informant uses a stone that belonged to her great-grandmother, although her favorite is a small one which she found herself at a nearby ruin. Stones are sometimes lent by one potter to another, but they very seldom find their way outside the family group.
[20]

Paint Brushes

These brushes are narrow slivers of the leaves of yucca or soapweed, from five to six inches long, and from an eighth to a quarter of an inch wide. For a distance

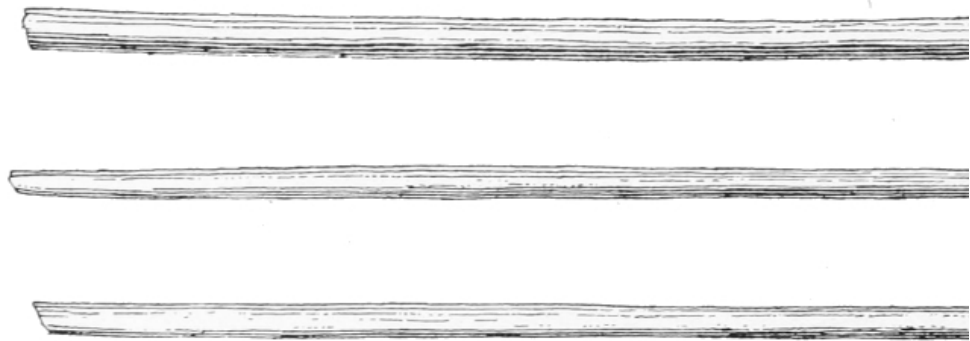


FIG. 5. Slivers of yucca leaf with shredded ends, used as paint brushes for decorating pottery.

of about an inch from one end, the fibres are chewed and thus separated. Most of the chewed fibres are then cut off, and the number remaining determines the fineness of the point (fig. 5). For very thin lines a brush of only one fibre is used. When out of service, the brushes are kept in some receptacle which will protect the shredded ends. When the implement is dry the chewed fibres become stiff and rather brittle, and must be handled with care; before using they are soaked in water for two or three minutes in order to soften them.
[21]

SECONDARY PARAPHERNALIA

Carrying and Storing Receptacles

The receptacles used for bringing clays and other ingredients from the pits to the village vary somewhat according to the distance that the material must be carried. If the beds are close at hand, the material is placed in a shawl spread flat on the ground. The shawl is then picked up, the corners are tied, and the bundle is carried on the back. Sometimes gunny-sacks and cement-sacks are used instead. If the beds are at some distance from the pueblo,

the material is placed in sacks and brought home in the body of a wagon. As a rule the ingredients brought home in shawls are transferred to sacks or boxes for storage. Paints are kept in ollas, with the exception of the guaco paint, which is allowed to remain in the bowl in which it is dried.

Mixing Surfaces

Whenever ingredients are to be mixed or crushed, the work is done upon some sort of extra surface laid upon the floor. This surface is about a yard square, and may be, apparently, of any suitable material—a bit of old canvas, an opened-out cement-sack, or a small skin such as that of a goat or dog. Similar surfaces also serve for mixing clay and temper, for cleaning temper, for mixing the paste with water and kneading it, and finally for holding the kneaded clay during moulding. Partly finished vessels are placed upon such a surface at various stages of construction. For winnowing clay, the women's shawls are used.

Boards

No boards are placed beneath the pukis, or moulding trays, while they are being used on the hard floor of the house, but in the courtyard boards are used under them in order that holes may not be worn in the ground during the constant turning necessary in moulding vessels. When small vessels are completed they are placed in rows on a board, where they remain during the early stages of the drying process. Rectangular, flat-bottomed vessels are built directly on boards, or on low footstools made of short boards with legs at each end.

Water Containers

While the potter is at work, she always has beside her a small lard-pail partly filled with water, which serves to keep the kajepes damp, to moisten parts of the vessel, and from time to time to clean her hands. When the kajepes are not in use, they are usually placed in the pail. At San Ildefonso such a pail is now the only form of water-container; formerly pottery bowls were probably used.

Mops for Slips

These mops are the same for all slips. Each consists of a rag, folded and refolded until it is about an inch or an inch and a half wide and from two to two and a half inches long. One end of the rag or mop is held between the thumb and the first two fingers of either hand; the other end serves as a wide paint brush. The potters state that in former times a piece of leather or skin served the same purpose as the cloth mop.^[22]

Paint Receptacles

Various receptacles hold the slips and paints that are ready for use. The white slips are usually mixed and kept in enamelled basins and pans, because a large quantity is prepared at one time. The red and orange-red slips may be in small pottery bowls, or in china sauce-

dishes. The guaco is invariably in an Indian bowl. Undoubtedly at one time native bowls were used exclusively, and there may even have been bowls of special shapes or designs for this purpose; at present, however, no trace of such a custom remains.

Wiping Rags

Wiping rags or, more simply, the aprons of the potters are used at various stages of the work—after scraping, after polishing, after the slip has dried, and after burning. Apparently any convenient piece of cloth is satisfactory.

Accessories in Firing

When the vessels are to be fired, various accessories must be at hand. Four or more half-bricks or stones are needed upon which to set the grate. The latter may be either a discarded stove-grate, or a collection of strips of iron, metal barrel-hoops, and odds and ends, placed together to form a grill.^[23] Small stones and tin cans—such as small condensed milk cans and meat cans—are needed as props to keep the fuel from touching the vessels during the burning. Flat pieces of tin and the covers of large lard-pails serve at times to close spaces between the fuel cakes, and to cover the oven-top, to support which metal rods are often used. Lastly, pokers are needed to handle the fuel and to remove the hot vessels. These may be wooden or iron rods, broomsticks, old shovels, or even pitchforks. Each potter has her own collection of the above materials and implements, stored away ready for use. These collections of apparent refuse vary to a surprising extent.

MOULDING

For the purpose of this report the vessels may be divided into four large groups: bowls, ollas, cooking vessels, and unusual shapes. With respect to size there are two classes of bowls: those less than four inches in diameter, and those larger than that size. In shape bowls may be either wide-mouthed (pl. 6, b, g) or constricted-mouthed (pl. 6, a, c). The ollas all have full, globular bodies and relatively small orifices with or without flaring lips (pls. 7, 15, a). As a rule, the cooking vessels are shaped like small ollas. Under unusual shapes fall all the types and forms which are not of the first three classes, such as terraced (prayer-meal) bowls, double-mouthed vessels, double bowls or "baskets", and vessels with handles.

The moulding of each of the above forms may be said to consist of four principal steps: the making of the base, the building of the walls, the shaping, and the finishing. Very briefly, the process is as follows: The potter first forms a pancake-shaped pat of paste from six to eight inches in diameter; this she presses into the mould, or puki, to form a base. Then the walls of the vessel are built up by the addition of successive ropes, or rolls, of paste laid one upon another. The small bowls are the only exception, for they are formed in the hands of the potter from a single lump. In some cases the building is done all at one time; in others, and always with the larger vessels, a few rolls are added, then the piece is set aside to dry a little before the addition of a few more rolls. The potter usually builds two or more vessels at once in order to permit work upon one while the other is undergoing a brief period of drying. The preliminary shaping of the vessel is done either in the course of the building or after the building has been completed. The obliteration of the junctions between the rolls and of finger-marks is accomplished with the kajepe, or gourd-spoon, and further use of this implement aids in giving the vessel its shape. The final step, the finishing, consists of going over the entire vessel carefully, first with the kajepe, then with the fingers, to remove slight irregularities to even the lip and rim. The finishing is a slow, exacting process, and the difference between the artist and the mere pot-maker comes out at this stage of the work.

The moulding, although usually done in the house, may also take place in some shady spot outdoors. If a woman begins her moulding in the morning, she will commonly devote the entire day to it. By evening she will have completed from five to thirty vessels, the number depending partly upon the size of the pieces and partly upon the rapidity with which she works.

The potter sits upon her heels on some kind of mat; it may be a sheepskin with the wool side up, a piece of canvas, or a small woven rug. Her attitude is changed only when stiffness requires it. Thereupon she sits with her feet out in front of her; or with one leg bent, the foot flat on the ground, the knee beside her shoulder, and with the other leg under her or fully flexed in front with both the knee and foot upon the floor. Toward the end of the day the potter usually sits against a wall to rest her back, for any one of the postures assumed is tiring. At the end of the day the women frequently complain of stiff backs and sore abdominal muscles, caused by the almost constant stooping posture. Some potters cover

their laps with shawls, others wear aprons, and still others use no protection whatever for their clothing.

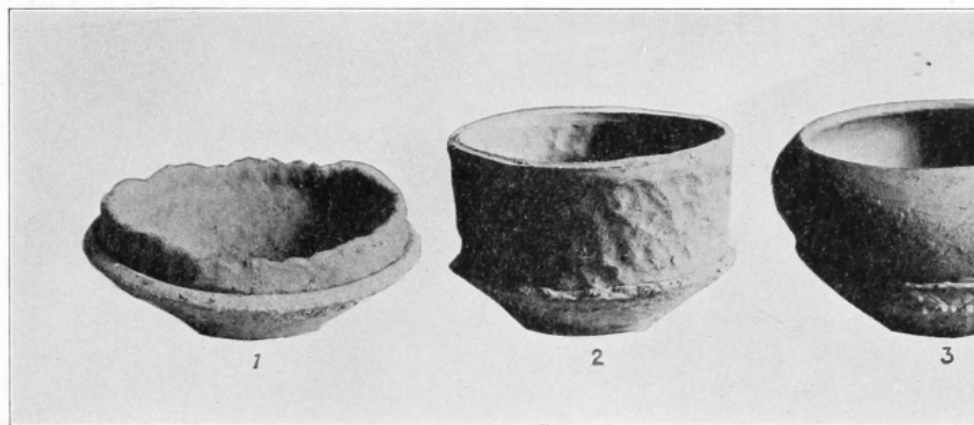
When the potter is sitting upon her heels, the mould, or puki, is laid directly in front of her upon the floor. If she is sitting with her legs extended, it is either held in the lap or placed close to her right side. In the house the puki rests directly upon the hard earthen floor; outdoors, a short board is placed under it to prevent its constant turning from wearing a hole in the relatively softer ground. In the puki is sprinkled a thin layer of temper or wood-ashes to prevent the vessel from sticking to it. At the right of the potter, within arm's reach, on a canvas or a piece of cloth, is a mass of prepared clay. Rarely this reserve paste is on the left. If the paste is in constant use, it is not covered unless exposed to the sun; when, however, the moulding is abandoned for even a few minutes, a corner of the canvas or cloth is thrown over it. A small lard-pail partly filled with water is an indispensable accessory during moulding. This is usually kept between the potter and the paste; occasionally it is on the left, sometimes near the puki beside the potter's right knee. The gourd moulding spoons, or kajepes, are also placed within easy reach, on the floor near the puki, beside the waterpail, or in the pail itself. The potter is then ready to begin moulding.

The first step is to ascertain by a little handling whether the paste is of the proper consistency. If it was kneaded the day before, it may prove to be too dry; in such a case the hands are well moistened and perhaps a little water is sprinkled on the paste; it is then rekneaded for a few minutes. Then a handful is scooped out of the mass with the right hand. If the first handful is not sufficient, a second scoop gives the required amount. It is next worked with the hands for half a minute to five minutes; if it is of just the right consistency, it is hardly worked at all; if it is too dry, the hands are occasionally dipped in water while it is being worked; if it is too wet, kneading continues until it is sufficiently dry. There are three distinct movements in this final working of the paste. First, the handful is pinched and squeezed by opening and closing the hands; if any gritty particle is encountered, it is picked out with the thumb and forefinger of the right hand and snapped upon the floor. The second movement consists in rolling the paste between the two flattened hands. When the roll is about six inches long, it is doubled and rerolled. This may be repeated from two to half-a-dozen times. Occasionally this step is entirely omitted, especially when a pat is being made. In the third movement the paste, now a spherical lump, is slapped back and forth between the cupped hands, which are held from six to eight inches apart. Sometimes this third movement precedes the second. The handful of paste is now ready to be made into a pat.

There are two ways of forming a pat. In the first one, the spherical lump is held on the open left hand and pounded flat with the right fist; from time to time it is turned a few degrees by a slight tossing motion. It is then finished by patting



a



b

PLATE 12

a. Kneading clay preparatory to modeling a vessel.

b. Bowl-moulding, primary stages. 1, Base formed in mould, edge crimped to receive first roll (cf. pl. 13, a); 2, Walls built to full height; 3, The same

piece after shaping and smoothing.



a



b

PLATE 13

a

Applying the first roll of clay to the base. The roll is held in the left hand, and is flattened and pinched into place in the inside of the rim by the fingers of the right hand.

b

A bowl-base to which four rolls of clay have been added; the potter is making a new roll. Note the junctions of the roll on the interior of the vessel.

it with the flat right hand until it is fairly smooth and of uniform thickness; in doing this the pat is turned by slapping it from the left hand to the right and back again. During this interchange, through the natural placing of the hands, the pat is turned through an angle of ninety to one hundred and twenty degrees. When the pat has been completely formed by this method, it is like a large pancake, six to eight inches in diameter and from three-eighths to half an inch thick. It is then pressed firmly into the puki, and the edge is turned up and crimped with the thumb and first two fingers of each hand to form a slight rim (pl. 12, b, 1).

In the second method, the spherical lump of paste is first pounded with the heel and then with the upper knuckles of the right hand; it is turned meanwhile by the tossing motion of the left hand. The next movement is the same as in the corresponding stage of the first method, namely, patting the paste with the flat right hand while turning it by transferring it to the right hand and back. Finally, before the pat is put into the puki, the edge is bent up

and smoothed off a little, so that a shallow flat circular tray is formed instead of the slab obtained by the first method. This tray is then placed in the puki, where it is pressed down very firmly with either the heel of the hand or the balls of the fingers, in order to insure a solid uniform base and to expel any air-bubbles in the paste.

Another handful of clay is now scooped out of the mass on the canvas to the right and worked to the proper consistency as before. From the resulting spherical lump the potter very skillfully forms a roll of uniform diameter by a backward and forward motion of her two hands placed palms together (pl. 13, b).^[24] There seems to be a tendency for the first part of the roll to be of slightly smaller diameter than the last part; if this occurs, the roll is reversed and partially rerolled. One end is then taken in the right hand, while the left hand holds it near the other at a point chosen to prevent too much sagging at the middle. The potter then places the end which is in her right hand against the inside of that part of the edge of the pat nearest her.^[25] The roll is pressed against the edge just enough to hold it in place (see pl. 13, a). The thumb is on the exterior, the fingers on the interior of the edge. Usually only the first two fingers are used, although the third finger is occasionally brought into contact with the paste. Then, as the puki is revolved counterclockwise,^[26] the rest of the roll is pressed against the edge of the pat. If the roll is not long enough completely to encircle the pat, another is made and placed on it in a similar manner. When the edge has thus been completely encircled, the unused remainder of the roll is pinched off and tossed back on the mass of paste on the canvas. The potter then pinches the roll more firmly to the edge of the pat. This is done with both hands. The thumbs, almost touching, are on the exterior, the fingers on the interior of the vessel. The puki is usually moved counterclockwise, although the same potter will sometimes revolve it clockwise. During this process of pinching, the roll is flattened until it assumes in cross-section the shape of a very much elongated ellipse (fig. 6, b). An attempt is made to keep the thickness of the side as nearly uniform as possible, and in order to accomplish this it is often necessary for the hands to be moved back over a part already flattened.

The diameter of the rolls, except for cooking-pots, varies from half an inch to slightly over an inch, according to the size of the vessel being built, and the individual doing the building. When a roll is pressed against the interior of the rim of the growing vessel,^[27] from a half to three-quarters of it lies below the level of the rim (see fig. 6, a). Then, when the roll is flattened, the junction-plane between it and the preceding roll will not be horizontal, but will slope downward sharply

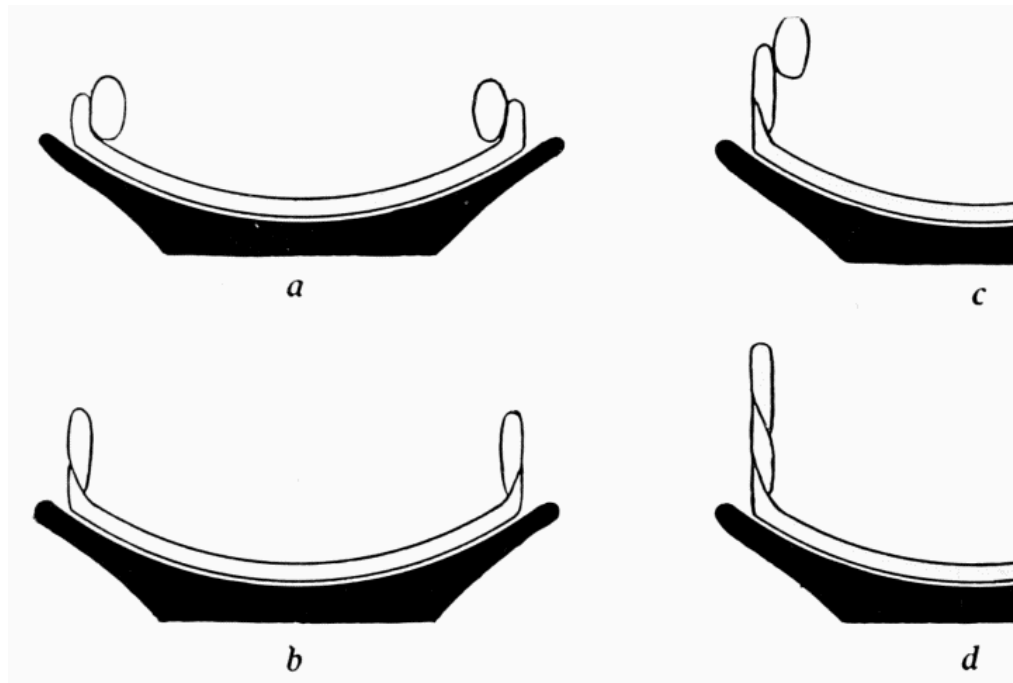


FIG. 6. Diagrammatic sections of a bowl on the mould.
a. Base made and first roll applied to inner side of rim. b. First roll flattened and welded on. c. Second roll applied. d. Second roll flattened and welded on.

from the exterior (see [fig. 6](#), b, d), thus producing a broad holding surface and minimizing the risk of the vessel's cracking along the junction-plane. In other words, the junction-line between any two rolls on the outside wall of the vessel is considerably higher than the corresponding line on the inside. The flattened rolls, in all but cooking-pots, are from an inch and a half to two inches wide, but because of the overlapping just described the distance between the junction-lines is considerably less (see [pl. 13](#), b, and [pl. 15](#), a).

During the process of flattening the roll, some potters obliterate the junction-line on the exterior by rubbing it with the first and second finger of the right hand, and by the addition of small pellets of paste in the more conspicuous indentations. Others consider it unnecessary to obliterate this line during the building of the vessel, in spite of the fact that the roll may break away because of not being firmly welded. Certain women smooth the temporary rim after flattening each roll, others do not. When the flattening has been finished, the puki has made a second complete revolution, and the junction of the two ends of the roll is again directly in front of the potter; but if a few additional touches are required she may again shift the puki in one direction or another. No potter pays any attention whatever to the location of the junction points of the ends of the rolls; in successive rolls, therefore, they may lie directly above one another, or they may be distributed about the circumference of the vessel ([pl. 13](#), b). When one roll has been completely welded on, flattened out, and incorporated into the vessel, another roll is

formed, and is applied in exactly the same manner. Thus the building proceeds to the height at which the shaping is begun. The steps so far enumerated are the same for both bowls and ollas. The further treatment of the vessel varies slightly, according to the shape desired.

Stevenson, describing the moulding of vessels at Zuñi, in 1879, wrote: "When the clayey dough is ready to be used, a sufficient quantity is rolled into a ball. The dough, if worked by a careful artist, is first tested as to its fitness for molding by putting a piece of the paste to the tongue, the sensitiveness of which is such as to detect any gritty substance or particles, when the fingers fail to do so. The ball is hollowed out with the fingers into the shape of a bowl (see [pl. 29, a](#)), this form constituting the foundation for all varieties of earthenware, and assumes the desired form by the addition of strips of clay; all traces of the addition of each strip are removed before another is added,^[28] by the use of a small trowel fashioned from a piece of gourd or fragment of pottery, the only tool employed in the manufacture of pottery. The bottoms of old water jars and bowls form stands for the articles while being worked by the potter" (see [pl. 29](#)).^[29]

Mrs. Stevenson, writing later, also of the Zuñis, said: "In beginning the work a sufficient quantity is first made into a ball and then hollowed out with the fingers until it assumes a conventional bowl shape, which serves as the foundation to be afterward built up and elaborated into any desired shape. The vessel is then formed by the successive additions of strips of the paste long enough to encircle the bowl, each layer being pressed on the brim with the fingers and accurately fitted, the trowel being then skillfully used to finish the joining and to remove all traces of the original separation of the strips. Most of the work of remodelling the vessel into its final shape is done on the inside with a trowel, this implement being used on the outside chiefly to smooth the surface. The clay, if it has been properly worked, possesses sufficient tenacity and plasticity to admit of being pressed and scraped without cracking."^[30]

The following quotation is of interest as showing how closely modern studio-practice, undoubtedly evolved experimentally, resembles Pueblo Indian methods: "The clay for building should be rather soft as it is apt to dry quickly on handling. A plaster bat (mould) should be made. It is first necessary to roll out the clay into cords which should be a little thicker than the proposed walls are to be. These cords should be as uniform as possible and should be rolled quickly to avoid undue hardening. It is best to roll them as required. A roll of clay is taken, one end laid in the center of the bat and the rest is coiled around it in a spiral line. When the disc so formed has reached the proper size, the coils are gently rubbed over with the fingers until they have thoroughly united and the lines of the spiral have disappeared. The clay disc may now be turned over and the rubbing continued on the other side. The circle is cut true and a new coil is laid on the outer edge, thus making a shallow circular tray. In raising the walls it is best to pinch off the roll of clay when one circle has been completed, and the new roll should be begun at another point so that all the joints will not be at the same place. This plan is better than coiling a long roll in a spiral, for in this case one side of the piece will be higher than the other.

“After three rolls have been laid in position, the wall, both inside and out, should be worked like the bottom so that the rolls will disappear and the clay be welded uniformly together. This should be done without water or with as little as possible. The use of water is very tempting. It makes the clay so smooth and seems to help, but it will inevitably make the work sloppy and will tend to soften the walls.

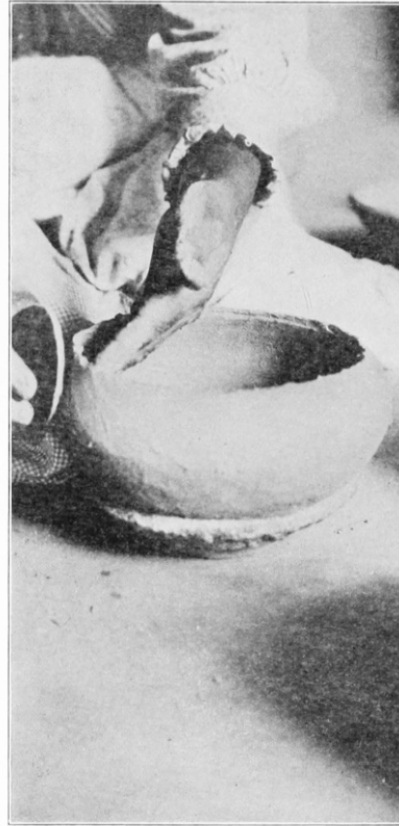
“After three or four rolls have been worked in, the piece should be laid aside for some hours to stiffen. If this be not done the weight of the second building will cause the work to sag and fall out of shape. For this reason it is well to have two or three pieces in hand at once so that there need be no waiting. When the cylinder is of sufficient height it should be allowed to become quite stiff and then the irregularities should be corrected with a little soft clay which is worked into the joints. The whole surface may now be gone over with tools and brought to the required finish.”^[31]

Before describing in detail the moulding of the various sorts of vessels, the accidents which are associated with this process should be discussed. These may result from one of two causes, the presence of foreign matter in the paste or imperfect moulding.

If the clay has been carefully sifted, as it usually is, there should be very little foreign matter in the paste. It is, however, impossible to remove entirely all gravel or stones fragment. If gravel is allowed to remain, it will cause the vessel-walls to scale during firing because of the difference in the rates of expansion under heat of stone and paste. In order to prevent this, the potter while moulding is constantly on the lookout for these small particles, which may be discovered at any stage of the work. They are at once picked out of the vessel, even if it is all but finished. The irregular hollow formed is then filled with a small pellet of fresh paste and smoothed over. The women say that a bit of hard clay does not cause flaking as does a piece of gravel or a stone splinter. Another kind of foreign matter is an air-bubble, which if left in the vessel will cause the same form of accident, because of the difference in the rates of expansion of air and clay. In order to eliminate these small air-bubbles, the pat is pressed down hard upon the puki



a



b

PLATE 14

a

Levelling and smoothing the rim of a bowl after it has been scraped with the kajepe, or gourd spoon. The left hand serves to keep the piece turning and to support the plastic wall.

b

Use of the kajepe in shaping a vessel; the left hand supports the wall within, while the kajepe, held in the right hand, thins the top of the wall and gently presses it inward.

and the rolls are carefully pinched at every point about the vessel. A skillful potter is able to feel an air-bubble even if it is covered with a layer of clay: such a bubble is pricked and the resulting hollow is filled as before.

If the moulding is imperfect, that is, if the rolls are not carefully welded, a weak spot occurs at the junction, which probably contains a thin layer of air. When the vessel is fired, it cracks along this line and is ruined. Such accidents seem to occur most commonly at the bases of large vessels, either because the first roll was not sufficiently welded to the base, or because the pat itself was not pressed into the puki with sufficient force. Since flaws between the rolls cannot be discovered until the firing, they cannot be corrected as in the case of foreign bodies in the clay.

BOWLS

From two to eight rolls may be used to form a bowl, according to its destined size and shape. When the building has been completed, the piece is in the form of a cylinder, with sides either vertical or very slightly flaring (pl. 12, b, 2, and pl. 13, b). As a rule the fingerprints and the irregular horizontal junction-lines may be seen both on the interior and exterior. The next step is the finishing of the temporary rim. This is done with the right hand, the fingers of which assume much the same position that they would for holding a pencil (pl. 14, a). The thumb is on the interior of the rim, the forefinger on the top, and the middle finger on the outer edge. The thumb and middle finger are used as guides to help keep the rim uniform in thickness. The forefinger removes the paste from the higher parts of the rim and transfers it to the hollows. The fingers do not move with respect to each other except to exert slight pressure. The play of the right hand during this process is a gliding backward-forward motion repeated many times, while at the same time the puki is usually kept revolving counterclockwise by the left hand, which also is generally but not always, used to steady the side of the vessel. Occasionally the same potter will turn the puki clockwise. In the former case the right hand is kept in the twelve-to-two-o'clock sector; in the latter in the six-to-eight-o'clock sector. If the rim is smoothed hastily, as it sometimes is at this stage, the puki may not be turned at all. This completes the building of the bowl, which is then set aside for a few minutes, while another is begun.

The next step is the shaping. This is done with a kajepe, or gourd spoon. The kajepe should always be damp when used, and for that reason is often kept in the water-pail. It is somewhat difficult to describe in detail the use of the kajepe, for no two potters work with it in exactly the same way. Often the same potter will show variations in her method of handling the tool.

Each woman has from four to a dozen kajepes beside her while she is moulding. These implements vary much both in size and shape (pl. 11). Most of them are circular or elliptical with all edges convex, although every potter has at least one with a concave edge. Generally the smaller, circular kajepes are used on the interior of bowls, the oval ones on the exterior, and the one with a concave edge on the exterior at the shoulder, where it would be difficult to use the other types. As a rule the larger kajepes serve for the larger vessels. A potter occasionally changes kajepes in the midst of her work, usually because of the edge, which may be rougher on one than on another. The kajepes are held at an angle of from thirty to sixty degrees to the tangent of the bowl at the point of contact, according to the curvature of the part of the vessel being worked upon.

The kajepe is always used first on the inside. This is done in order to press out the sides and to give the bowl something of its final shape before the paste becomes too dry. The work begins at the bottom of the vessel and slowly advances toward the top; the bowl gradually assumes a spherical shape, the sides being forced farthest out about half way between the base and the rim. During this process the outside surface, since it is being expanded, begins to show innumerable small cracks. The strokes of the kajepe on the interior are short and always nearly horizontal, the implement being pulled towards the body. During this first use of the kajepe all traces of the junction-lines between the rolls are removed from the inside of the vessel. The tool is always held in the right hand,

and the left hand, which is constantly used on the exterior as a brace and stop, is placed directly opposite the spot on the inside which is being worked. One woman scrapes in the seven-to-ten-o'clock sector, turning the bowl clockwise; that is, working from the scraped to the unscraped surface. The vessel is turned through about sixty degrees at each change of position, and about a third of the surface just scraped is gone over again by the strokes on the new sector. Another works in the nine-to-twelve-o'clock sector, turning the bowl counterclockwise, that is, working from unscraped to scraped surface; in doing this she turns the puki five or six times through an angle of between sixty and eighty degrees for each turn, before the entire interior has been gone over.

The kajepe is then applied to the exterior of the bowl (pl. 14, b). The concave surface of the kajepe is preferred for the first of the exterior scraping, although the convex surface is not infrequently used. The strokes begin again at the base. At first, when the kajepe comes in contact with the puki, the strokes are very nearly vertical. A little higher on the bowl they become diagonal. Near the rim the strokes are almost horizontal, but still have a perceptible downward slant. The kajepe is always pulled toward the body. The stroke on the exterior is from four to five inches, generally a little longer than that on the interior. During this step one woman turns the puki counterclockwise, working in the two-to-four-o'clock sector; another turns it in the same direction, but works in the four-to-six-o'clock sector. In both cases the strokes are from the scraped to the unscraped surface. Although the larger portion of the actual shaping of the bowl is done from the interior, the drawing in of the upper part to form a constricted mouth is done with the kajepe on the exterior, when the step last described is nearing completion. As when working on the interior, the left hand again acts as a brace and a stop, but this time on the inside of the bowl.^[32] Usually only the ends of the fingers rest against the side. Upon the completion of the use of the kajepe on the outside for the first time, all traces of the coil-marks and of the fine cracks developed during the interior scraping are removed, but the surface is still rather rough (pl. 14, b; also 12, b, 3). After both surfaces have been gone over for the first time with the kajepe, the rim is once more trued up in the manner already described. Then, depending upon the stiffness of the paste, the piece may be set aside to harden a little or the work may continue without interruption.

The women hold the kajepe with fingers either slightly bent, or nearly fully flexed; in the latter position the tool is grasped between the thumb and bent forefinger. Just before using, the kajepe is dipped in water; and the edge is then drawn across the palm of the left hand to remove excess water before it is applied to the bowl. After perhaps a dozen strokes, the paste which has collected on the tool is removed by a dexterous motion: the inner side of the thumb of the left hand is run along the edge of the kajepe to collect the paste; then the inner side of the forefinger of the same hand is run from the base to the tip of the thumb. This brings the paste to the ends of the two fingers in the form of a pellet, which is thrown upon the mass on the canvas. Since the paste has a tendency to stick to the fingers, the potter must throw it with a jerk or snap.

When the kajepe has been used once over the entire surface, it is usually reversed so as to present its convex side to the bowl, made a little damper than for the previous scraping, and the whole process repeated in detail for both the interior and exterior. This

results in better symmetry and a smoother surface. The rim is again finished in the manner already described. The walls of the bowl have by this time been thinned to about one-half or two-thirds of their original thickness, all major irregularities and rough spots have been removed, the vessel is fully shaped, and only the finishing touches are needed before it is ready to be sun-dried. The paste now has the consistency of soft thick leather.

The first step in the finishing sometimes consists in again going over both surfaces with a wet kajepe, to smooth them still more. Occasionally only the exterior is so treated. Then the bowl is turned repeatedly, while the potter touches small areas with the convex surface of the kajepe. At this point particular attention is given to the bottom of the interior; it may be retouched with the kajepe, or simply smoothed over with the ball of the thumb or forefinger. If the surface seems slightly uneven, small pellets of paste are rubbed into the bottom and then levelled down with the kajepe. The bowl on the puki is then placed upon the palm of the left hand, held at eye level, and revolved slowly, while the potter examines it for rough spots and slight irregularities of contour, which when found are smoothed over with the forefinger of the right hand. After being replaced upon the floor, the bowl may be gently pressed between the hands in order slightly to improve its symmetry. Finally the rim is smoothed for the last time, little pellets of paste being added, if necessary, to fill in shallow depressions, and great care is taken to make it as regular and smooth as possible. It is obviously impossible to describe in detail the exact procedure followed in this finishing process, for the method adopted is chosen according to the needs of the individual vessel. After all this is done, the piece is ready to be sun-dried. Its present appearance is very different from that which it had when the building was completed (compare pl. 12, b, 2, and b, 3).

The moulding of bowls less than four inches in diameter is a very simple matter compared with that of the larger ones, for there is no use of puki, rolls, or kajepe. After working the paste in the hands the resulting lump is formed into a flat spheroid. The forefinger of the right hand is inserted in the centre of this, the clay held horizontally and slowly revolved by a slight tossing rotary motion of the left hand. The forefinger of the right hand enlarges the hole and gradually shapes the mass into the desired form, thinning the sides at the same time. If the resulting bowl is too high, the rim is pinched off with the thumb and fore-finger of the right hand. When the proper shape and thickness have been attained, the little bowl is held upright in the left hand, while the fingers of the right smooth the sides and the rim. When completed it is set on a board to dry. Sometimes one of the larger of these bowls is placed in a small puki and reshaped with the kajepe. One woman used the kajepe on the interior of the vessel while it rested upon her left hand. The bowl, when treated in this way, and finished in the same manner as the larger bowls, was noticeably a better piece of moulding than the average small piece.

The time consumed in moulding bowls is variable, depending on the dexterity and speed of the potter and to a somewhat lesser degree upon the sort of vessel being moulded. Comparisons between potters are apt to be misleading, for no two make bowls of exactly the same size or exercise the same amount of care in finishing, which, from the point of view of time consumed, is the most variable phase of the moulding. One

potter moulded a wide-mouthed bowl of four rolls in twenty minutes; another finished two spherical bowls of six rolls each in forty-eight minutes; a third potter, the swiftest of the three, finished a wide-mouthed bowl of two rolls after it had been under her hands just fourteen and one-half minutes. The first potter moulded six of the small bowls (less than four inches in diameter) in fifty-five minutes, averaging nine minutes to each, but to refinish one of them with a kajepe she took thirteen minutes. The third potter moulded one of the small type of bowls in four minutes.

In considering in detail the time consumed in moulding, the work of Maria Martinez was studied. She is the swiftest and most dexterous potter of San Ildefonso, and moulds in a quarter to a third less time than the slowest of the others. In three hours she turned out ten bowls, averaging about seven inches in diameter, of the constricted-mouthed and similar types; this is an average of one in eighteen minutes. She was working constantly, first on one, then on another. Subsequently she moulded six two-roll constricted-mouthed bowls in one hour and twenty-six minutes, an average of fourteen and a third minutes each. The various steps in the moulding of two bowls are recorded in Table I, which gives the actual time consumed in minutes and seconds from the beginning of the first of the pair until the two were finished and set aside. In Table II the growth and treatment of a single bowl are given in more detail, with the time in minutes and seconds that it was under the hands of the potter. This piece was one of five constricted-mouthed bowls of three rolls each which were moulded in sixty-four minutes, an average of twelve minutes, forty-eight seconds for each vessel.

TABLE I

Bowl A, constricted-mouthed bowl, diam. 8 in.; Bowl B, wide-mouthed bowl, diam. 7 in.

<i>Min. Sec.</i>		
4.30	0.00	Bowl A— clay picked up from pile
	2.00	“ “ — pat in puki and pinched up
	3.00	“ “ — first roll on and pinched flat
	4.00	“ “ — second roll on and pinched flat
	4.30	“ “ — rim smoothed and bowl set aside
<i>(no pause)</i>		
2.50	4.30	Bowl B— clay picked up from pile
	5.30	“ “ — pat in puki and pinched up
	6.20	“ “ — first roll on and pinched flat
	7.10	“ “ — second roll on and pinched flat
	7.20	“ “ — rim smoothed and bowl set aside
<i>(pause of 40 seconds)</i>		
8.00—	8.00	Bowl A— picked up again

8.30	“ “	— interior scraping stopped
9.15	“ “	— exterior scraping stopped
9.45	“ “	— first rim smoothing stopped
11.00	“ “	— first interior smoothing stopped
11.30	“ “	— first exterior smoothing stopped
12.45	“ “	— second rim smoothing stopped
13.45	“ “	— second interior smoothing stopped
14.30	“ “	— second exterior smoothing stopped
15.15	“ “	— third rim smoothing stopped
16.00	“ “	— finishing touches stopped; bowl completed

(pause of 15 seconds)

	16.15 Bowl B	— picked up again
	16.45	“ “ — interior scraping stopped
	17.00	“ “ — exterior scraping stopped
	18.00	“ “ — first rim smoothing stopped
	18.30	“ “ — first interior smoothing stopped
5.05—	19.00	“ “ — second rim smoothing stopped
	19.45	“ “ — second interior smoothing stopped
	20.45	“ “ — touching up stopped
	21.00	“ “ — third rim smoothing stopped
	21.15	“ “ — third interior smoothing stopped
	21.20	“ “ — bowl completed

Total, Bowl A—12 minutes, 30 seconds

Total, Bowl B—7 minutes, 55 seconds

TABLE II

Small, constricted-mouthed bowl

Min. Sec.

00.00 — clay picked up from pile

00.15 — pat put in puki

01.00 — clay for first roll picked up

01.15 — one end of first roll pinched on pat

01.50 — first roll flattened, and clay for second roll picked up

02.10 — one end of second roll pinched on pat

02.40 — second roll flattened, and clay for third roll picked up

02.50 — one end of third roll pinched on pat

03.30 — third roll flattened, and rim smoothing begun
03.50 — bowl set aside

(14 minutes, 10 seconds, elapsed)

03.50 — interior kajepe-scraping begun
04.25 — exterior kajepe-scraping begun
04.55 — rim smoothing begun
05.20 — bowl set aside

(18 minutes, 30 seconds, elapsed)

05.20 — interior kajepe-smoothing begun
06.00 — small roll placed around edge of interior base and patted down
06.40 — interior kajepe-smoothing begun
06.55 — exterior kajepe-smoothing begun, lip turned in
07.45 — rim smoothing begun
08.50 — exterior kajepe-smoothing begun
09.25 — interior kajepe-smoothing begun
10.05 — rim smoothing begun
10.50 — bowl held at eye level, exterior kajepe-smoothing begun
11.17 — bowl placed on floor, interior kajepe-smoothing begun
11.45 — moulding completed, bowl set aside

OLLAS

In the moulding of ollas the first step is the building of the lower part of the body to a height of four or five rings in the same manner as in bowl-construction. Since ollas have a larger diameter than bowls, more than one roll is needed to form a complete ring about the growing edge. The number of rolls per ring varies from one and a half to four, depending upon the size of the olla and the part worked upon. When this first building has been completed (pl. 15, a), the rim is smoothed somewhat and the scraping and shaping are at once begun.

The use of the kajepe in the shaping of ollas is identical in all details with that described above in discussing the moulding of bowls. The roll-junctions and the fingerprints are entirely removed and the sides of the vessel are flared a little. Great care is taken not to flare the sides too much, for the paste at this stage is still rather soft, and if the sides are flared unduly they collapse outward of their own weight. When the preliminary shaping has been finished the rim is smoothed a

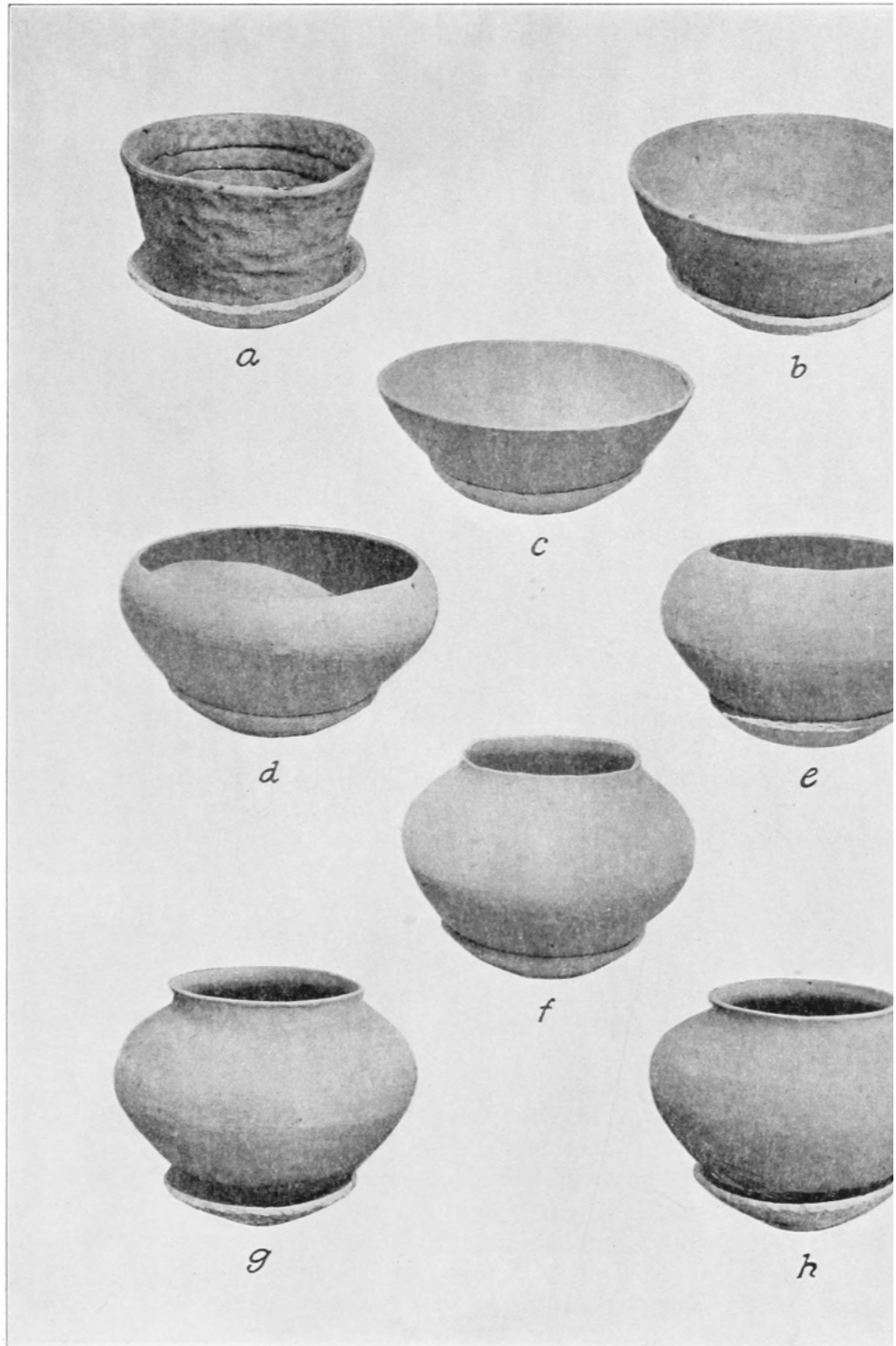


PLATE 15

SUCCESSIVE STAGES IN THE MOULDING OF AN OLLA

a. Lower wall built up of rings of clay. b. Sides smoothed and flared by use of the kajepe. c. Sides further flared and more carefully smoothed. d. Shoulder

partly formed. e. Shoulder finished. f. Neck built. g, h. Neck flared and final smoothing completed; the vessels have been shifted in the moulds to allow work on lower sides.

little, and the olla is placed in the sun to stiffen. The moulding has now reached the second stage (pl. 15, b).

The vessel remains in the sun from twenty to thirty minutes. This period may be extended to as much as an hour and a half, if the vessel is in the shade or if other things occupy the potter's time. During all the drying periods, at intervals varying from five to fifteen minutes, the potter dips her right hand in the water of the lard-pail, and moistens the rim of the partly finished olla in order to keep it soft enough to permit perfect welding when additional rolls are added. While one vessel is drying, the potter is usually occupied with another. In case the work on the latter is finished before the former is dry enough, the potter simply waits, or finds some other work with which to occupy herself until it is ready.

The test of the condition of the olla after this short drying is to pat the side gently with the balls of the fingers. The clay is now very much like soft leather. The degree of resistance offered by the side indicates the condition of the paste. Of course the value of such a test depends entirely upon the knowledge and sense of touch possessed by the potter, two qualities which can be gained only through long experience.

When the paste in the olla, which now has the shape of an open-mouthed bowl, has reached the proper degree of firmness, the vessel is brought in and the shaping continued. By using the kajepe first on the interior and then on the exterior, the sides are flared still more until the desired shape for the lower part of the olla has been obtained. Then, after a hasty smoothing of the rim, the vessel, now in the third stage (pl. 15, c), is again set aside to dry. When it has become firm enough to support the weight of the new rolls to be added, the building is continued. The temporary rim has been kept soft by the application of water from time to time. The method of preparing this rim for the application of the rings varies slightly with different potters. The rim may be pinched into scallops between the thumb and forefinger of either hand, usually the right, or it may be roughened by vertical strokes of the kajepe on the exterior. In the former case the scallops may touch one another (pl. 16, a), or they may be separated by a quarter to a half inch of unpinched rim. When the kajepe is used, it also destroys the original smoothness of the rim, producing an irregular, broken surface to which the new roll can easily be welded.

The potter must now build that part of the olla which has the greatest diameter. Each new ring of clay must therefore be of greater diameter than those which preceded and those which will follow. Three or four rather slender rolls are used to make such a ring. After two complete rings have been placed on the vessel and flattened, the shaping is begun. Gradually the new rim is drawn in until the shoulder of the olla has been formed. Again the rim is smoothed before the vessel is set aside. The moulding of the olla has now reached the fourth stage (pl. 15, d).

The olla is once more allowed to dry for about the same length of time as before. During these successive dryings the presence or absence of wind plays a considerable part in the length of time the vessel is allowed to remain in the sun; for a stiff breeze will dry the paste almost as quickly as the sun itself. The potters do not like to mould ollas when there is much wind because the paste dries so quickly that the vessel is ready to be taken up again before the second vessel, upon which the potter is working, is ready to set aside. Occasionally, but by no means always, the drying olla is turned so as to present a fresh section of the side to the sun.

After testing with the hand the newly built part of the olla, to determine its firmness, the vessel, if sufficiently dried, is taken to the work-board and the building continued. The pinching of the rim is done as before, two more rings are added, and their shaping is begun. The rolls decrease very slowly in diameter. In shaping with the kajepe only the newly added section of the vessel is treated. At this stage the work becomes more difficult, for the two rings just added form the lower part of the neck of the olla, and therefore constrict the mouth of the partly finished vessel. If the paste is a little too soft, the newly built section will sag when the shaping is begun. The vessel must then be set aside at once until it reaches the proper degree of stiffness. While using the kajepe on the exterior, during this stage of the construction, the left hand is held inside the vessel not only as a brace or stop, but also as a support for the incurving side. When the scraping and smoothing have been finished, and the side has been given the proper curve, the rim is smoothed a little before the vessel, now in the fifth stage (pl. 15, e), is again set aside.

Half an hour or so later the building of the olla is completed by the addition of one more ring. The rolls composing this ring are distinctly more slender than the first rolls used in the vessel. When this ring has been applied the rim is cursorily smoothed and the last shaping begun. The final delicate contour of the vessel depends to a large extent upon this final shaping near the rim. The kajepe is used with painstaking care, and the work progresses slowly. A slight outward flare is given to the lip by careful manipulation of the kajepe on the interior. When this is completed, the olla has reached the sixth stage (pl. 15, f).

The finishing touches consist largely in going over the rim carefully and adding pellets of paste when necessary, thus making its curve as nearly uniform as possible. This process consumes a considerable amount of time, for the work is done very painstakingly and slowly (pl. 17, a). The vessel has now attained its final shape (pl. 15, g and h), and for the last time is set out in the sun to dry.

The time taken to mould an olla is far greater than that necessary for a bowl. The type of olla shown in the illustrations is about fourteen inches in diameter at the shoulder. The potter who made these counted on moulding two such ollas each day. On some days a little time was left in the afternoon in which to make some smaller vessels, but no olla was ever begun in the afternoon. Ollas were usually started between ten and eleven o'clock in the morning and finished between half-past-three and half-past-four in the afternoon. During the early morning the household tasks had to be attended to; at

noon about an hour was used in preparing and eating lunch; and at various times during the day the children, especially the baby, made it necessary for the potter to leave her work. Table III, columns A and B, gives in detail the various steps in the construction of two ollas, together with the time each vessel was actually under the hands of the potter and the length of the intervals in which the vessel was drying; columns C and D record two other ollas on which the observations were less complete. As a rule it requires one and a half



a



b

PLATE 16

a

Potter moulding an olla. The first roll at the shoulder has just been added. The temporary rim has been pinched up with the thumb and forefinger to provide a firm seat for the succeeding roll.

b

A potter of San Ildefonso. The costume is typical of the Rio Grande Pueblos. With the exception of the shawl it is entirely of native manufacture.

hours' work to mould an olla. In the case of olla A, the potter was unfortunate in misjudging the proper consistency of the paste, which delayed the construction of this vessel an entire half-hour, omitting the periods of additional drying. The two ollas were begun at eleven o'clock in the morning; the second of the two was set aside completed at eight minutes past four in the afternoon.

It is said that it requires one whole day, and sometimes two, to mould a single very large olla. The process is identical with that just described. In the case of small ollas (such as those seen in pl. 18, a) a single period of drying elapses between the construction of the body and that of the neck. These small ollas can be moulded in a little less than twice the time taken for a bowl of about the same diameter.

TABLE III

	A	B	C	D
	<i>Min.</i>	<i>Sec.</i>	<i>Min.</i>	<i>Sec.</i>
First clay picked up	00.00	00.00	00.00	00.00
Pat placed in puki	04.00	04.00		
Second handful of clay picked up	04.30	05.30		
First roll added	05.00	06.00		
First ring flattened, second begun	06.45	08.30		
Second ring flattened, third begun	09.00	10.00		
Third ring flattened, fourth begun	11.30	12.30		
Fourth ring flattened, fifth begun	15.00	16.45		
Fifth ring flattened (Stage I, pl. 15, a), inside scraping begun	18.30	20.00		
Exterior scraping begun	21.00	21.30		
Rim smoothed and interior scraping begun	23.00	24.00		
Rim smoothing begun	25.00	27.30		
Set aside (Stage II, pl. 15, b)	26.00	29.00		
<i>Time out</i>	34.00	24.30		
Interior smoothing, then exterior	26.00	29.00		
Set aside (Stage III, pl. 15, c)	33.00	32.30		
<i>Time out</i>	53.00	59.00		
Pinching of rim begun	33.00	32.30	33.00	33.00
Pinching of rim completed	34.15	33.30		
First roll made	35.00	34.15		
First roll pinched on, second begun	35.30	34.30		
Second roll pinched on, flattening begun	39.45	36.30		
First ring flattened	41.00	37.30	37.00	
Second ring flattened, rim smoothing begun	45.00	41.30	41.00	
Interior scraping begun	45.30	42.00	42.00	
Exterior scraping begun	50.00	45.00	45.00	
Rim smoothing begun	52.00	47.00		

Interior smoothing begun	53.30	48.30	47.00	
Exterior smoothing begun	55.00	49.30	52.00	
Rim smoothing begun	57.00	50.15		
Set aside (Stage IV, pl. 15, d)	58.00	50.45	54.00	55.00
<i>Time out</i>	27.00	20.45	28.00	68.00
Pinching of rim begun	58.00	50.45	54.00	55.00
Pinching of rim completed	58.45	51.15		
First roll completed	59.30	52.00		
First ring flattened, second begun	64.00	56.45		
Second ring on	66.15	59.45		
Second ring flattened, rim smoothing begun	67.15	60.30	61.00	
Exterior scraping begun	68.00	60.45	61.15	
Interior scraping begun	70.00	omitted	omitted	omitted
Set aside (clay too soft)	70.30	“	“	“
Rim smoothing begun	omitted	63.00	“	“
Set aside	omitted	63.45	“	“
<i>Time out</i>	13.45	17.00	“	“
Interior scraping begun	70.30	63.45	64.00	
Exterior smoothing begun	75.15	69.15		
Rim smoothing begun	77.15	71.45		
Set aside (Stage V, pl. 15, e)	79.15	73.30	69.00	67.00
<i>Time out</i>	18.00	15.15	68.00	42.00
Interior smoothing begun	79.15	omitted	omitted	omitted
Exterior smoothing begun	83.45	“	“	“
Rim smoothing begun	84.15	“	“	“
Set aside	84.45	“	“	“
<i>Time out</i>	33.00	“	“	“
Interior smoothing begun	84.45	“	“	“
Exterior smoothing begun	88.00	“	“	“
Set aside	89.45	“	“	“
<i>Time out</i>	21.00	“	“	“
Pinching of rim begun	89.45	73.30	69.00	67.00
First roll put on	92.45	75.00		
Ring flattened, rim smoothed, exterior scraping begun	94.45	77.30		73.00
Interior scraping begun	97.45	79.00	76.00	
Exterior smoothing begun	100.15	81.30		

Interior smoothing begun (Stage VI, pl. 15, f)	101.45	83.00		
Exterior smoothing begun	102.45	omitted	omitted	omitted
Rim smoothing begun	103.30	84.30	84.00	83.00
Set aside (clay in rim too soft)	105.15	omitted	omitted	omitted
<i>Time out</i>	16.30	“	“	“
Interior and exterior smoothing begun	omitted	“	“	87.00
Rim finishing begun	105.15	90.30		91.00
Vessel completed (Stage VII, pl. 15, g, h)	119.45	94.30	93.00	98.00
Total elapsed time from beginning: Ollas A and B, 5 hours, 7 minutes, 30 seconds; Ollas C and D, about 5 hours.				

COOKING-VESSELS

The clay forming the paste of these pieces is called Apache clay, and is very different from that employed in making ordinary ware (see p. 22); it is used without the addition of any tempering material.

Just before moulding is started, the clay is carefully gone over in a final search for small pebbles. The moulding differs only in detail from that of vessels made from other kinds of paste. A pat very much thinner than those described above is made and placed in the puki. The vessel is built by the addition of rolls which are of smaller diameter than usual—about three-eighths of an inch. When the body has been built to the desired height, the scraping and smoothing with the kajepe are begun. At first the vessel, like the others, is cylindrical; but the scraping, first on the



a



b

- a. Putting the finishing touches on an olla. The potter's left hand is supporting the soft neck while she shapes it from the outside with the kajepe. In front is a pile of base-moulds (pukis) for small bowls; at the potter's right is a lard-pail of water and a reserve supply of clay wrapped in a canvas.
- b. Applying a horizontal handle to a small olla; two gourd spoons (kajepes) are lying on the table.

interior and then on the exterior, thins the sides and gives it a spherical shape. After the smoothing the sides are from one-eighth to three-sixteenths of an inch thick, much thinner than the walls of vessels made of other kinds of paste. When the body has assumed the desired shape, additional rolls are added to form the lip, which is scraped, smoothed, and flared with the kajepe. During the entire period of moulding small pebbles are constantly being found and removed, for Apache clay contains a great many more pebbles than the other clays. After the finishing touches have been completed, the rim is indented, and a vertical handle is added as described below (see p. 50).

The indenting of the rim is done with the thumb and forefinger of the right hand. The thumb is placed under the flaring rim, with the nail vertical; the forefinger rests upon the lip, the nail horizontal. The fingers are in about the position they would assume were the potter holding a pencil. A gentle downward pressure of the hand indents the rim slightly, the nail of the thumb leaving a small mark on the under side of the lip. This pressure is repeated at intervals of about three-quarters of an inch until the entire lip has been given an indented or undulating edge.

The construction described above is that of a small olla about six inches in diameter. Larger ollas are built in several stages in the same manner as ollas made of the other forms of paste.^[33]

The time consumed in the moulding of cooking vessels was noted in only one case. Six rolls were used for the body and two for the lip. Before the kajepe was used for the first time the cylindrical body was five inches in diameter and four and a half inches high. Table IV gives the time of construction, divided according to the various steps of the work.

TABLE IV

Min. Sec.

- 00.00 — Kneading of small handful of clay to eliminate pebbles begun
- 04.00 — Pat placed in puki, kneading of clay for rolls begun
- 12.00 — Making of first roll begun
- 19.00 — Six rolls added and flattened, interior scraping begun
- 20.30 — Exterior scraping begun, later interior smoothing begun, followed shortly by exterior smoothing
- 26.00 — Seventh roll added
- 29.00 — Eighth roll flattened and interior smoothing begun

29.30 — Exterior smoothing begun
32.00 — Rim smoothing begun
38.00 — Finishing touches begun, followed by indenting rim
46.00 — Moulding completed, vessel set aside
9.00 — *Time out*
46.00 — Work upon handle begun
49.00 — Handle finished, vessel completed

PRAYER-MEAL BOWLS

Prayer-meal bowls are of two sorts, bowl-shaped with a terraced elevation on one side of the rim, and rectangular, or box-like, with a terraced elevation at either end. In the construction of a rectangular prayer-meal bowl, no puki is employed. The pat, either circular or rectangular in shape, is made as usual. It is placed upon a board, care being taken to press it down hard, and the rectangular shape is accentuated by the pinching up of the edge. A thin layer of temper may, or may not, be sprinkled on the board before the pat is placed upon it. To this pat are added two or three rolls in the manner already described. When it is desired to turn the vessel, the board upon which it rests is turned. The board is either held in the lap or rests on the top of a low stool. In scraping and shaping the vessel one potter began to use the kajepe on the exterior first, then proceeded to the interior; another reversed the process. The stroke on the exterior is vertical at the corners and nearly horizontal on the sides; on the interior it is nearly horizontal. When the building of the vessel has been completed, the rim is smoothed, and the corners

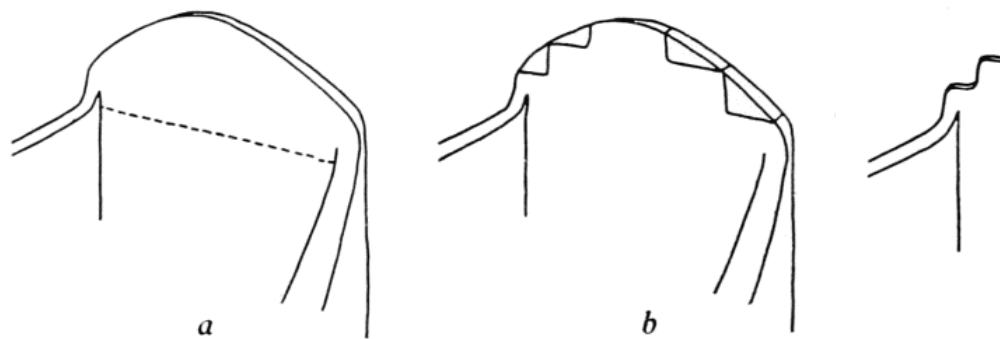


FIG. 7. The three stages in making the terraced end of a prayer-meal bowl:
a. End built to full height. b. Notches cut with a taut string. c. Bits of clay removed from the notches and edges of terracing smoothed down.

made straight. The bowl may be pressed lightly between the hands to improve the symmetry and to make all the corners more nearly right angles; it is then set aside for a while.

The next step is the construction of the terraced ends. A single short roll is attached to the interior of the rim at one end. This roll is pressed flat, and smoothed with the *kajepe*, first on the interior and then on the exterior. The sharp edges are softened, the rim is smoothed, and the junctions of this new portion with the sides are rounded off. The result is a flattened semicircular vertical projection at the end (see [fig. 7, a](#)). The other end is then treated in the same manner. The bowl is now ready to have the terraces put in. There are two methods of doing this. One potter uses a long string in making the first marks. The string is stretched taut horizontally, parallel to the length of the bowl. It is then pressed lightly on both ends at the same time, first on one side of the centre, then on the other. The distance between the hands is next shortened, and two notches are cut on either side of the centre of the end with this string, which is still held horizontally. Similar notches are cut in the other end. Each end of the bowl is now as shown in [fig. 7, b](#). Another potter does not make the initial marks in the two ends simultaneously. Holding the end of the bowl towards her, she lightly marks the lines to be cut with her fingernail. Then, picking up the string and following the marks made by her fingernail, the notches are cut in the way described.

As soon as the notches have been cut, the small triangular pieces of clay are removed, the edges rounded with the thumb and forefinger of the right hand, and the corners made into curves. Small pellets of clay may be added if necessary. After a few finishing touches with the forefinger the terracing of the ends of the bowl is completed (see [fig. 7, c](#)).^[34]

The time consumed by the swiftest potter of the village, in moulding an unusually large prayer-meal bowl, is recorded in Table V. The piece was seven and a half inches long, four and three-quarters inches wide, and three and a half inches high. The terraces raised the ends one and three-quarters inches more.

TABLE V

Min. Sec.

00.00	— Moulding of the bowl itself completed, after some time spent in pressing down the pat on the interior of the bowl. Vessel set aside
22.00	— <i>Time out</i>
22.00	— Interior and exterior smoothing begun
26.00	— Paste picked up for roll to be placed on one end of bowl
29.45	— Roll flattened and smoothing of edges begun
30.30	— Interior scraping begun
31.10	— Exterior scraping begun
31.45	— Smoothing of connection with sides begun
32.30	— Roll placed on other end
33.45	— Roll flattened and interior scraping begun
34.15	— Exterior scraping begun

- 34.45 — Very small roll placed on first end to improve curve, followed by a little smoothing with kajepe
- 35.45 — Finishing of end begun
- 37.30 — Marking of one end with fingernail begun
- 38.15 — Notches cut with thread, pieces removed, and finishing of terrace begun
- 40.15 — Marking of other end with fingernail begun
- 40.35 — Cutting of notches with thread begun
- 41.05 — Pieces removed and finishing of terrace begun
- 43.15 — Finishing completed

DOUBLE-MOUTHED VASES

The first step in the construction of this type of vessel is the moulding of an open-mouthed bowl. Across the mouth of this bowl a clay bridge is placed. Finishing touches are given at this stage of the work, just as if the moulding had been entirely completed. The piece is set aside for a while to stiffen.

Later, additional rolls are added to each half of the mouth of the bowl by attaching them to the rim proper and also to the bridge. In this way two necks are gradually built up. At regular intervals the work is set aside to stiffen in order that it may support the additional weight to be added. The shape and height of the two necks depend entirely upon the whim of the potter. The moulding of the upper parts of the two necks is a delicate process, and requires considerable skill. The roll-marks on the interior of the neck, if obliterated at all, are destroyed by the fingers only, since the diameter of the neck is too small to permit the use of a kajepe. After the moulding has been entirely finished, a handle connecting the two necks is usually put across the top at right angles to the bridge which forms the base of the necks. Obviously this handle has a structural as well as ornamental purpose.^[35]

HANDLES

In all cases in which handles are to be applied the vessel is entirely finished before they are put on. The potters of San Ildefonso make three types of handles: (1) passing across the top of an open-mouthed bowl, (2) attached vertically to the side of a vessel, (3) attached horizontally to the side of a vessel. All three types consist of a short roll of paste of the proper length, usually somewhat smaller in diameter than the rolls used in the body of the vessel. This roll is slightly flattened before it is applied.

The first two types of handles are placed on the vessel in the same manner. The ends of the flattened roll are pinched still flatter. In the case of the handle over the top of a bowl, first one end of the roll, then the other, is attached by pressure to the exterior of the rim. The vertical handle on the side of a vessel is attached first to the exterior of the rim, then bent over and attached to the side just above the shoulder. The left hand is used as a stop on the inside of the vessel in order that the shape of the side may not be altered

by the pressure exerted in applying the handle. The junction-lines between the handle and the vessel are then obliterated with the forefinger of the right hand. Small pellets of paste may be added to the side of the contacts, especially on the inner side of the handle where it makes an acute angle with the side of the bowl. In this way the welding is made solid, and the curves regular. Finally the curve of the handle itself is made symmetrical. A few finishing touches, such as the obliterating of fine cracks in the handle and the smoothing of its surface, complete the process. Such a handle may be constructed in three to five minutes.

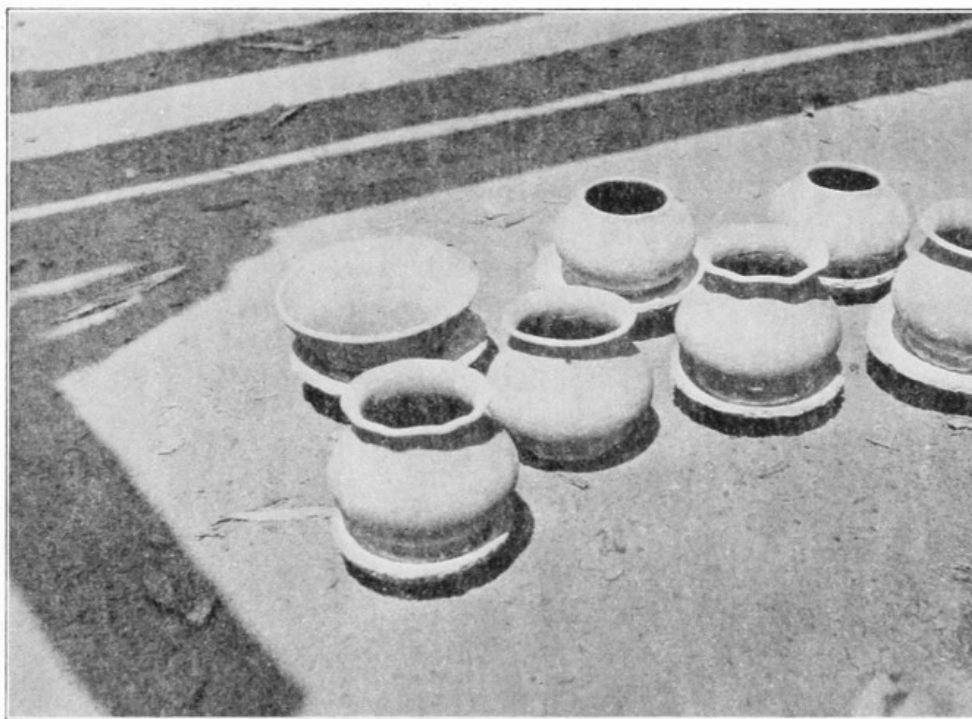
Horizontal handles are attached to the sides of vessels in an entirely different manner, being keyed or riveted in, rather than merely welded on. Handles of this type are usually put on in pairs. The position of one handle is chosen and the rim of the finished pot is marked to show its location; by sighting across the top of the vessel a point exactly opposite is also marked to give the location of the other. For each handle two holes, about three-eighths of an inch in diameter and fairly close together, are cut through the side of the vessel from the exterior with a small stick. One end of the roll which is to form the handle is inserted in one of the two holes. Then the other end is similarly inserted. The left hand holds the first end in place on the interior, while the other end is being put in ([pl. 17, b](#)). Next the junctions between the ends of the roll and the interior surface of the vessel are smoothed over and obliterated with the fingers of the right hand. The handle itself is flattened a little, and small pellets of paste are added at the junctions between it and the exterior of the vessel. These are smoothed over with the fingers, so that superficially the handle looks as if it had been pressed upon the surface in the same manner as a vertical handle. A smoothing of curves, and touching up of the handles to make them symmetrical, form the last stage of the construction. It took one potter just an hour to place a pair of handles of this type on a globular bowl.^[36]

SUN-DRYING

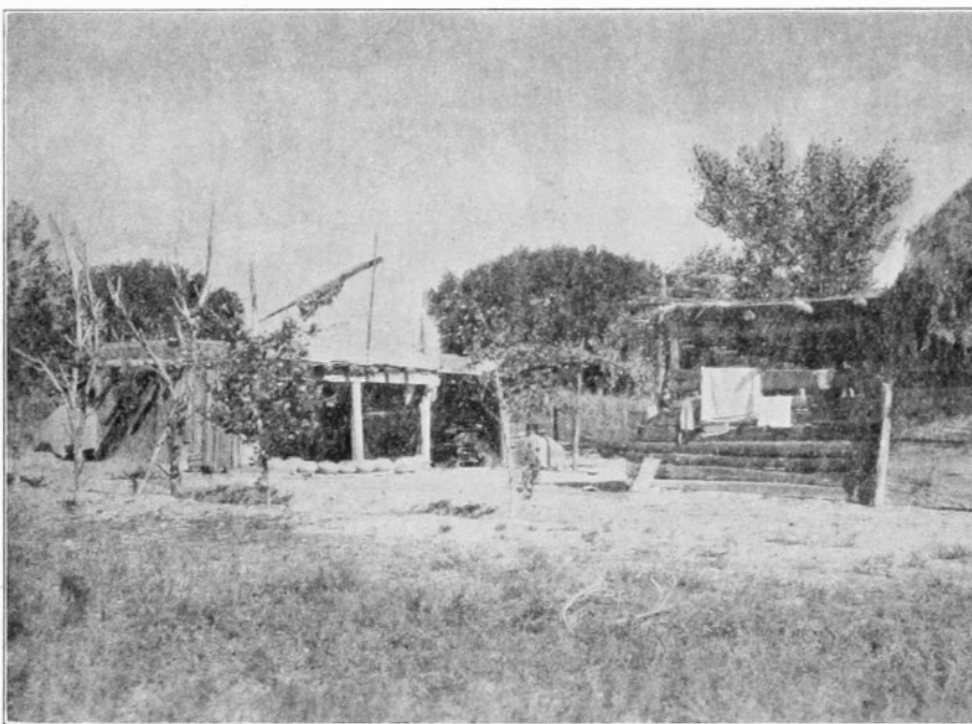
The purpose of sun-drying is to allow the vessels to harden, and to remove all moisture before the work on them is completed. During sun-drying, which immediately follows moulding, the ability of the vessels to withstand some heat is also tested. Pieces made of improperly mixed clay are eliminated at this stage of the work, because of the cracks which develop (see under Temper, p. 21). The length of time allowed for sun-drying depends upon the weather and also upon the place where the vessels are exposed.

During the dry months of the year, notably in May and June, vessels placed in the sun will dry completely in less than a day, often in half a day (pl. 18, a). In the fall it requires an entire day under the same conditions. When the sky is cloudy or showers threaten, the drying is done in the house. If time is not pressing, the pots are placed on a table, or in the corner of the room, and allowed to remain there three or more days. At the end of that time they are usually sufficiently dry. Vessels moulded one afternoon and placed on the table for the night are dry enough the following morning to permit lifting them from the pukis, which can then be used again. If, during cloudy weather, the potter desires to dry the vessels quickly, they are placed in the oven of the small wood-stove with which most San Ildefonso houses are nowadays equipped. Sometimes a piece of corrugated cardboard from a carton is placed on the floor of the oven, the door of which is left open to allow the evaporating moisture to escape. Only a low fire is built in the stove, for a hot one would cause the vessels to dry too quickly. Under such treatment a batch of pottery can be dried in two days, more or less. The length of time depends very largely upon the number of pieces to be dried, for the potter tries to keep all the vessels at about the same stage of drying; this of course requires frequent relays in the oven. The most common, and most natural, method, however, is to place the vessels in the sun and, when showers threaten, to carry them hurriedly into the house. In this way pottery is dried in a day or a day and a half. The loss of moisture in the paste changes its color from a dark reddish or greyish brown to a light reddish or whitish grey, and in the early stages the variations in the color of the paste serve as indications of the dryness of the vessel. Later the color-change is very difficult to detect. Various parts of the vessel dry at different rates; the rim always first, then the body, and last the base, both because it is thicker and because it is usually in the shadow of the vessel itself; in spite of this the vessels are never inverted while they are drying.

The occurrence of cracks in the paste is the only form of accident which takes place during drying. There are three sorts of cracks: those in the bottom of the vessel, which usually pass through or very near the centre of the base; rim-cracks; and vertical cracks in the body. All are caused by the contraction of the paste in drying. The part of the vessel which receives the greatest strain is the base, which,



a



b

- a. Vessels on their base-moulds drying in the sun; the pottery is always thoroughly sun-dried after it is moulded and before it is decorated.
- b. Summer house of Antonita Roybal, a woman who specializes in the manufacture of large red ollas: a row of these may be seen drying before the house.

because of the amount of paste in it, has a tendency to contract more than the bottom of the side. The next greatest strain is at the bottom of the side, where the paste must adjust itself to the contraction in the base as well as to that in the body of the vessel. The body and lip have the least strain, for the contraction may be compensated for by a slight settling of the paste. It is in the base, therefore, that cracks usually occur, and then near the centre of it, since that is the point of greatest strain. The size and quantity of the cracks are an indication of the amount of temper still needed to make the paste of just the right consistency. If there are many small cracks, or a single serious one, the piece may be discarded entirely, to be broken up later, and re-used in moulding. If a crack is not serious, it is filled as follows. With a small sliver of wood, or the end of a case-knife, the paste on the edge of the crack is forced down into it, first on the interior and then on the exterior; little pellets of paste are then added and pressed into the crack until it is filled. Final smoothing with the finger or the kajepe completes the obliteration.

Small vertical cracks near the base of the body probably also indicate faulty mixture of the paste. The potters, however, say that these fine cracks are caused by a vessel's drying too quickly in the sun, and insist that if it had been allowed to dry slowly in the house, they would not have appeared. Such cracks are too fine and too numerous to warrant the careful filling of each one; so accordingly the surface is merely dampened, some soft paste is added and rubbed in with the fingers or the kajepe.

Cracks running downward from the rim are exceedingly rare. The women of San Ildefonso make no attempt to repair them, but occasionally cut down the vessel to a smaller size. In the single instance noted^[37] the crack was first traced downward from the rim until its end was located upon the neck, then the upper part of the neck was cut off below the end of the crack. It was thus possible to use the base of the olla as a large bowl. A line parallel with the rim was marked about the circumference of the neck with a lead-pencil. This line was incised with a penknife, and gone over again in order to deepen it. A second line was incised around the neck about three-sixteenths of an inch nearer the rim. Then the paste between these two lines was dug out with great care, forming a V-shaped groove (pl. 20, a). This groove was cut through about one-half the thickness of the wall all the way around the neck. Then at one point the wall was pierced, and, from there around, the groove was deepened by long and short strokes of a knife pulled toward the body until the wall had been cut through for about three-fifths of the circumference. The remainder broke away easily. The raw surface of the new rim was then softened by the application of water and smoothed with the fingers. The potter said this was the usual method of treating a rim crack.

SCRAPING

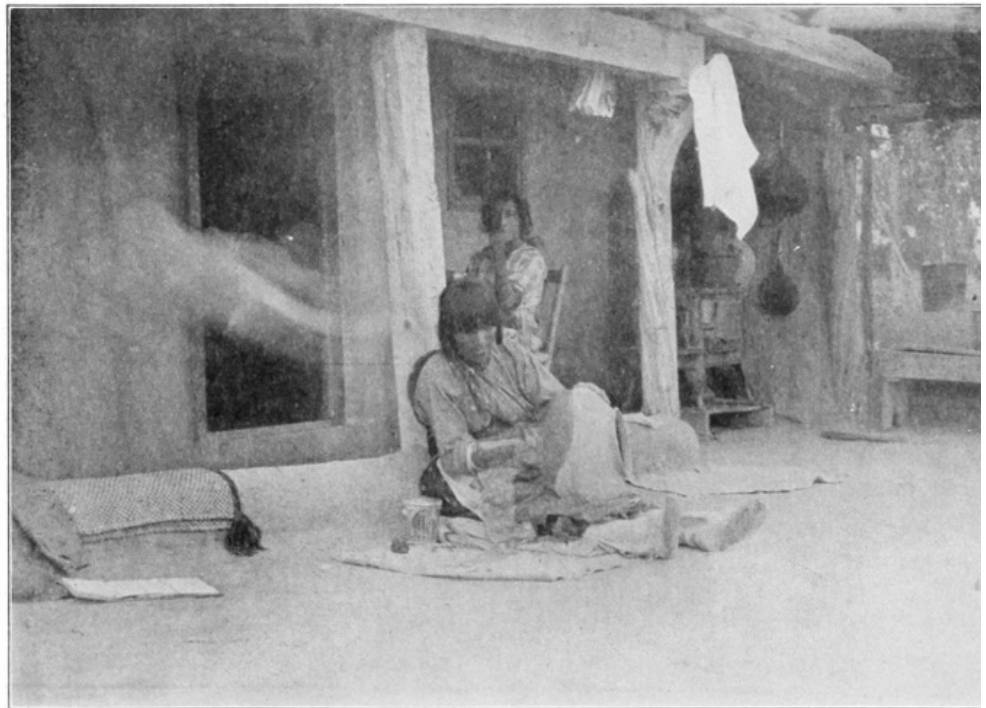
The purpose of this process is twofold, to improve the surface of the vessel by removing the marks left by the kajepe and the puki, and to thin the sides, thus reducing the weight of the finished piece. It is usually begun the day following the completion of the moulding and sun-drying, unless other duties, such as planting, harvesting, and the like, force a postponement. Large vessels such as ollas are allowed to dry nearly forty-eight hours before they are scraped. At San Ildefonso vessels are made in quantities ranging from a dozen to fifty pieces. The moulding and drying of the entire group, a process which may extend over a period of several days, is entirely finished before the scraping is begun.

There are three steps in the scraping: the wetting, the actual scraping, and the smoothing of the surface. The implements employed are a wet cloth and a scraper. The latter is either the top of a baking-powder can, or a kitchen case-knife. The can-top seems to be the more popular because it may be used on nearly any type of curve on the vessel. The informants said that their people formerly used potsherds, stones, or broken animal-bones as scrapers, in fact anything that had a suitable edge; the potsherds were sharpened and straightened by rubbing them on coarse sandstone.

When the vessel is brought in from drying, it is easily lifted from the puki, since the layer of ashes or temper prevents sticking. A small olla or bowl is held upon the left knee, with the mouth tilted to the left and away from the body. A bowl is held with the left hand, the fingers on the interior, the thumb on the exterior of the lip. The exterior surface of the vessel is then softened by wiping it with a wet rag. This step is omitted by some potters always, by others only when the vessel needs a small amount of scraping. The scraping itself is begun while the surface is still damp. The scraper first touches the vessel near the shoulder. The work then continues toward the base by means of short, quick strokes taken toward the body. As the work advances the upper part of the bowl is turned away from the body, that is, the vessel turns counterclockwise. When the ridges marking the former position of the edge of the puki are being destroyed, care is taken to keep the curve of the side uniform from the base to the shoulder. If the scraping discloses an impurity, such as a stone fragment in the paste, it is removed; the resulting irregularity is filled with a pinch of soft paste, and smoothed over. The strokes of the scraper are usually approximately parallel to the rim of the vessel. The upper part is scraped as far as the surface remains convex. In bowls the scraping is done to the very rim. In both large and small ollas with flaring lip, the scraping continues only to the base of the flare. The interiors of vessels, even of wide-mouthed bowls, are never scraped. When a bowl requires little scraping to make the surface uniform, it may be finished in from three to five minutes. Those which are too



a



b

- a. Scraping an olla after sun-drying; the implement used is an ordinary kitchen case-knife.
- b. Scraping; the dark part of the olla has just been moistened to soften the clay.

heavy, and therefore need thinning, take considerably longer. It may even be necessary to moisten the surface a second time. The potters judge the proper thickness of the bowls by weighing them in the two hands with the elbows unsupported.

Large ollas, while being scraped, are held in the lap, with the mouth inclined to the left and away from the body. The left hand supporting the olla is placed palm down upon the neck (pl. 19, b). Because of the extent of surface to be gone over only that part of the olla about to be scraped is moistened with the cloth. When a case-knife is used instead of a can-top, it is held either at right angles to the surface worked or at an angle of about sixty degrees with the surface, the upper edge of the knife tilted away from the body (pl. 19, a). In all vessels, bowls as well as ollas, the angle with the horizontal made by the mouth varies considerably, according to the part of the vessel being scraped. After the scraping proper is completed, one potter sometimes rubs the surface, including the rim and lip, with a little steel wool, which noticeably improves the smoothness. The final step is to go over the entire surface carefully with the palm of the right hand in search of irregularities or small uneven spots.

When the scraper is laid aside for the last time, the surface of the vessel is again softened with a wet cloth. By means of vigorous rubbing the thin film of moist surface-paste is redistributed over the entire exterior of the vessel, filling the small scratches made by the scraper, and softening the edges of the larger ones. If the vessel is a bowl or a large olla, its position is then changed so that its mouth is tilted to the right and towards the body, whereupon the interior is treated with a wet cloth. The finishing touches consist in smoothing, either with a wet cloth or the ball of the finger, small areas of the surface which are not quite to the potter's satisfaction. The use of the wet cloth gives a uniform smooth texture which is a distinct improvement over the scraped surface. The paste resumes its dry color in three to five minutes after the wetting. This process actually amounts to the same thing as putting a thin slip of paste upon the vessel, but the potters do not think of it in that light. It is said that some women entirely omit this final smoothing step. Formerly a wet cloth was not used; the fingers alone, a piece of sandstone, or a corncob served, according to the informants, to smooth the surface after scraping. As a matter of fact, however, cloth was used in old times for smoothing pottery, as clay-smeared rags are occasionally discovered in the rubbish of ancient cliff houses. Ground potsherds, on the other hand, were more often used than gourd-rind kajepes, the latter being very rare even in the dry deposits found in caves. At Pecos well-worn fragments of the spongy interior parts of large animal bones have been found which may well have been employed for smoothing.

The time element in scraping is a variable quantity. Table VI on the following page is the record of one potter. A, B, and C were large ollas; D, E, and F were small globular ollas six inches in diameter and five inches high; G was a shallow, wide-mouthed bowl.

TABLE VI

	A		B		C		D		E		F		G	
	<i>Min.</i>	<i>Sec.</i>	<i>Min.</i>	<i>Sec.</i>	<i>Min.</i>	<i>Sec.</i>	<i>Min.</i>	<i>Sec.</i>	<i>Min.</i>	<i>Sec.</i>	<i>Min.</i>	<i>Sec.</i>	<i>Min.</i>	<i>Sec.</i>
Wet cloth applied	00.00		00.00		00.00		00.00		00.00		00.00		00.00	
Scraping begun	00.30		01.00		00.30									
Wet cloth applied again	02.30		08.30											
Wet cloth applied again	04.00		11.00											
Steel wool rub begun	05.30		16.30											
Scraping done, exterior smoothing begun	08.00		18.30		18.00		08.30		03.00		05.00		08.00	
Interior smoothing begun	14.30				26.00		11.00		05.30		07.30		11.00	
Finishing begun	17.00													
Set aside	17.30		31.30		35.00		13.00		08.30		08.30		15.30	

The treatment of the vessels after scraping varies considerably among different potters. One places the scraped vessels in the sun for a period of not less than three days in order to test them for cracks; another begins the next step in the work, that of slipping, almost at once, sometimes the same day; a third after scraping ollas replaces them in the sun for a day or so, in order to allow them to “get warm” (that is, dry thoroughly),^[38] before the slipping and polishing are begun.

SLIPPING AND POLISHING

Up to this point all vessels are made in the same general manner. In the succeeding stages the treatment differs according to the type of decoration which is eventually to be applied. The making of the vessel has been completed, and the finishing is now begun. Just as the making is divided into three general processes—moulding, sun-drying, and scraping, so the finishing is similarly divided into slipping, painting, and firing.

Slipping is the application of a very thin layer of clay to the surface or surfaces of the vessel to produce a smooth texture, uniform in color, which gives the pottery a pleasing appearance; it also serves as a background upon which designs may be painted. The slip further acts as a sizing. San Ildefonso slip is a saturated solution of a colored clay in water; it is very little thicker than water, and is applied by means of a small piece of cloth, used in much the same way in which a painter would handle a brush an inch and a half or two inches wide. It is said that formerly a small piece of skin was used instead of a cloth mop. The treatment after application depends upon the slip used; some slips are merely wiped vigorously with a cloth, others must be polished with smooth, fine-grained stones.

At San Ildefonso the potters use slips of four different colors—white (of two varieties), orange-red, red, and dark-red. The red and dark-red slips and one variety of the white must be polished; the other white and the orange-red do not need it. The white slips are principally used as backgrounds in polychrome ware (see [pl. 6](#)), the red for undecorated red ware and polished black ware (see [pl. 8](#)), and the dark-red for decorated red ware (see [pl. 7](#), a, b). The orange-red slip is the only one of the four which is not used on the body of vessels, it being confined to the bases of ollas and of some bowls. The white and the orange-red are used also in the elements of designs.^[39]

WHITE SLIP

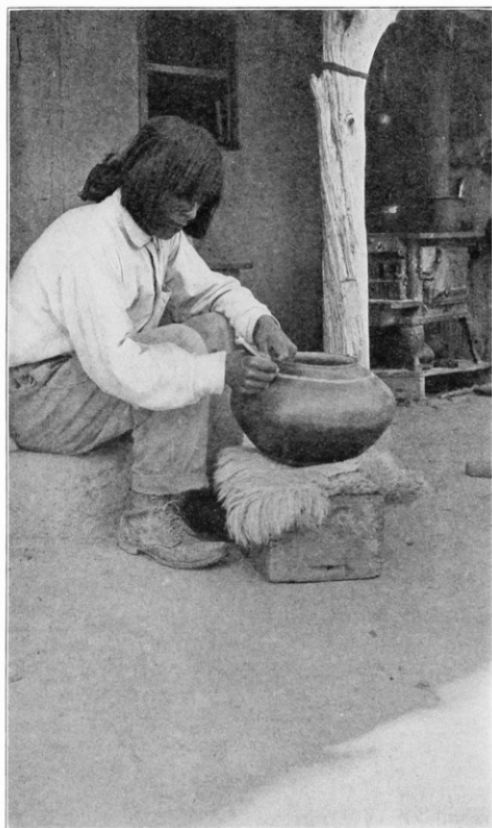
The native white slip (see p. 23) is applied and polished in the same manner as the red slip (see p. 23). It has been very largely supplanted by the Santo Domingo white slip, which does not require polishing.

The Santo Domingo white slip (see p. 23) is a soapy clay which the San Ildefonso potters obtain from the Indians of Santo Domingo and Cochiti. It is mixed with water in enamelled pans and basins or in china dishes, although formerly pottery vessels were used to hold it. Although undissolved lumps of the clay remain in the bottom of the vessel containing the solution, the latter is not appreciably thicker than water. The mop with which it is applied is a folded cloth, about two inches wide and three, or three and a half, inches long. This is held at one end, between the thumb and the first two fingers of the right hand, the other end of the cloth acting as a two-inch wide paint-brush, which is manipulated with an easy, backward-forward stroking motion, parallel to the rim.^[40]

Before the slip is applied, the hand is rubbed over the surface in order to remove any dust or powder which may remain after the scraping. Some vessels, while being slipped, are held with the mouth vertical and to the left, supported by the bent fingers of the left hand against the interior of the rim. Others are held upright, resting upon the palm of the left hand (pl. 20, b). In both cases the vessels are revolved counterclockwise as the slip is applied. In the container the slip is creamy white, but when first laid on it turns a muddy yellowish-white, because the clay below darkens as it absorbs the moisture. Within two or three minutes the under clay dries, and the surface becomes dead white. Five or six coats are applied, the vessel being allowed to become completely dry after each one. One potter rubbed the surface vigorously with a dry cloth after each application and before the slip had thoroughly dried. Another rubbed the vessel with a cloth only after all the coats had been put on, but before the last had dried. Usually the work is done in the sun, at the place where the pieces have been drying, but occasionally it is carried on in the house, and between coatings the vessels are placed in an oven heated by a slow fire. The length of time, five or more minutes, during which the vessel remains in the oven is determined by testing the warmth of its surface. When the work has been completed, the very faint marks of the mop are visible.

Pieces of pottery of various shapes are slipped on different surfaces. Small ollas and constricted-mouthed bowls are coated with white only on the upper two-thirds or three-quarters of the exterior. No particular care is taken to keep the lower edge of the slip regular. Of smaller vessels the entire exterior, including the base, is usually covered. Open-mouthed bowls are coated with white both in the interior and the upper part of the exterior. Shallow open-mouthed bowls may be coated only on the interior. The exterior surfaces not treated with white slip are later coated with orange-red slip (see below).

The actual length of time that each vessel is in the hands of the potter while it is being slipped is very short. It requires one-quarter to one-half a minute to apply a coat of slip to a constricted-mouthed bowl six to eight inches in diameter. The six coats could be applied in less than three minutes. Another half minute is needed for the rubbing with the cloth at the end. Four minutes in all is a generous estimate. A coat of slip is applied to the entire group of vessels at one time (pl. 20, b). Less than fifteen minutes are required to give a single coat of slip to a group of from twenty to thirty pieces. After this step has been completed, the vessels are placed in the sun for an hour or more before the painting of the design is begun, the length of time depending upon attendant circumstances.



a



b

PLATE 20

a

Cutting the top from an olla which had cracked slightly at the rim during the final sun-drying. The raw edge was afterwards worked down to form a new rim, and the vessel was successfully fired.

b

Applying white slip with a cloth mop. The enamelled pan containing the prepared slip is on the bench at the potter's side. A number of vessels are slipped at one time.

ORANGE-RED SLIP

This slip is applied only to the bases of vessels (ollas and certain bowls) and to the interior of the lips of ollas. Before burning it is mustard-yellow in color, but after firing it becomes orange-red. A cloth mop is used for application in the same manner as with the white slip; one coat only is applied to each surface.

As a rule the vessel being slipped is held in the lap, the mouth vertical and turned to the left (pl. 21, a). The left hand is placed in the interior to support it and keep it turning counterclockwise, that is, the upper part is turned away from the body, as the strokes are taken towards the body. The formation of the junction-line between this base-slip and the body-slip is very carefully handled. No attention is paid to the lower edge of the body-slip, except to notice its general location, for the base-slip covers it and itself produces the edge. This upper line, made by long, careful, slow, trailing strokes of the

mop, is drawn entirely free-hand, yet the potters succeed surprisingly well in keeping it always the same distance from the centre of the base. Three or four strokes are more than sufficient to encircle the vessel (pl. 21, a). It is often possible in the finished piece to trace the lower edge of the body-slip, under the orange-red coating. After the junction line has been drawn, the rest of the base is covered with shorter forward-backward strokes. The application of the orange-red slip to the interior of the lips of ollas is done with the same careful, slow technique used in forming the junction-line near the base of the body; during the process the vessel is held on the palm of the hand right-side up. The exterior of shallow bowls which receive no white slip is completely coated with the orange-red, the bowl being inverted over the left hand during the application.

The orange-red slip may be put on either before or after the painting of the design; if before, the painting may be begun at once; if after, the application may be delayed until within an hour or two of burning. If the vessels are set aside for a while at this stage, they are covered with a cloth to keep the dust and flies away. The time element is practically the same as in the case of the white slip.

RED SLIP

The red slip, as was said above, serves to cover the surfaces of undecorated red-ware. It is always polished with the rubbing-stone after application. When given a certain special firing (see p. 74) it turns a lustrous black and so produces polished black ware (pl. 8, b). This slip is a saturated solution in water of the red clay just as it was obtained from the claypits (see p. 24). Although undissolved lumps remain at the bottom of the solution, the latter is no thicker than water. The containers are either open-mouthed pottery bowls or china dishes. The slip is applied as usual with a small folded rag. While it is being put on, the vessel may be hung from the fingers of the left hand with the mouth vertical, or held in the usual way upon the palm of the left hand, its position depending upon whether or not the base is to be slipped. The surface is covered two or three times with the cloth mop, in a rather haphazard manner, so that certain portions where the strokes overlap receive as many as four coats. The potter herself is uncertain regarding the number of coats the surface has received, as they are applied one immediately after the other, with no wait between. The mopping is stopped when the surface seems uniformly covered with just the right shade—a rather bright red. San Ildefonso potters usually cover with this slip the entire exterior of small ollas and bowls, including the base.^[41] In some cases shallow open-mouthed bowls are slipped only on the interior.

Before the slip dries, the rubbing with the polishing-stone is begun. The stone is held between the thumb and first two fingers of the right hand. The entire surface is gone over several times with a backward-forward motion in strokes about three or four inches long.^[42] Each stroke is made with the entire forearm, there being no noticeable play in either the wrist or the fingers. This is a rather fatiguing and exacting process, for to obtain the best results all parts of the surface must receive equal attention: a definite system, however, has been developed by means of which the entire surface is gone over.

In polishing the exterior of a small olla or of a bowl, the vessel is first placed upon the lap in an inverted position, tilted slightly to the right. It is held at the rim by the left hand, which rests upon the left thigh. The polishing is begun on the right-hand side of the base near the body, and proceeds diagonally across the base to the far left-hand side. The vessel is then tipped up with the mouth to the left and the rim nearly vertical. The polishing is begun at the shoulder and continued down to within a very short distance of the centre of the base (pl. 21, b). At this stage some potters polish from the base to the shoulder. As the work continues, the upper part of the bowl is turned counterclockwise away from the body. Usually the stroke is parallel to the rim, but occasionally a diagonal motion develops. The strokes across the bottom are then made at right angles to the previous series. A constricted-mouthed bowl is then turned so that the mouth, still vertical, is to the right, and the section from the lip to the shoulder is polished; the bowl is now turned clockwise, that is, the top still turns away from the body. Small ollas may be held either in this position or with the mouth tilted at an angle of about forty-five degrees to the left and away from the body, in which case the polishing is done from the shoulder to the lip (see pl. 30, b, which shows a Zuñi potter using the polishing stone).

After the surface has been completely covered in this manner once, and none too carefully, the rim is coated with the slip, applied by the forefinger of the right hand, and gone over with the stone. The vessel is then wiped with a cloth upon



a



b



c

PLATE 21

a. Applying orange-red slip with a cloth mop; the beginning of the stroke may be seen just above the lap of the potter. b. Polishing a bowl with a small stone after red slip has been applied. The marks of the stone are visible at first, but quickly disappear as the slip dries. c. Women working with the polishing stone. Note the light line about the shoulder of the large olla, where a space is always temporarily left unpolished.

which a little lard has been rubbed. The lard may be kept near the potter in any convenient receptacle, like a sauce-dish or the bowl of a spoon. A clean cloth is then immediately used to distribute the lard evenly upon the surface, and to remove any surplus. Some potters do not use this second cloth at all; others apply the lard to the vessels with their fingers, wiping it immediately afterwards with a cloth. In some cases, the application of the lard is withheld until the polishing is entirely completed.

The polishing is then continued in exactly the manner just described. The surface is covered more carefully and more slowly, the areas worked upon often overlapping considerably. A cloth, sometimes a part of the potter's apron, is now between the left hand and the vessel in order to protect the smooth surface. A considerable pressure is exerted. Some women use a short, very quick stroke; others a somewhat longer, slower stroke, hence taking more time for the work. In this way the surface is covered several times. Finally the routine treatment is dropped, and finishing touches are given by polishing small areas here and there which do not show the required amount of lustre. The strokes used at the end are usually slightly longer than the previous ones. Finally, after the stone has been laid aside, the surface is wiped once with a clean cloth before the vessel is set aside; it is usually placed upon a mat or cloth in order that the base may not be scratched by contact with the earthen floor. When the polishing of several vessels has been completed, they are gathered in a corner of a room on a mat or rug, and covered with a cloth to keep the flies and dust away, for it is said that fly-specks leave a black mark upon the burned vessel.

In the case of large ollas one half of the vessel is doubtless coated with slip and polished before the other half is slipped. This is the process employed in polishing large ollas slipped with dark-red (see p. 63). No opportunity presented itself to see a large polished olla being made.

In polishing the interior of a shallow bowl, the vessel is held in its normal position, tilted slightly to the right, and resting upon the right knee and the palm of the left hand, which in turn, rests upon the left knee. The polishing is done from the lip to the centre of the vessel. As the work advances, the bowl is turned counterclockwise.

The degree of polish obtained by different potters varies considerably, and yet in even the very finest examples of polishing the marks left by the stone may be faintly seen in certain lights in the form of exceedingly low ridges, generally running roughly parallel to the rim and too low to be felt with the fingers. The surface has a lustre almost equal to that of burnished metal. In pieces which are not as well polished, the ridges are higher and may be felt with the fingers; indeed they are often prominent enough to make the lustre uneven. Maria Martinez, who does the best polishing, is also the swiftest worker in the village. From the moment the vessel is picked up to apply the slip until the completion of the polishing, it is not laid down for an instant. Her strokes are quicker than those of the other potters; she covers the surface in much less time and therefore polishes a given area more often before the slip dries. Continued polishing tires the wrist and hand very quickly and is apt to produce cramps. Other potters are inclined to rest for a few minutes from time to time, especially after the application of the slip and again

after the lard has been put on. One woman complained of the humid weather, because it did not allow the slip to dry quickly enough. The potters themselves say that streaky polish is due to lack of persistence on the part of the polisher. Maria attributes her success to the fact that she uses a faster stroke and puts on a little more lard than the others. Large vessels usually have a poorer polish than small ones. The secret of good polishing seems to lie in the ability of the potter to go over her work as many times as possible after the slip has been applied and before it becomes too dry.

The time during which the vessel being polished is actually in the hands of the potter is given in Table VII. The first column (A) represents a constricted-mouthed bowl, about eight inches in diameter at the shoulder and four inches high, which was polished by Maria. In one hour and fifteen minutes elapsed time three such bowls were done, an average of twenty-five minutes for each. The second column (B) represents the polishing by another potter of the interior of a shallow wide-mouthed bowl. The actual time from the moment the bowl was picked up for the application of the slip until the polishing was done was forty-one minutes. A rest of one minute was taken after the slip had been applied, another of thirteen minutes after the application of the lard, in order to let the bowl “dry”, and a third of one minute when the final polishing was about half done. The interior of this bowl was a very fine example of polishing.

TABLE VII

	<i>Min.</i>	<i>Sec.</i>	<i>Min.</i>	<i>Sec.</i>
First application of the slip	00.00	00.00		
First polishing begun	02.50	03.00		
Slipping of rim begun	04.45			
Polishing of rim begun	05.30			
Application of lard begun	06.05	10.00		
Wiping with dry cloth begun	06.40			
Final polishing begun	07.05	11.00		
Polishing finished	24.05	26.00		

DARK-RED SLIP

The dark-red slip is a mixture of red slip, native white slip, and tempering material. It is applied to vessels which are later to receive a black design (see [pl. 7](#), a, b). It is mopped on with a cloth in exactly the same manner as the red, and is also polished with a stone but never acquires as high a lustre as does the red. There seems to be a tendency for the first coat of this slip to dry more readily than the first coat of the red slip.

While applying the slip to small ollas about six inches in diameter, the vessel is held by the rim with the left hand. Either the entire exterior may be slipped, or the base may remain uncoated, to be covered later with the orange-red slip. The slip is first applied to

the rim, the forefinger being used instead of a mop. During this process the vessel is right-side up either in the lap or on the floor. The body is then slipped with the cloth mop, and the polishing done in two parts, with the shoulder as the dividing line. In polishing the lower part the strokes begin at the base, or at the lower edge of the slip in case the base has not been coated, and proceed upward to the shoulder. The direction of the strokes is practically parallel to the rim. The upper part is polished from the shoulder to the lip. On the base, if it has been slipped, the strokes are first made in one direction, then at right angles to that direction. From time to time, as the polishing proceeds, the vessels are set aside in order to allow the slip to dry somewhat. They may even be placed in the sun or near a fire on the hearth. The potters explain that if the work is completed while the slipped surface is very damp, the resulting polish will not be as high as when the slip is just drying. Apparently, the work done just before the surface becomes dry plays an important part in securing a high lustre. One of the potters working on this ware applied a little lard to the surface after the polishing had been completed. Another potter used no lard whatever for this class of ware.^[43] When the polishing has been completed the vessel is placed in the sun or near a fire to dry thoroughly before it is put away to await decoration. While the olla is drying it is covered with a cloth to protect it from flies.

In polishing larger ollas one half of the vessel is coated with the dark-red slip and polished before the other half is slipped. Either the half above the shoulder or the half below the shoulder may be polished first. Rather quick strokes, three or four inches long, are made with the stone. Sufficient pressure is exerted to cause motion in the entire body of the worker. The strokes, as before, are usually parallel to the rim. Because of the larger surface to be covered in these vessels, the position of the olla on the lap of the potter is changed frequently. It is supported with the left hand, which is placed palm down upon the exterior surface. While the lip is being polished, the mouth is turned toward the body and to the right, making an angle of about sixty degrees from the horizontal. In polishing the upper part of the shoulder the potter turns the mouth to the left and away from the body (pl. 21, c). Usually the mouth is more nearly horizontal in this position than in the former.

Before applying the slip to the lower half of the olla, a faint line is drawn with the polishing-stone around the lower part of the body to define the lower edge of the body-slip. During the slipping and polishing of this portion of the surface, the olla is held with the mouth nearly vertical and to the left, directed away from the body. When the polishing proper is completed, the finishing touches are given to the entire exterior by rubbing small areas here and there which do not entirely satisfy the potter.

As in the case of the red slip, the polishing must be completed before the slip becomes too dry. When there is a large surface to be polished, as in the case of these ollas, it is of course necessary that the work upon any given section of the surface be completed as quickly as possible. Therefore when two potters are working together, the swifter of the two does the polishing on the larger vessel, regardless of which of the two applies the slip. After the polishing proper has been completed, the olla may be turned

over to the slower worker for the finishing touches. This necessity for speed in polishing is probably the reason for slipping only a part of the surface at one time.

Now and then, as the polishing proceeds, the potter changes stones. An important factor in this change is the desire to rest the fingers through the slight alteration in grip afforded by the different shapes of the stones. Occasionally the change is due to a wish to obtain a stone with just the proper shaped surface for the section of the olla being worked upon. There is also a tendency to use a larger, rougher stone at the beginning of the work, for which is substituted later for the final finish a smaller, finer-grained, and therefore smoother, stone.

Because of the practice of slipping and polishing only half the surface at a time, there develops around the shoulder a narrow line of dried slip, which is naturally lighter in color than the worked areas. (This line may be seen encircling the olla in pl. 21, c). After the polishing of the upper and lower areas has been completed, the potter's attention turns to this line, and since dry slip cannot be polished, the rubbing stone is either wet with the tongue or dipped into the liquid in the slip-container before being used on the line. The latter method appears to give the best results, but in either case, the moistening of the stone is done very frequently, and the process of eliminating the lighter line is a painfully slow one. Even when the best results have been obtained, the line is not entirely obliterated. Occasionally the stone is moistened with the tongue while giving final touches to other parts of the surface.

When the polishing with the stone has been entirely completed, a little lard is rubbed on the surface either with the forefinger or with a greasy cloth; the surface is then vigorously rubbed with a chamois.^[44] This distributes the lard evenly and very noticeably improves the lustre. Then the vessel, after being covered with a cloth, is put in the sun to dry thoroughly.

The great difference in the time required for polishing large and small ollas requires two tables to present the details. Table VIII gives the time during which a large olla, about fourteen inches in diameter at the shoulder, was under the hands of the potters. The time devoted to obliterating the light junction-line, about fifteen to twenty minutes, was considerably longer than usual. Table IX gives the time required for three small globular ollas, about six inches in diameter at the shoulder. The surface of each of these was approximately the same as that of the polished black constricted-mouthed bowl recorded in Table VII.

TABLE VIII

Hrs. Min. Sec.

- 00.00 — Drawing of line about lower part of body begun
- 01.00 — Slipping of lower half of olla begun
- 08.30 — Slipping finished, polishing begun
- 23.30 — Vessel transferred to a slower worker

40.00 — Set aside
 04.45 — *Time out*
 40.00 — Polishing by slower worker continued
 47.45 — First application of lard begun
 48.45 — Polishing with chamois begun
 50.45 — Lower part finished
 ? — *Time out*
 50.45 — Slipping of upper half begun
 55.15 — Slipping of body done, slipping of rim begun
 56.45 — Polishing begun
 1.00.45 — Vessel transferred to the swifter worker
 1.20.45 — Vessel transferred to the slower worker, obliteration of junction-line
 1.53.45 — First application of lard begun
 1.55.45 — Polishing with chamois begun
 1.59.45 — Polishing of vessel completed

TABLE IX

	<i>Min.</i>	<i>Sec.</i>	<i>Min.</i>	<i>Sec.</i>	<i>Min.</i>	<i>Sec.</i>
Slipping begun	00.00		00.00		00.00	
Slipping completed	03.00					
<i>Time out</i>	01.00		omitted			
Polishing begun	03.00		04.00		04.00	
Set aside (second one beside fire)	27.30		31.30			
<i>Time out</i>	03.00		34.30		omitted	
Polishing continued	27.30		omitted			
Lard first applied, chamois used later	30.30		31.30		32.00	
Work on vessel completed	31.45		33.00		36.00	

PAINTING

Three varieties of paint are employed in making the designs—the black, or guaco, which is used on red and polychrome wares, the orange-red (also used as a slip), which serves as a paint in filling certain spaces in the designs upon polychrome ware, and the black ware paint which produces the designs upon polished black ware.^[45]

The consistency of the guaco, when dissolved in water and ready for use, varies from that of water to that of thick cream. The solution is sticky and has a characteristic odor. A thin solution when applied to the surface of a vessel is yellowish-green, and has much the same appearance as a fresh coat of the orange-red paint. After drying the two are easily distinguishable. A thick solution of guaco is dark brownish-green when applied to the vessel, and on drying has a noticeable “body”, so that the lines appear slightly raised and glossy. It is not possible to draw as fine a line with the thick solution as with the thin one.

The orange-red is never used in painting lines, but only for filling areas, and is therefore always applied with a medium or heavy brush.

In preparing the black ware paint, a slight amount of guaco is added as an adhesive, and the solution is often stirred with the fingers in order to insure complete dissolving of the coloring matter. When first applied to the vessel, this paint is almost transparent; as it dries it becomes yellowish.

The principal precaution taken when painting is begun is to guard against flies. They eat the moist guaco, causing blank spaces in the lines, and also make fly-specks on the slip which turn black in the firing. The room is therefore cleared of flies as far as possible, and great care then exercised to keep the door shut. The painter watches the pot constantly, and often interrupts her work in order to brush a fly from the design. If it is necessary to do the painting out of doors, the services of an assistant are required to wave a cloth back and forth across the top of the vessel to keep the flies away.^[46] When the work has been completed the vessels are immediately covered with a cloth.

The receptacle for paint may be either a small open-mouthed pottery bowl or a china sauce-dish. During the painting it is on the right side of the potter within easy reach, either on the floor or on a low stool. In it is always a stirring-stick.



a



b

PLATE 22

Painting designs on small vessels. The decorator's right hand does not come in contact with the pot, and the tip of the brush is trailed rather than stroked across the surface. a. Maria Martinez. b. Maximiliana Martinez.

The brushes are usually kept in the paint, or occasionally beside it, and sometimes a pencil is also near at hand (pl. 23, a).

In decorating the exterior of a constricted-mouthed bowl, the vessel is held upright and inclined very slightly towards the body; it rests on the left knee and is steadied by the pressure of the left hand against the interior of the rim (pl. 22, a). The potter sits in a position which allows the light to come over her right shoulder directly upon the section of the bowl being painted. As the work proceeds the vessel is turned counterclockwise. When the design extends slightly below the shoulder, the bowl is inclined to the left and away from the body. If a considerable portion of the design is below the shoulder, the vessel is inverted and is supported by the fingers of the left hand against the interior base of the bowl.

If the interior of an open-mouthed bowl is being painted, the enclosing lines near the rim are made first. During this process the bowl is held with the mouth vertical and to the right, while the lines are drawn on the lower part of the interior rim. The vessel moves counterclockwise. The bottom of the interior is painted with the bowl standing in its normal position on the left knee, the mouth inclined to the right and toward the body.

When a large vessel, such as an olla, is to be painted, it is placed upon a box or table of the proper height, which has previously been covered with a rug or mat to protect the base (pl. 23). The painter chooses her position so that the light will fall over her shoulder upon the vessel. During the painting of the upper part of the vessel it is touched with the left hand only when it needs to be turned. In working below the shoulder the vessel is tilted slightly to the left, and away from the painter.

The paint-brushes are made of slivers of yucca (fig. 5). They may be roughly grouped into three weights—fine, medium, and heavy.^[47] The fine ones are used in making outlines on the smaller vessels; the medium ones either for line work or for the filling of small areas; and the heavy ones for filling only. During the painting there are from three to six brushes in the paint vessel. These brushes are of such durability that the potters have no definite idea of the length of their usefulness. Since the fibres are brittle when dry, brushes are occasionally broken by accident; to prevent such breakage they are soaked in water for a few minutes to soften them before they are used.

The brush is held in the right hand, with the fingers in the position used in holding a pencil or pen. The ends of the fingers are two or three inches from the tip of the brush (pl. 22, a, b.; pl. 23, a). The fingers of the right hand do not rest upon the surface of the vessel and the entire right arm is unsupported (pl. 23), although at times the elbow may be unconsciously steadied against the body. The straightness and evenness of the lines drawn under these conditions are remarkable.

There is a slight personal variation in the method of contact of the brush with the vessel. One potter uses only the tip of the brush; another first places the tip upon the surface, and then makes the line with the body of the brush; a third places the body of the brush upon the vessel at the first contact. In making a line a trailing stroke is invariably used; its direction is usually toward the body, but occasionally away from it. The brush is always moved slowly, and the angle at which it is held varies according to the part of the surface over which the line is being drawn, and upon the part of the stroke which is being made. At the beginning of a stroke the angle is an acute one, varying from forty-five to seventy degrees (fig. 8, a); about the middle it approximates ninety degrees (fig. 8, b); while at the end it is often an obtuse angle of as much as one hundred and twenty degrees (fig. 8, c). On small vessels the strokes seldom exceed four inches in length; on larger pots they are sometimes as much as six inches long. After each stroke the brush is dipped into the paint vessel; it is then always drawn across the stirring stick, which is kept

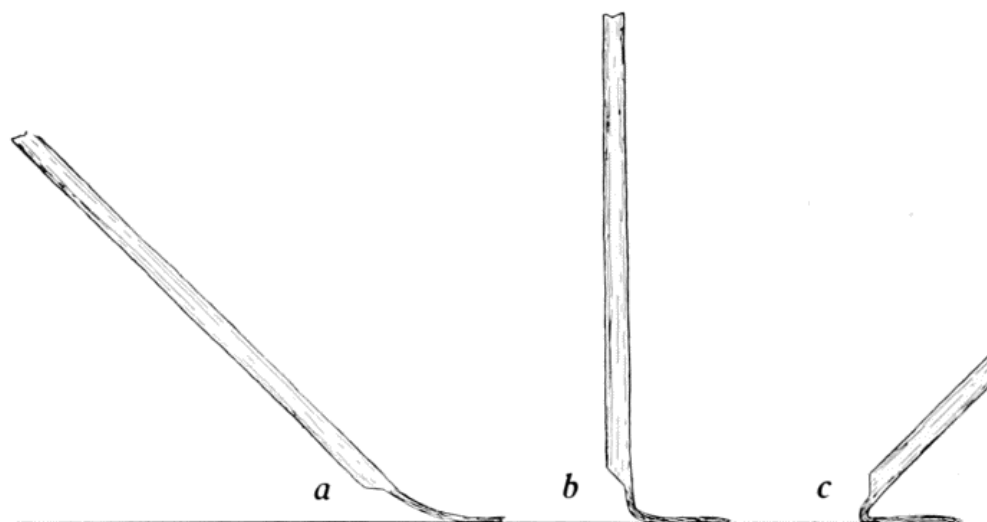


FIG. 8. Angles at which paint-brush is held during a long stroke; a. Beginning. b. Middle of stroke. c. End.

there, in order to remove excess paint before being applied again to the surface. Even after this precaution more paint occasionally remains on the brush than is needed; in such a case the excess is removed by touching with the tip of the brush various larger areas which will later be covered with paint.

The width of the lines drawn is largely dependent upon personal variation, although the type of design also governs their width to some extent. Exceedingly fine lines, for example, would obviously look out of proportion in a bold design upon a large olla. The lines vary in width from one-thirtieth to one-tenth of an inch; the great majority are between one-fifteenth and one-twentieth. Lines one-eighth or one-quarter of an inch wide are in reality double, that is, they actually consist of two lines painted so close together that they touch along their entire length. Still wider lines are made by two parallel lines, with the space between filled up. Some potters simply make the line once,

retouching it only in small sections which most plainly require such treatment. Others go over each line at least twice in order to insure a constant width.



a



b

PLATE 23

Decorating large ollas. Note the unsupported arm of the painter and the length of the brush. On the bench in front is the paint-cup with stirring-stick and extra brushes (Antonita Roybal).

From the point of view of technique, the lines upon the exteriors of small vessels, such as constricted-mouthed bowls, may be divided into two groups—the long enclosing lines and those within panels. In forming the enclosing lines, especially the long horizontal ones, the vessel is turned while the brush remains almost stationary. There are two methods of forming such long horizontal lines, which of necessity are made of a series of relatively short strokes. In one case the growth of the line is in the same direction as the strokes, that is, each stroke begins at the point at which the previous stroke ended. In this method the vessel is turned counterclockwise. In the other case the growth of the line is opposite to the direction of the stroke, that is, each stroke is begun a short distance beyond the end of the line and is drawn to meet the beginning of the previous stroke, the vessel being turned clockwise. In either case the fingers of the left hand, which hold the pot by the interior of the lip, are spread far apart in order to insure the constant and regular motion of the vessel. When the lines within a panel are being drawn, the procedure is just the reverse. The bowl remains stationary while the

brush is drawn across it. Occasionally lines which are nearly horizontal are drawn first in one direction, then in the other. In some cases, when long diagonal or curved lines are being made, the vessel is turned very slightly counterclockwise.

Large vessels, such as ollas, remain stationary while the brush is being drawn across the surface, and are turned only when a new area is to be worked upon. Lines within panels are usually made with a single stroke of the brush, for it is seldom that a diagonal or curved line within a panel is more than four or five inches long. In large open designs the long lines, both straight and curved, are often drawn in two parts. Crosshatching is made by two sets of parallel lines crossing each other. Dotting is done with the tip of the brush touched lightly to the surface. Spaces are filled with a heavier brush by means of many short strokes. As a rule the strokes begin at the upper right hand corner of the area and proceed downward. In filling spaces, as in drawing lines, some women go over their work only once, others apply two or three coats in order to insure even distribution of the paint.

Some potters exhibit much sureness of touch and confidence in their own ability; their designs are executed without hesitation, but at the same time slowly. Others make hard work of the painting, seem uncertain, and occasionally stop to rest and inspect the symmetry of the pattern.

There are two general points in the painting of vessels which should be emphasized. The first is the remarkable steadiness of the painter's hand. It seems almost incredible that such straight, unwavering, even lines can be drawn by a hand which is supported only by a completely free arm, and guided merely by the end of the brush. The second is the fact that the men often assist in the painting of designs. They may either execute a complete design upon a vessel, or relieve the painter from time to time during the construction of large patterns. This is the only phase of the actual manufacture of pottery in which the men of the village take part.^[48]

FIRING

The firing of the vessels is not only the last major step in the making of pottery, but it is also the most critical one. The supreme test of the potter's work is to subject it to the fire, for many forms of accident occur during the process, some of them due to careless handling of the vessels or of the firing materials, others to defective workmanship or to hidden flaws in the paste. It is only natural that the potters should exhibit considerable excitement and nervousness during the firing. Some women proceed calmly and methodically, and take occasional accidents as part of the day's work; others are exceedingly nervous, and show exasperation when things go wrong. The latter, as a rule, have more accidents than the former. One potter, in six burnings, lost two large ollas through breakage, and had innumerable smoke-clouds, several of them rather severe ones.

The firing falls naturally into four distinct phases: the preparation, the building of the oven, the burning, and the treatment after burning. Each of these phases will be considered separately.

PREPARATION

The oven is usually situated behind the potter's house. All that is required is a level space free from vegetation; no attention is paid, as a rule, to the position of the sun or to the direction of the prevailing wind, and the same spot is used time after time. If on the evening before the firing there are any signs of rain, or if a heavy dew is expected, the area on which the oven is to be built in the morning is covered with a thick layer of dead ashes to protect it from moisture. It would be a fatal mistake to make the oven on ground which would give off steam during the burning.

About six o'clock the next morning a fire is started on the area over which the oven is to be built. While this preliminary fire is thoroughly drying the ground, the accessories are collected ([pl. 24, a](#)). Slabs of dung are brought out from the storage places; some of them are placed against the house wall to dry in the sun, others remain in the washtubs in which they have been carried. The material for the grate, and other iron bars, are placed nearby. Pokers of wood and iron, shovels, and sometimes pitchforks, which are to be used later in handling the burning fuel and hot vessels, are placed within reach. When a collection of small tin cans, pail-tops, and small stones saved for this purpose has been added, the workers are ready to begin the burning. Usually a supply of cedar kindling has been split, but sometimes this is done as the wood is needed.

BUILDING THE OVEN

When the preliminary fire has died down to a mass of coals, these are levelled to form an area of hot ashes about three feet in diameter. On this the oven is built. The surface upon which the vessels are placed must be raised a few inches



a



b

a. Drying the oven site with a preliminary fire before burning the pottery. At the lower right-hand corner are an old iron grate which will later support the vessels, two iron rods to hold up the roof of the oven, and a wooden poker. Cakes of dung for the oven walls are drying in the sun against the house.

b. The vessels on the grate, and the oven wall of dung cakes begun. Small tin cans support the cakes and keep them from touching the vessels. The grate rests on stones, and split cedar kindling is placed below it on the embers of the preliminary fire.

above the ground to permit the introduction of fuel below the pottery. Such a surface is usually formed of iron rods of one kind or another. One potter built up a grate of a varied assortment of iron junk supported on tin cans and odd-shaped iron fragments. Another potter used a worn-out stove-grate supported on four half-bricks.

The vessels are then placed upon the grate in an inverted position. No attempt is made to keep them from touching one another. In fact, they are crowded together in order that the greatest possible number may be burned at once (pl. 24, b). In firing red ware and polychrome ware only one layer of vessels is placed upon the grate. When polished black ware is to be burned, two layers may be made. The vessels in the second layer are always placed carefully between those of the bottom layer, in order that there may be free circulation of air on both the interior and exterior of all the vessels (pl. 25, a). As a rule the larger bowls are placed in the lower layer.

The number of polychrome vessels burned at one time varies from half-a-dozen to twenty, according to the area of the grate and the size of the vessels. At one burning there were eleven pieces, eight medium-sized and three small; in another there were twenty, eight of which were medium-sized, and the other twelve very small. It is possible to burn as many as thirty-five polished black bowls at once. Some potters burn both polychrome and red pottery in the same oven; others insist that this should not be done, because red ware requires far less time and heat than does polychrome.

After the pots have been placed on the grate, pieces of split cedar six to fifteen inches long are inserted underneath it. Piñon is never used, but the potters could not explain this, saying they had never tried it. A wall-like ring of dung-cakes, placed on edge but not set as snugly together as they might be, is then built around the grate (pl. 25, b). Small tin cans, held in place by the weight of the dung, are used to prevent the cakes from touching the vessels (pi. 24, b, shows one of these cans very clearly). Sometimes small stones and broken bricks are used in place of cans. While the ring is being completed more kindling may be added. The Zuñi oven differs from that of San Ildefonso in being built of smaller pieces of manure which are laid up horizontally instead of on edge (see pl. 31, b, c).

The fire may now be lighted before the oven is finished, but sometimes the potters wait until the roof is nearly complete. Cedar-bark shreds are generally used to start the fire. The lighting is done through the spaces between the dung-cakes, usually at more than one place, sometimes in as many as five. No attention is paid to the wind in this process. If the fire does not start quickly enough, cloths are used to fan the flames.

The roof or vault of the oven is then built. Some potters place iron bars across the top of the dung-ring, keeping them from touching the vessels by inserting tin cans and stones. On these bars flat cakes of dung are laid, and the larger holes are covered with smaller pieces of dung. Other potters dispense with the bars and rest a large circular cake of dung upon cans and stones set in the centre of the pile of pottery. The space between this central cake and the top of the dung-ring is bridged by other cakes which rest on the dung alone. In this manner a low vault of dung-slabs is formed, roughly square in shape, two and a half to three feet on a side, and about a foot and a half high (pl. 25, c). There are still left numerous spaces, both large and small, between the cakes. These are covered with smaller pieces of dung, and with old tin tops of lard-pails or discarded tin plates (pl. 26, a). Even after this, however, there remain enough small spaces to allow free circulation of air, and to prevent the smothering and consequent smoking of the vessels. Through these smaller spaces the flames and pots may be seen. The building of the oven is now complete; by this time the fire has usually been started, and the flames are well under way.

As a rule more than one burning is undertaken during a morning. The building of the oven for the second burning differs slightly in its preliminary stages from that described above. After the previous oven has been destroyed, the ashes are smoothed out, and the grate set straight and carefully dusted so that no ashes remain (pl. 26, b). The bowls are then placed on it as before. Partly burned cakes of dung, which remain from the first burning, are re-used. If, in placing them on the oven, ashes drop on the vessels, care is taken to blow them off.

Some potters stand the vessels which are to be burned on the warm cakes of dung for a few minutes before placing them on the grate (pl. 26, b), a process which is supposed to heat them slightly and thus prevent breakage during firing. Other potters would not think of doing this, for vessels treated in this manner always have discolored bases, where the warm dung has touched them.

Ollas twelve to fourteen inches in diameter are burned in pairs, with a few smaller pieces beside them. The oven is built in the same manner as before. Very large ollas are usually fired one at a time; the oven, except for its greater size, is said to be of the usual type.

BURNING

As was said above, the firing of the pottery is done in the morning. The Indians say that they wish to get it over with "before the sun grows hot". Their own personal comfort probably plays a more important part in this than does the effect of the sun upon the pottery. As a rule the firing is withheld until a sufficient number of finished vessels have accumulated to permit three or four burnings in one morning. The entire forenoon is spent at this work, and sometimes it drags over until early afternoon. The actual burning of the vessels occupies a period of about half-an-hour for each oven; it is said that the length of time has not been appreciably shortened in the past forty or fifty years.

The black ware is always saved for the last oven, since the process employed completely precludes the possibility of rebuilding the oven at once.

At the beginning of the burning white smoke is apt to be given off, but too much smoke is regarded as a bad sign, since it signifies that the dung cakes are not sufficiently dried. After fifteen minutes or so, that is when the process is about half finished, the amount of smoke has greatly diminished ([pl. 26, a](#)). About this time more kindling is added through openings made by temporarily removing a piece of tin, and new cakes of dung are added where necessary, usually on the top of the oven.

The Indians say that they judge of the condition of the pottery in the oven by its color; the paste and slip must be of just the proper shade before the vessels are



a



b



c

PLATE 25

The building of an oven. a. A double layer of bowls is on the grate; the woman is placing a can between a dung-slab and a bowl. b. The wall of dung-cakes completed. c. The oven finished, but the chinks not yet filled.



a



b

a. The oven fired. The chinks between the dung-slabs have been partly closed by smaller pieces of dung. b. A firing completed and a grate prepared for a second firing. Two vessels to be burned are warming on hot dung-slabs; the pots that have just been taken from the fire are cooling in the shadow of the house.

thoroughly burned; if they are too dark the burning must continue.^[49] Some potters seem to pay little or no attention to the color of the vessels in the oven.

When the potter considers that the burning has been completed, the cakes of dung on the top of the oven are lifted off with a pair of pokers, and those in the ring around the grate are tipped outward. Thus the vessels on the grate are exposed, and their removal is begun at once. Some are tipped on shovels or hay forks; pokers are inserted under and into others. They are then deposited upon tins which have been placed on the ground within six or eight feet of the oven, and are left there from ten to twenty minutes to cool. The women often complain about the heat from the oven during the removal of the vessels. As soon as the grate has been cleared, preparations are begun for the next burning (pl. 26, b).

In one instance a potter decided that a certain vessel which had been removed from the grate was not sufficiently burned. Another fire was already in progress, but when the third was built the vessel was replaced on the grate and burned again. When it came out, however, one side was badly overfired.

The time occupied by various burnings is given in the following table. The first four columns represent burnings by Maria Martinez; the last three by Antonita Roybal. The seventh column records the oven in which the partly fired piece was reburned, and in which a cooking vessel was also placed.

TABLE X

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	<i>Minutes</i>						
Preparation and building of oven begun	00	00	00	00	00	00	00
Fire started	25	15	06	12	15	15	09
More kindling and dung added	45	25	25		57		
Oven broken up	52	37	36	41	72	40	27
Last bowl removed from grate	58	41	42	47	77	45	30

The time required for burning the different wares varies only slightly. Some potters say that red ware takes less fuel, less heat, and less time than polychrome ware; others place both wares in the same oven. The difference in the time required for these two wares is probably no greater than the chance variations in the length of burning as given in Table X. Cooking vessels need to be fired only from one-half to two-thirds as long as the two wares just mentioned (cf. Table X, column 7). The polished black ware will be considered later.

The changes in color before and after firing, and when hot and cool, are very noticeable in some pigments and slips. The orange-red paint is yellow before firing, firing turns it orange-red, almost the color of burnt sienna; cooling produces no noticeable change in this shade. Burning has only a fugitive effect on the red and dark-red slips. While still hot, upon removal from the oven, these are both a dark chocolate shade. As the vessels cool the different reds gradually reappear, until when cold they are the same color as before firing. In some cases, these pigments seem to be just a shade darker after firing than before. The two kinds of white slip are a dead white when applied; after firing they take on a slight pinkish-brown tinge, more cream than white. Cooling produces no further change.

The most interesting change caused by firing is in the black vegetable pigment (guaco). It has already been pointed out (p. 66), that when this paint is laid on thickly, the line made is a little raised and glossy, like a glaze. This "body" to the paint is probably formed by vegetable matter in suspension in the solution. When the vessels with such decorations are removed from the oven after burning, these lines are seen to be white or grey. The material which formed the body of the pigment has been reduced to white ash, which adheres to the surface, but can be rubbed off with the finger or a cloth, while the black color itself has been burned into the clay. Where a thin layer of the pigment was applied, the line, after burning, is light grey, because the black under the thin coat of ash shows through. In those vessels on which a thick solution of pigment was used, the decorations are dead white, and the surface of the ash is often crackled. The thickness of this ash seems to have some effect upon the manner in which the black pigment is burned into the clay, for in decorations covered with a thick layer of ash, the black is of an irregular color with streaks of grey in it. When the thin pigment is used, the resulting black on the finished vessel is of a uniform shade. The glossiness and raised character of the pigment is of course completely destroyed during burning. Guaco lines do not change in shade while the vessel is cooling.

The color of the paste is little affected by burning. In both the red and the white clays, the change is simply one of tone, that is, the clay is lighter after burning than before. The cooking-vessel clay before burning is greyish yellow, but after it has been removed from the oven it is a rich orange-yellow. These cooking vessels eventually become blackened by use over an open fire.

Vessels destined to be polished black ware are treated in quite a different manner during the firing stage, for they are to be subjected to a smothered fire, which will result in the deposition of carbon, thus turning them from red to black. The oven is built in the same way, but greater care is taken to fill gaps, so that more of the heat may be retained. Enough spaces remain, however, to permit free circulation of air. More kindling is used, for a hotter fire is necessary. When the fire has reached the stage at which other wares are removed, it is smothered with new, pulverized, loose manure. Just before the smothering the vessels have the dark chocolate color typical of the red wares while hot. The potter always attempts to smother the entire oven at once by dumping upon it a whole washtubful of fine loose manure. If there is a wind, flames are apt to break out in one or two places. They are, however, hurriedly extinguished. When the loose manure is

added, the arch of the oven is, of course, broken, and both cakes and loose manure come in contact with the vessels. For this reason the potter does not exercise particular care in building the oven to keep the cakes from touching the vessels, as they are to be entirely black in the end and a little premature smoking does no harm. After the manure has been added, the mound is continually prodded with a poker to redistribute the loose manure and make certain that all the pieces are equally covered. An extra supply of loose manure is at hand, and this is added in large and small quantities from time to time, as occasion demands. Two washtubsful are generally used, sometimes three. The mound of manure gives off a dense white smoke after the smothering has begun ([pl. 27](#), a). Great care is taken to prevent flames from appearing, since these would remove the carbon from the vessel.



a



b

PLATE 27

a. A fire smothered with loose manure to produce polished black ware. b. Wiping vessels that have become cool enough to handle. One woman wipes off the ashes with a dry rag, then passes them to her companion, who goes over them with a slightly greasy cloth.

Ten or fifteen minutes after the smothering the first piece is dug out of the mound with the poker. It is placed a foot or two from the oven and hastily wiped with a dry cloth to remove all manure. The surface of the vessel is a beautiful shiny black color, which does not change at all on cooling. If black ware paint (see p. 24) has been used, it changes from the greenish-yellow color that it had before burning to a dead matte black which contrasts very effectively with the polished surface of the vessel (see [pl. 8, a](#)). At first the vessels come from the smoking mound slowly, and all holes left in the pile by the removal of vessels are carefully filled again, but later they are taken out as quickly as possible, and toward the end no attempt is made to keep the pieces still in the pile thoroughly covered. If, when a vessel is removed, its condition does not quite suit the potter, it is replaced in the smoking manure and completely covered.

Certain polished black vessels are further manipulated in such a way as to produce an irregular red blotch upon them.^[50] Specimens destined to receive this red blotch, which is usually placed near the rim, are reburied in the hot manure lying round the edge of the pile with that portion which is to have the blotch left uncovered. Against this exposed portion is placed a smoking fragment of a dung-slab. The action of the heat and air results in the removal of the carbon from the surface, so that that part of the vessel which is not covered with manure and ashes regains its former color, the red of the pigment. Since the pot is pushed about a good deal during this process, the line between the red and the black surface is not always very definite, which improves the appearance. The Indians themselves cannot tell beforehand just what shape the blotch is going to take, and must therefore watch the vessel continually. Sometimes burning shreds of cedar bark are placed against the exposed surface to hasten the process, but actual flames are not necessary in order to obtain the desired result. If the red blotch is too large, part of it is simply recovered with hot manure-ashes, and a few minutes later will again become jet black. So far as looks are concerned the success or failure of this red blotch upon black ware depends largely upon the artistic sense of the potter making it. In one group of thirty-three polished black vessels fired together, sixteen were given a red blotch; eight or ten of these were excellent pieces.

The time consumed in burning polished black ware is shown in Table XI. It should be noted that before the loose manure was added, the vessels were subjected to the fire about the same length of time as those of other wares.

TABLE XI

	<i>Minutes</i>
Preparation and building of oven begun	00 00
Fire started	16 18

More fuel added	41 39
Oven smothered with loose manure	46 45
First bowl removed	59 59
Last bowl removed	74 99

ACCIDENTS

The accidents which may occur in firing are of two general classes—those which result from faulty treatment during the construction of the vessel itself and those which are due to careless manipulation in burning. The latter are the more frequent.

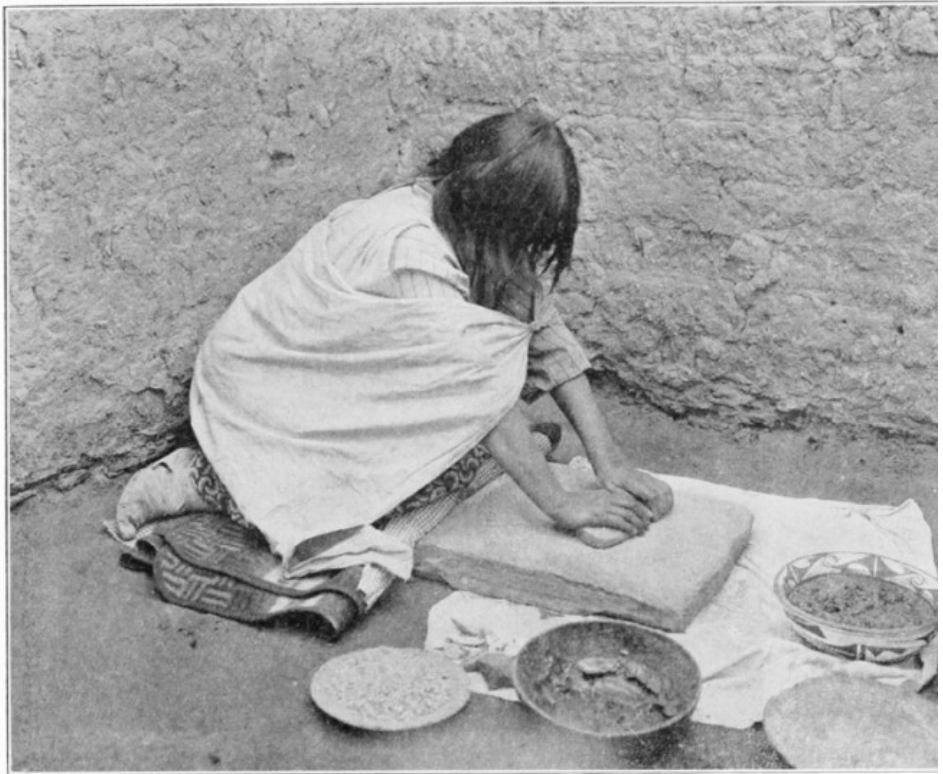
A vessel may crack or flake badly in the oven; often the entire base breaks off. The fault in this case lies in the original moulding, because small stone fragments or air-bubbles were allowed to remain embedded in the paste. Since the rate of expansion under heat varies for different substances, a great strain develops about such stones or air-bubbles during firing, and the natural result is cracking or flaking of the clay. In some larger vessels the slip flakes away around the shoulder, where the two areas of the surface met when the polishing was done (see p. 64). In the potter's opinion these flakes, which are exceedingly small, are probably caused by minute air-bubbles that had lodged under the slip while the line about the shoulder was being polished. A third type of accident is merely a blemish; it is due to flies settling on a vessel before burning; the resulting fly-specks are burned into the clay, and the surface of the pot is sprinkled with dots quite as black as guaco decorations.

The most common form of blemish caused by careless manipulation during firing is the "smoke-cloud" or "fire-cloud", a circular blackened area, the darkest part of which is at the centre. There is no question but that this is caused by contact between the vessel and a dung-cake, and the result is probably due to moisture in the dung which brings about a deposit of carbon on the surface. If, during the burning, a cake of dung falls, it will probably come in contact with the surface of some bowl. If it is removed at once there is little danger of a smoke-cloud, and the potter, when she notices such an accident, immediately tries to secure the fallen cake with two pokers. With much skill she carefully extricates it from within the oven without disturbing the other cakes about the place from which it fell. Fire-clouds may be produced upon the bases of vessels by setting them on hot cakes of dung before firing; or upon damp, or relatively damp, ground immediately after firing. Excessive dampness, such as steam from wet ground under the oven, may in addition to discoloration, produce warping of the vessels. Another sort of blemish is occasionally caused by rust from a tin can resting upon a vessel during firing; such a spot is usually reddish brown in color, and covers a very small area.

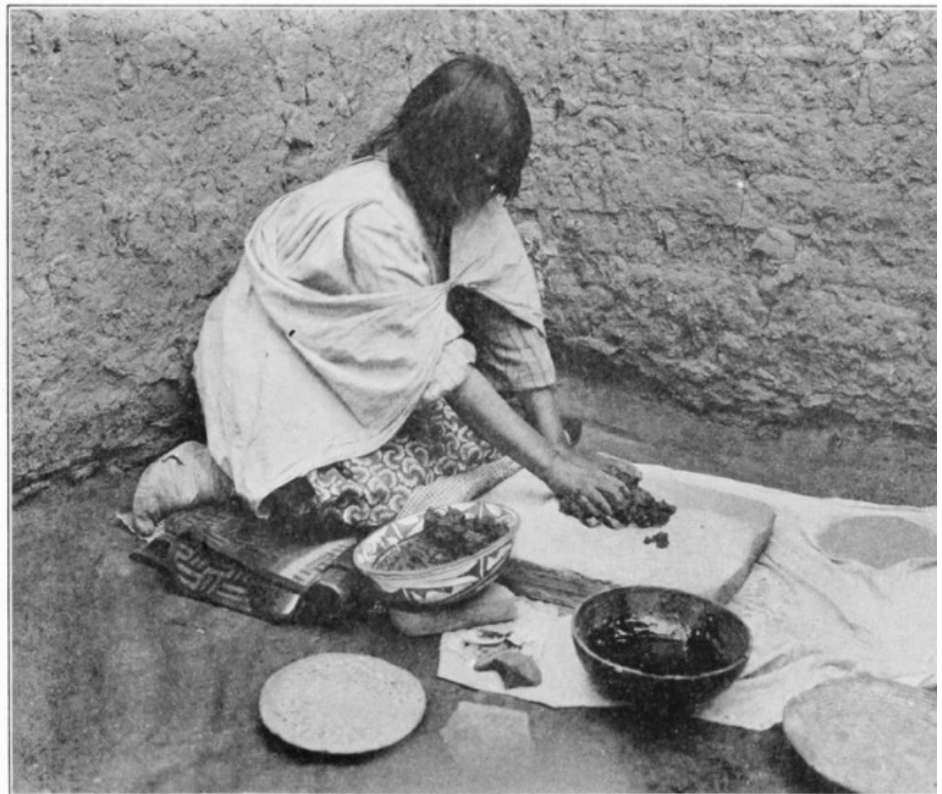
If light areas appear on polished black ware, it is safe to assume that they were brought about by lack of proper ventilation within the oven during the preliminary burning, such as would occur if two vessels were so close together as to prohibit the free passage of air. If white slip is touched just before burning, the oil from the fingers is apt

to be fired in, thus ruining the uniform appearance of the surface by the potter's indelible fingerprint. To avoid such an accident, polychrome vessels are usually handled only by the interior of the rim on the day they are burned.

Underfiring and overfiring, with consequent damage to the ware, are usually due to gross carelessness on the part of the potter in the irregular placing of kindling. The wind also has something to do with it. If there is a stiff breeze blowing, the probabilities are that the windward surfaces of vessels within the oven will be underfired, while those in the lee of the vessels will be overfired.



a



b

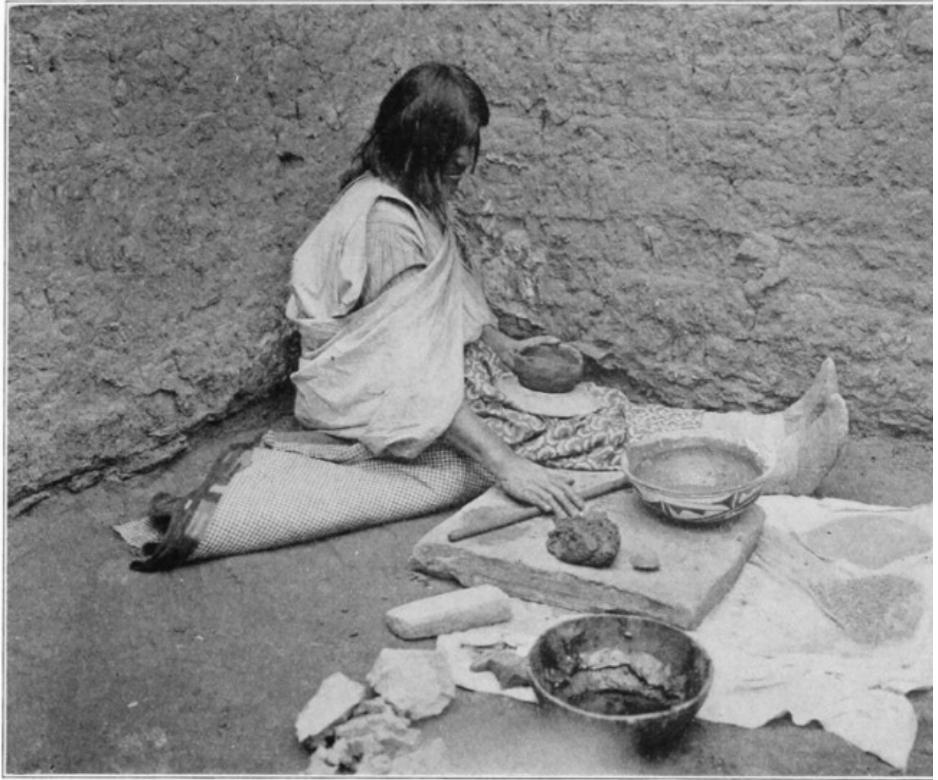
Courtesy Amer. J.

PLATE 28

Courtesy Amer. Mus. Nat. Hist.

ZUÑI. PREPARING CLAY

- a. Pulverizing dry ingredients on flat stone slab with a *mano* or hand stone.
- b. Wetting, mixing, and kneading clay; water-container in foreground; prepared clay in bowl.



a



b

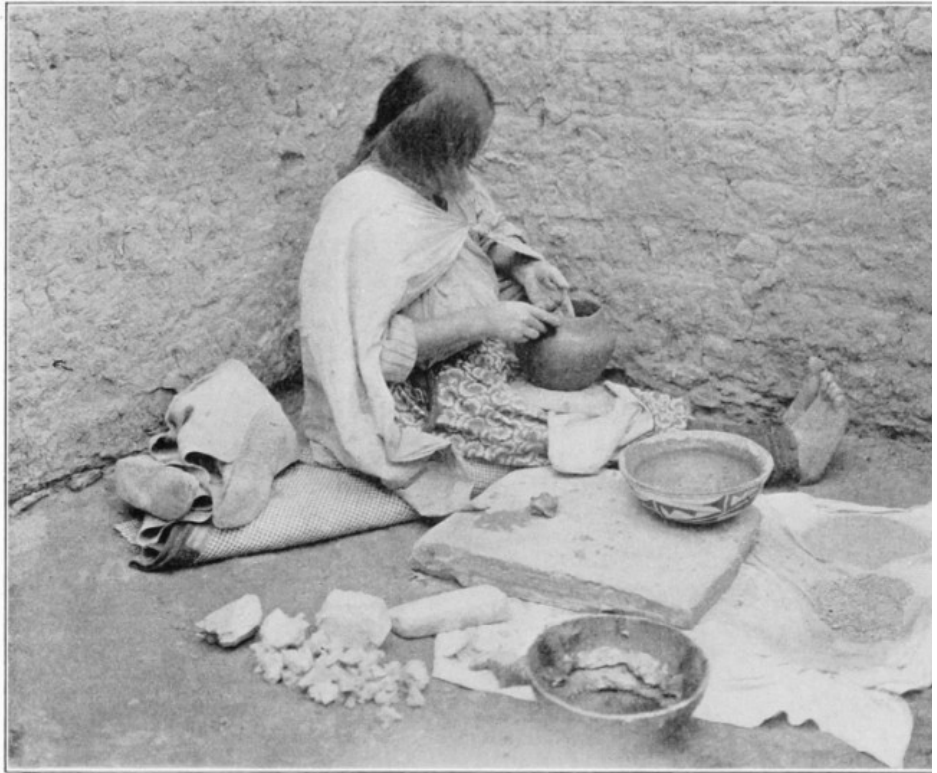
Courtesy Amer. M

PLATE 29

Courtesy Amer. Mus. Nat. Hist.

ZUÑI. MOULDING

- a. Lower part of vessel completed; potter forming roll of clay with which to continue building.
- b. Applying roll of clay; left hand guides roll; right hand welds it to side of vessel.



a



b

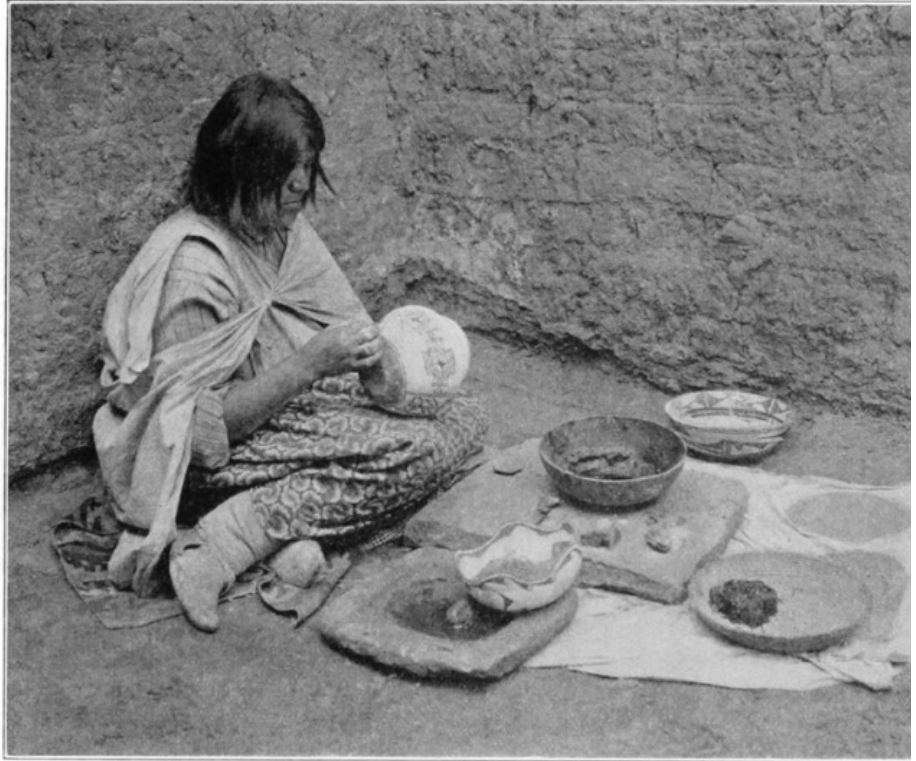
Courtesy Amer. Mus.

PLATE 30

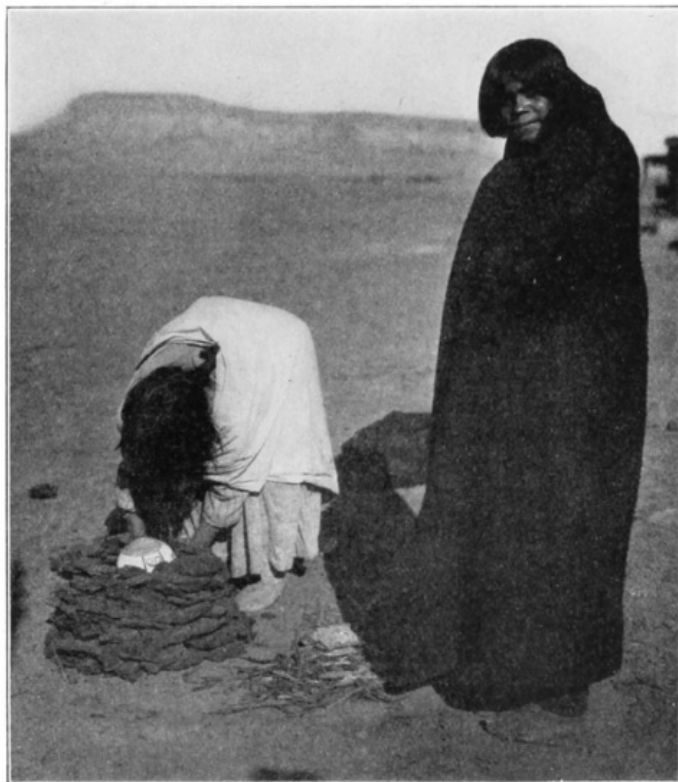
Courtesy Amer. Mus. Nat. Hist.

ZUÑI. SURFACING

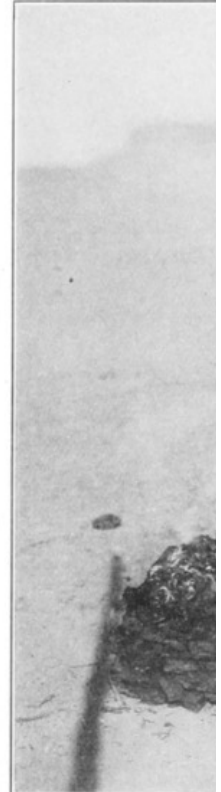
- a. Going over outside of vessel with moulding tool to smooth out irregularities.
- b. Vessel has been removed from base-mould and covered with white slip from bowl in foreground. Potter now polishing slip with rubbing stone.



a



b



c

PLATE 31

ZUÑI. PAINTING AND FIRING

a. Applying design with yucca-leaf brush. Black pigment in shallow stone mortar, red in small white bowl. b. Building oven of dung cakes—note kindling at feet of figure at right. c. Oven domed over and set afire.

(Photographs from American Museum of Natural History).

In underfired vessels the color of the pigments is not true, being usually midway between the unfired color and the fired color; as a rule the paste is also dark and friable. Overfiring is more easily detected than underfiring. A slight overfiring may be first noticed in the black guaco paint, which has a tendency to become bluish and light if burned too much. Other bad results follow in quick succession as overfiring is increased: at an early stage the white slip becomes smoky; in severe overfiring it may turn black, as in a smoke-cloud; the paint of the designs is apt to flake off; and finally, in a bad case, the guaco will turn nearly white.

TREATMENT AFTER BURNING

Polished black vessels, when removed from the fire, are set directly upon the ground within a foot or two of the oven. They are at once wiped hastily with a dry cloth to prevent any fragments of the new manure from sticking to the surface. When the vessels are sufficiently cooled they are again wiped with a clean dry cloth and taken into the house for storage. Occasionally the first wiping, while the bowl is still hot, is omitted.

When pieces of polychrome ware or red ware are removed from the oven they are placed on the ground some six or eight feet from the fire, resting on tins in order to prevent possible discoloration from contact with the damp soil, as well as to prevent dust and dirt from sticking to their bases. The vessels are piled on these tins in any manner, often three or four on top of one another. When the pottery first comes from the fire, it is still much too hot to touch, and radiates heat copiously. Ten or fifteen minutes later, when the vessels are cool enough to handle, they are removed from the tins and collected at some convenient place to await wiping.

First the pieces are gone over with a clean dry cloth to take off the guaco ashes and any dust that may remain from the oven (pl. 27, b). Each vessel is then wiped with a slightly greasy cloth, which gives a faint sheen to the surface and removes the blue tinge which is apt to be found in the black guaco-covered areas. Some potters have substituted a chamois skin for this second cloth. It took the two women shown in pl. 27, b, just fifteen minutes to wipe with the two kinds of cloths the twenty-two vessels figured. When the wiping is done, the pottery is carried to the house and stored in one of the rooms. The pieces are placed on the floor, usually with a carpet or rug under them, and covered with a sheet to keep off dust and flies.

The time occupied in each of the individual stages of pottery making has been considered, but the total time, from the beginning of the moulding until the finished vessels are placed in the store room under a sheet, is far in excess of the mere sum of the

separate hours and minutes used in each process. Household duties and other tasks are constantly interfering with the work. About nine o'clock on a certain morning one potter began moulding the first of a group of perhaps forty pieces. The scraping was commenced on the morning of the third day, and the polishing on the fourth morning. Nothing was done on the fifth day, but early on the sixth the painting began. A fiesta interrupted the work for two more days, and the burning was done on the morning of the tenth day, although it could have taken place on the seventh. Another potter finished burning several large ollas at noon on the ninth day.

PAINTING OF DESIGNS

By means of the decorations on the vessels the potter expresses her personality. The type of design used by any one potter is very constant, and is distinctly individual. It is a comparatively easy matter, by an inspection of the design alone, to distinguish the vessels made by one potter from those of another.

The designs are planned in several different ways. Maria Martinez sits with the bowl in her hands for a few minutes doing nothing; apparently she is working out in her mind the combination of elements which she will use. Designs so conceived are generally simple. Maximiliana Martinez begins painting almost at once. While she is working ideas occur to her and are incorporated. Occasionally, after the painting has been completed and the vessel set aside, she will pick it up again to add some detail. This method of working is apt to cause somewhat involved figures. Antonita Roybal, in choosing her designs, refers to drawings of her own, or to photographs which have come into her hands, of old San Ildefonso vessels. This potter uses a pencil to outline very sketchily the design upon the vessel, as an aid in obtaining the proper symmetry. The figures obtained in this manner are usually very elaborate.^[51]

The first lines drawn in a design are almost without exception the enclosing lines under the rim. These are followed by the enclosing lines near the shoulder (in the case of an olla), or near the bottom of the interior (in the case of a bowl). When the design consists of panels, the vertical division-lines are then added. If there are to be four panels, one division-line is drawn, and then the one on the opposite side of the bowl, dividing the surface into halves. Each of these halves is then bisected. The next step is to double all the vertical division-lines. Occasionally each quadrant is judged by the eye only, and the division lines are drawn in sequence about the vessel. When there are to be either more or less than four panels, they are outlined one after the other. No measuring instrument of any kind is used.

After the skeleton of the design has been completed, the outlines of the figure within each panel are drawn. All the lines are first placed in one panel, then the second panel is finished, and so on, until all have been filled with the outlines. All the areas on the vessel that are to be colored black are then filled, followed by the areas which are to be red. The strokes taken in outlining are not always made in the same order in the various panels. Such variation is entirely natural and should be expected in work done without the use of a visible pattern.

When a design is attached to the lower enclosing line of a panel, it usually consists of a repetition of some small figure of one or two elements. The position of such added figures has absolutely no relation in the mind of the potter to the panel-design.

When a design consists of a repetition of figures not enclosed within panels, the painter always refers, before adding another figure, to the amount of the surface as yet unfilled. The correctness of the painter's judgment is therefore easily determined by the

proportions of the last figure in the design as compared with the remaining figures. As a rule, with the product of the present-day potters of San Ildefonso, it is exceedingly difficult to determine in a finished vessel, which of the figures was the last one drawn.

In designs which consist of a single figure, or of two or more figures, with several complex elements, the development of the design upon the vessel is necessarily at variance with that described above. As a general rule, each element is completed, including the filling of areas, before the next element is outlined. Similarly, each figure is finished before the next is begun.

In considering generally the ornamentation of San Ildefonso vessels, a distinct group-similarity can be observed. The polychrome vessels are painted in black and orange-red upon a white base. The red-ware vessels are decorated only in black. The designs usually consist of several almost identical figures, each composed of a small number of rather simple elements, in which curved lines are common. And yet, in spite of this almost indefinable similarity among the vessels, the differences between the designs made by different potters of the village are clear-cut and distinct.

Maria Martinez specializes in constricted-mouthed and similarly shaped bowls of polychrome ware. Her lines are relatively narrow, and the black of the design is very uniform in color. She uses panelled designs almost exclusively. The elements within the panels are simple, and few in number. Her work is easily recognized because of its simplicity and pleasing composition (see [pl. 6](#)). Maximiliana Martinez generally confines herself to red ware in the form of small ollas and small bowls with a slightly constricted lip. Her lines are somewhat wider and more uneven than those of Maria, and the black is of varying shades. As a result of her custom of developing the design as the work progresses, the figures are rather complex and intricate, or, as the Indians themselves say, "mixed up". The outstanding characteristic of her work is the use of one or more elements entirely detached from the figure proper. Antonita Roybal devotes most of her time to making large ollas of red ware. Her lines are relatively broad, often indeed of double width. Her blacks are also uneven. The designs upon her vessels are usually composed of two or four very large, complex figures, in which spiral curves are conspicuous. Dotted, crosshatching, and filling of many small areas characterize her work. The products of a man painter, Julian Martinez, are easily recognized by the abundant use of very narrow lines. The figures, which are usually composed of many intricate elements, impress one with the amount of detailed and careful work lavished upon them. Julian has obviously been strongly influenced by the technique of modern Hopi potters, such as the famous Nampeo, whose work he of course often sees in the curio-stores and in the State Museum at Santa Fe.

In the following pages typical designs of Maria, Maximiliana, and Antonita are discussed in detail, with the aid of figures upon which the direction and order of the strokes have been indicated. These drawings are given in order to amplify and clarify the general statements made above.

DRAWING NO. 1 ([fig. 9](#)); original by Maria Martinez. This design was on a constricted-mouthed polychrome bowl, and consisted of four identical panels, each four

and one-half inches long by two and one-half inches wide. The work was divided into four distinct sections, or stages, each of which was completed in all four panels, one after the other, before the next stage was begun.

First stage (fig. 9, a); construction of the framing lines. Lines 1-4 are drawn completely around the bowl; then lines 5-8. The corresponding two pairs of lines are then drawn upon the other half of the bowl, thus dividing the space into four sections.

Second stage (fig. 9, b); placing of the outlines within each panel. The first three lines drawn (9-11) divide the panel into three triangles. Then the details are

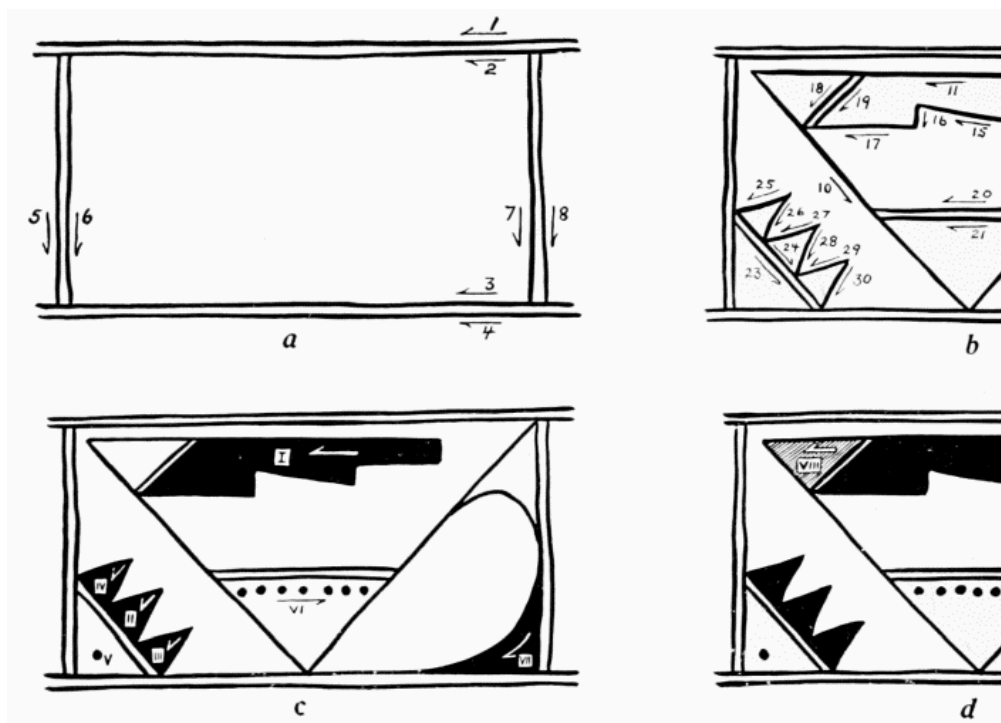


FIG. 9. The growth of a polychrome design as painted by Maria Martinez (the shading indicates red).

outlined in each triangle in turn. When the outlining of one panel has been completed, each of the other panels is treated in turn in the same manner. The work of Maria is noteworthy in that the order and direction of the lines and the filling of spaces is hardly ever changed from one panel to the next.

Third stage (fig. 9, c); filling certain areas with solid black paint. The largest area (I) is always filled first. No order is followed in filling areas II, III, and IV, and in making the dot (V). In one panel, IV is filled before II, in another V precedes the other three. After this the dots (VI) are placed below line 21. In the different panels the number of these dots varies from six to eight. The last area filled in this stage is the triangle VII. When the first panel has been completed the other three are treated in like manner before the fourth stage is begun.

Fourth stage (fig. 9, d); filling of areas with red paint (shown in the reproduction by shading). In some panels area VIII precedes area IX, in others the reverse is the

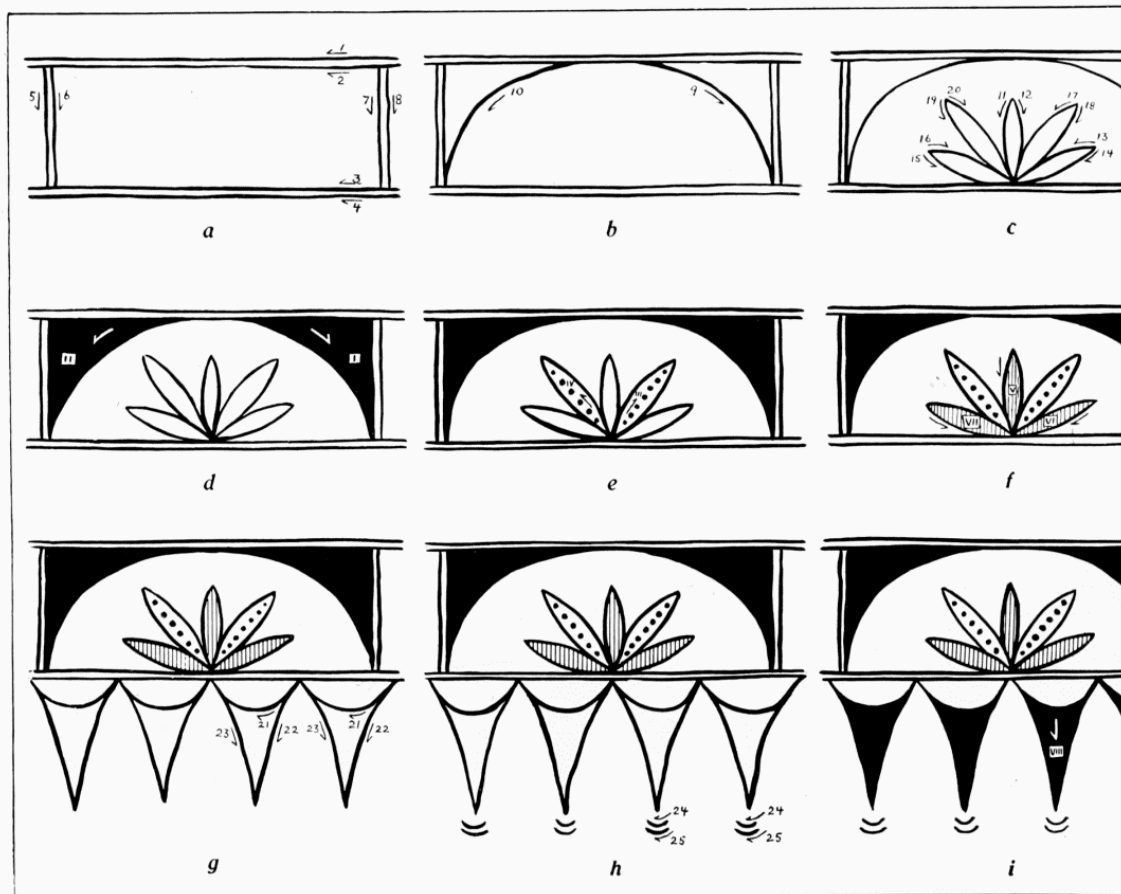


PLATE 32
Design by Maria Martinez

case. With the completion of this fourth stage in all the panels, the painting of this comparatively simple design is finished. The time consumed was twenty-five minutes.

DRAWING NO. 2 (pl. 32); original by Maria Martinez. This design may be placed upon either a constricted-mouthed bowl or an olla-bowl. It is made up of five identical panels. During the painting the bowl is turned counterclockwise. There are eleven distinct stages.

First stage (pl. 32, a); the enclosing lines of the panels are produced. Each panel is four inches long and one and a half inches wide. The horizontal lines (1-4) are drawn completely around the bowl before the dividing lines (5-8) are drawn.

Second stage (pl. 32, b); lines 9 and 10 are drawn, dividing each of the five panels into a semicircle and two triangles.

Third stage (pl. 32, c); the plumes are outlined in each of the panels by means of lines 11 to 20.

Fourth stage (pl. 32, d); the two triangles (areas I and II) are filled with black pigment. Sometimes area I is filled first, sometimes area II.

Fifth stage (pl. 32, e); when the triangles in all five panels have been filled, the dots are placed in two of the five plumes in each panel (areas III and IV). These dots are sometimes six in number, sometimes seven. They are painted from the base of the plume upwards.

Sixth stage (pl. 32, b); areas V to VII are filled with red pigment, thus completing the panel-design.

Seventh stage (pl. 32, g); now that the design in the panel has been finished, the second half of the figure, that below the bottom enclosing line, is begun. A series of small semicircles (lines numbered 21) are appended to the lowest enclosing line. The position of the semicircles bears no relation whatever to the series of panels. Three and a half to four semicircles fall below each panel or a total of from seventeen to twenty in the entire extent of the design as it encircles the vessel.

Eighth stage (pl. 32, g); by means of lines 22 and 23, a triangle is outlined below each of the semicircles.

Ninth stage (pl. 32, h); a very small semicircle (24) is drawn just below the point of each triangle.

Tenth stage (pl. 32, h); another similar semicircle (line 25) is drawn just below line 24.

Eleventh stage (pl. 32, i); the triangles (areas numbered VIII) are filled with black pigment.

Although this figure is based upon the panel-design, it differs from pure panel-decoration in the use of a series of simple figures appended to the bottom enclosing-line of the panel. The time required to paint this design was thirty-five minutes.

DRAWING NO. 3 (pl. 33); original by Maria Martinez. This design is of an entirely different type, in that there is no use of a panel. It was placed upon a constricted-mouthed bowl.

First stage (pl. 33, a); the enclosing lines (1, 2) are drawn around the rim of the vessel. These are immediately followed by the three lines (3 to 5) which form the outlines of the five spirals composing the design. The proportions of the figures were determined by constantly watching the position of the first spiral as the bowl was turned counterclockwise.

Second stage (pl. 33, b); the outlines of the details (lines 6 to 17) are drawn on each spiral in turn.

Third stage (pl. 33, c); the outlining is completed by adding to each spiral a horizontal triangle at the base of the curve. It is interesting to note that line 18, the first of the group 18 to 21, was the first one drawn, although it was entirely detached at that time from the main figure.

Fourth stage (pl. 33, d); the filling of areas is begun by painting the four small triangles on the upward curve of the spiral (areas I to IV). The order in which they are filled is apt to vary with the different spirals.

Fifth stage (pl. 33, e); the large triangle at the base of the spiral (area V) is filled with black pigment.

Sixth stage (pl. 33, f); the triangle at the bottom (area VI) is filled in each unit.

Seventh stage (pl. 33, g); this consists in placing a series of dots along the median line of the leaf-shaped area in the base of the spiral. In the various spirals these dots vary in number from eight to ten.

Eighth stage (pl. 33, h); the design is completed by filling the red areas (VIII and IX). In the first spiral so treated, area VIII was filled first, but in all the others area IX preceded area VIII.

The base of each spiral is four and one-half inches long. The width of the design is three inches.

DRAWING NO. 4 (pl. 34); original by Maximiliana Martinez. This rosette was placed in the interior bottom of a small constricted-mouthed bowl of red ware. A cloud-design which was painted on the rim of the same bowl is described and illustrated in fig. 6.

First stage (pl. 34, a); the outlines of one set of four leaves are drawn (lines 1-8).

Second stage (pl. 34, b); the areas within these enclosing lines are hatched (areas I-IV). These four areas are filled in clockwise rotation.

Third stage (pl. 34, c); the same areas (numbered V-VIII) are crosshatched, starting at the tip of each area and working towards the centre.

Fourth stage (pl. 34, d); four more leaves are drawn between the four already made. After lines 9 and 10 are drawn, the area between them is filled with black pigment, leaving an oblique white bar, before lines 11 and 12 are placed upon the vessel. Each of the four leaves is entirely completed before the next is begun.

Fifth stage (pl. 34, e); this consists in the placing of four lines (17-20) at the end of each of the four longer leaves. These four groups of lines are drawn upon the design in clockwise rotation.

Sixth stage (pl. 34, f); at this point Maximiliana glanced at the drawing which the writer was making in his notes. By accident the ends of the shorter leaves had there been made more pointed than the actual painting. When she saw this, she pointed the tips of these leaves with black to make them correspond in shape to the writer's drawing.

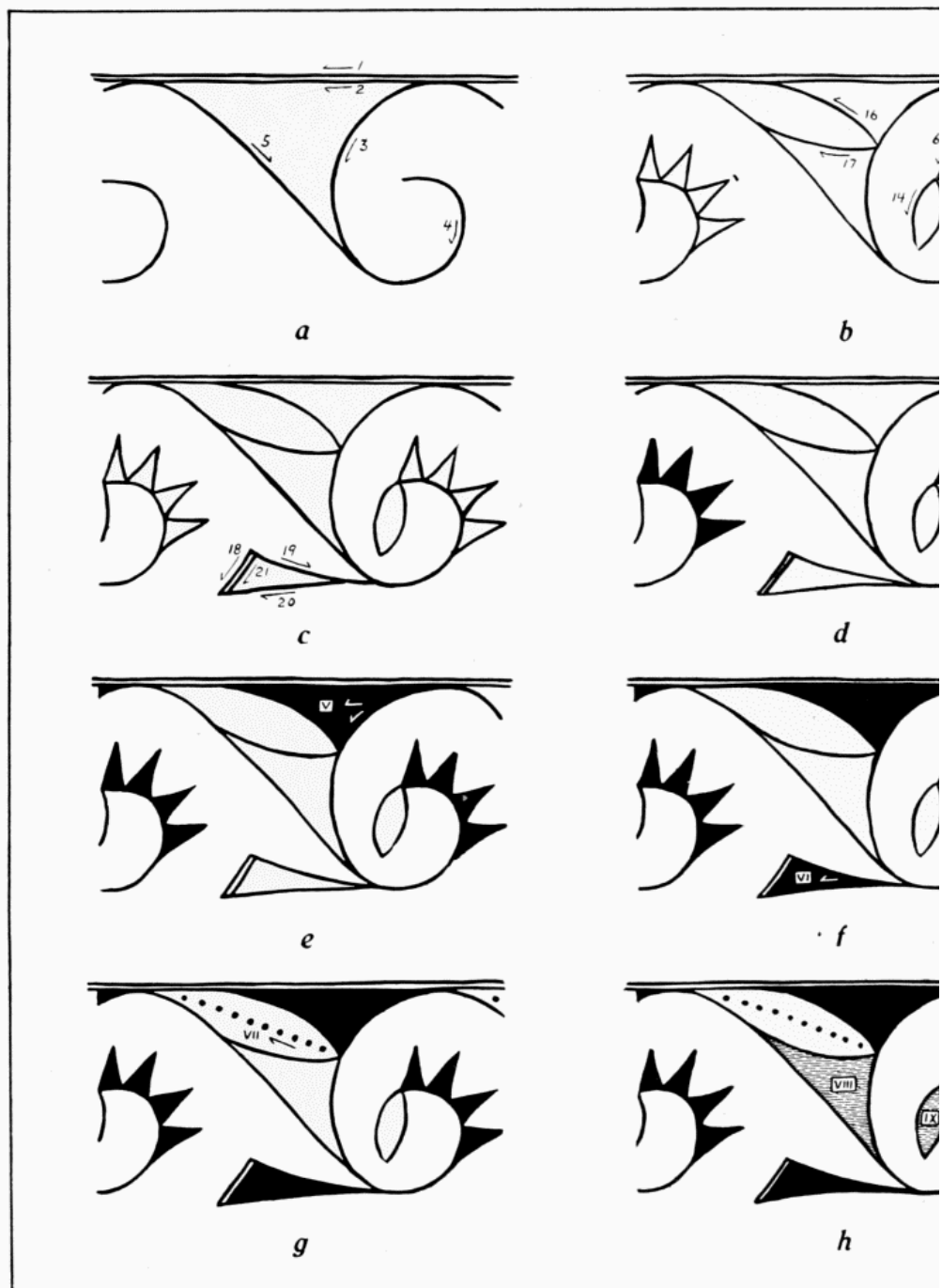
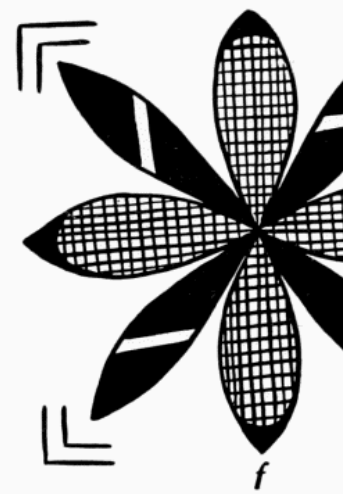
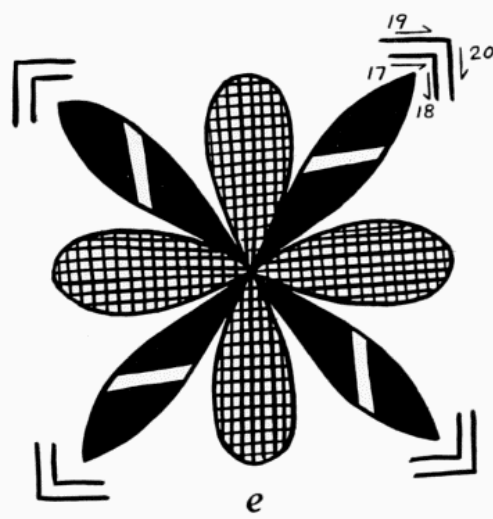
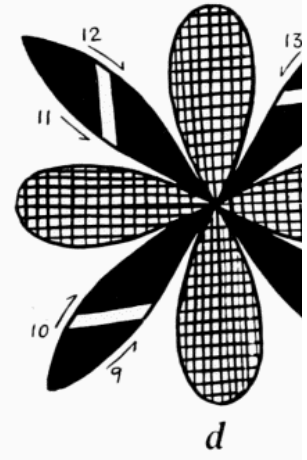
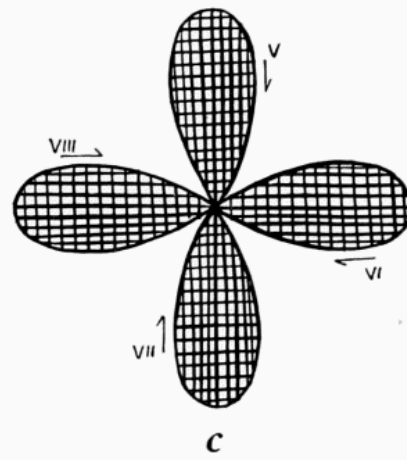
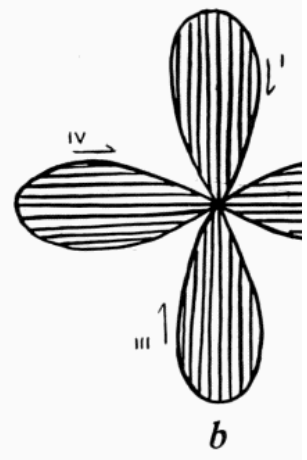
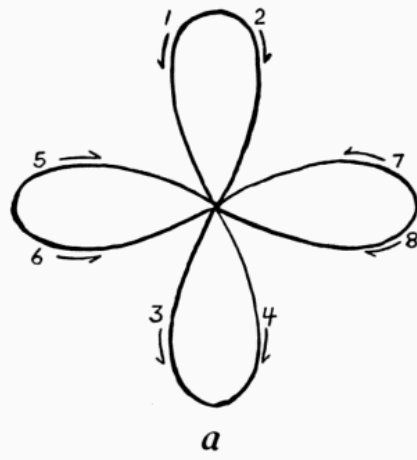


PLATE 33
Design by Maria Martinez



This is an excellent example of the type of work done by a painter who develops her pattern while drawing it. The contrast between this and the simple strength of the first three designs (fig. 9 and pls. 32, 33) should be noticed.

DRAWING NO. 5 (fig. 10); original by Maximiliana Martinez. This is a border decoration that was applied to the edge of the constricted-mouthed bowl on which was produced Drawing No. 4.

First stage (fig. 10, a); this is the drawing of the two enclosing lines (1 and 2) about the lip of the bowl. These are followed at once by 3 to 5, which outline three small semicircles. Eight of these triple semicircles complete the circumference of the bowl.

Second stage (fig. 10, b); the three semicircles are filled with dots of pigment, two in each of the upper ones, three in the lower.

Third stage (fig. 10, c); the three small lines are added to the lowest of the three semicircles.

The painting of the rosette (pl. 34) and the eight sets of semicircles took just sixteen minutes.

DRAWING NO. 6 (pl. 35); original by Antonita Roybal. This elaborate decoration was placed upon a large red ware olla. Antonita, it may be noted, specializes



FIG. 10. Raincloud design by Maximiliana Martinez
FIG. 10. Raincloud design by Maximiliana Martinez

in vessels of this sort, and her designs, while not slavishly repeated, all have a strong family resemblance, due to her bold use of scrolls and step-figures. The small upper drawing illustrates the method by which the design under consideration is applied to the vessel. In the larger one it is projected flat; the centre circle represents the mouth of the olla. The time taken for painting the various parts of the pattern is incorporated in the description; the total time at the end of each stage being given immediately thereafter.

Framework lines. The heavy double-width line forming the middle of the central square is drawn first. Then two narrower lines are drawn, one inside and one outside, thus completing the square (sixteen minutes; 00:00-16:00).

The two lines encircling the mouth of the olla are next produced (four minutes; 16:00-20:00).

Neck-design within the square. Each of the four corners of the square is ornamented with a small scroll-ended element. Each element is first outlined and then filled with black before the next is begun (twelve and one-half minutes; 20:30-32:00).

Small irregularly placed spots are then dotted into the remaining surface within the square (six minutes; 33:00-39:00).

The medallions. Appended to each of the four corners of the original square is a flower-like medallion. The drawing of each is done as follows. The outer line is first traced, then the inner one. Next the inner part is filled with black, except for a narrow band across the middle. To the exterior ring are added the outlines of the nine radiating elements. As a last step these are filled with black. The olla is then turned counterclockwise and the next medallion is begun.

Each medallion is completed more quickly than the last. The time: No. 1, ten minutes (39:00-49:00); No. 2, eight minutes (49:00-57:00); No. 3, seven and one-half minutes (57:00-64:30); No. 4, six and one-half minutes (64:30-71:00).

The side patterns. Each of the four sides of the square bears a design composed of two scrolls with a double stepped element and two long appendages between them. Each design is completed before the next is begun. The drawing proceeds as follows:

1. The outlining of the scrolls and the drawing of the crossbar between them.
2. The outlining of the central stepped figures.
3. The filling of the stepped figures with black.
4. The outlining of the two long appendages between the scrolls.
5. The filling of the appendages with black, leaving two narrow crossbars in the ground color.
6. The addition of the two small appendages between the stepped figures.
7. The outlining of a central streak ending in a small circle in each of the scrolls.
8. The filling of the scrolls with black, leaving the central streak and small circle in the ground color.

As in the case of the medallions, each one of the side patterns is completed more quickly than the last. The time: No. 1, twenty-six minutes (71:00-97:00); No. 2, twenty-two and a half minutes (97:00-119:30); No. 3, twenty-one and a half minutes (119:30-141:00); No. 4, fourteen and a half minutes (141:00-155:30). The entire time required for painting the design was thus two hours, thirty-five minutes, and thirty seconds.

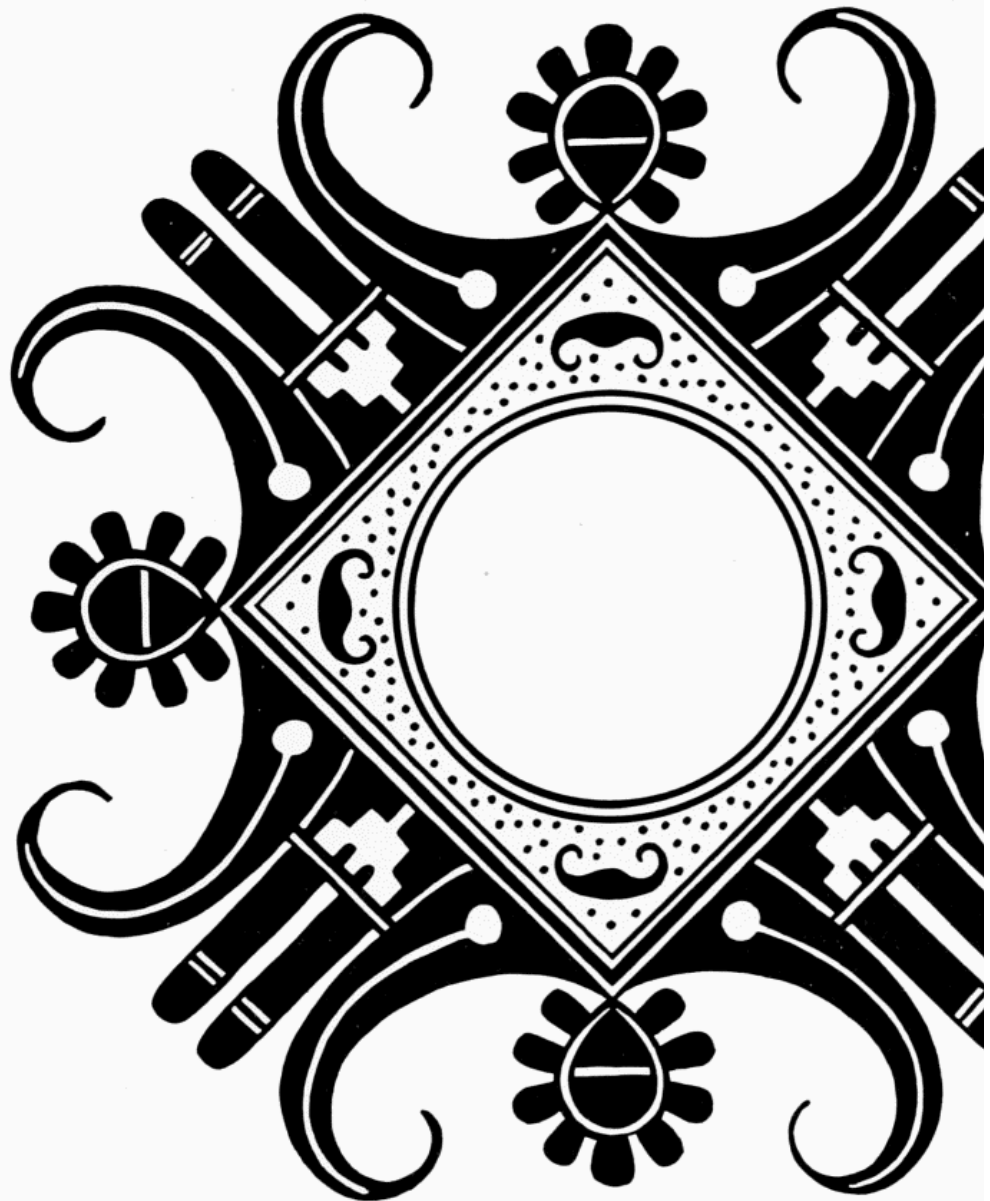


PLATE 35

Black design on red jar by Antonita Roybal (the small drawing above shows how this design is applied to the olla).

SYMBOLISM^[52]

Symbolism of one kind or another, plays a very important part in the existence of the Pueblo Indian. It is generally agreed that even the minor acts of everyday life have a certain religious symbolic meaning. For example, the gourd spoons, or kajepes, used in moulding pottery, must apparently be consecrated before they may be used. Similarly, a new technique in pottery making must pass through a period of consecration before it becomes completely established. On the other hand, white men often overemphasize the importance of symbolism in studying any civilization whose customs and philosophy differ radically from their own.

In the light of what is known of the mental attitude of the Pueblo Indian, it would not be unreasonable to assume that the designs on their pottery have some symbolical meaning. Whether this symbolism is in any sense religious, or is merely the symbolism of conventionalization of design, is an open question.^[53] It is probable that vessels made before the time when pottery became to some extent an article of commerce between the Indian and the tourist, bore designs of symbolical meaning. It may be that the vessels made today for their own use also have such meanings, but there is every reason to believe that much of the ware now turned out by the San Ildefonso potters for sale to tourists bears designs of no special meaning.

It is true that through bitter experience, the Indians have learned to guard carefully their religious and secular philosophy, and it would require many years of study, living with them, to gain an intimate knowledge of their beliefs. If, however, there are definite meanings associated with the designs upon commercial pottery, the inhabitants of San Ildefonso have become past masters of the art of concealment.

An attempt to obtain some idea of the general reason why vessels are decorated and of what was taking place in the mind of the painter while at work, proved wholly fruitless. Inquiries as to the meaning of whole designs, figures, and elements met with three kinds of response. One family, who have enjoyed considerable contact with investigators of Indian customs, were ready and eager with explanations of the meanings of various elements. They could not, however, explain the meaning of the entire design upon any vessel. Other potters met the inquiry with a frank statement of ignorance, such as "I don't know", or "Ask the men, the women don't know". A third group exhibited great uncertainty. These people would usually translate the questions into Tewa for the benefit of the rest of the Indians present. Then, after much laughter and discussion, the potter would sometimes offer a meaning, at other times say nothing. One old man explained with great gusto that the design which he had just finished upon a small globular olla represented four small clouds sailing across the sky, one behind the other, from west to east. His manner was very similar to that which one of us might adopt in telling an improbable yarn to a particularly guileless listener.

Some potters gave meanings for whole figures. Another potter, however, composed her figures as the work advanced, evidently with no clear idea, when she began it, of the ultimate form of the figures and therefore probably with no idea in regard to their meaning. It has been suggested that minor variation in similar figures on the same vessel have some meaning. The more probable explanation would seem to be that these variations are due to the lack of a visible pattern. At one time a potter was distinctly annoyed because she noticed, after completing a figure, that she had filled the wrong part of a small detail.

It is equally fallacious to lay too great importance upon alternating figures in a panel design. For example, one potter in filling the areas in a four-panel decoration filled the same areas in the first and third panels, and different areas in the second. When she began to fill the areas of the fourth panel in the same way as the first and third, her attention was called to the areas in the second, the order of which she obviously considered a mistake. The result was that the fourth panel was filled as the second had been, thus giving the vessel a panel-design with alternating figures.

The fact that in excavations no two vessels have ever been found with identical designs has been attributed to something in the Indian's way of thinking. Yet one potter at San Ildefonso placed upon one constricted-mouthed bowl out of a group of vessels a design identical with that on a bowl of the same shape in the previous group made.

Among older pots the "line-break", a small space left in the horizontal enclosing-line at the lip of the vessel, is a constant element. At San Ildefonso its use has become almost obsolete but one of the informants did employ it. Careful questioning on the subject with another potter as interpreter, brought out the fact that the line-break is called a "door", through which a spirit may enter or depart. Persistent questioning in regard to the nature of the spirit caused an animated discussion among those present. The potter was clearly puzzled by the questions. Then, after a single sentence by her, the Indians all laughed heartily. Finally, the interpreter stated that the potter had said, in effect, that if the door was going to cause all that trouble she would close it. The matter was finally settled by the interpreter, who pointed out a passage in the advertising pamphlet of a Santa Fe curio-dealer which explained that the line-break was a passage through which the spirits of the dead might pass.

The ready answers given by some potters in response to inquiries as to the meaning of elements of a design, seem to show that the elements are actually conventionalized symbols of definite objects. In one instance the potter, on noticing a dance-costume lying on the bed, pointed out certain parts of it, and then drew the elements representing those parts. Other potters simply deny all knowledge of the meaning of elements. Still others by their hesitancy, seem to grope for a plausible meaning and offer the first which occurs to them. Whether the elements do represent definite objects or not, it nevertheless remains a fact that the meanings given cause apparent contradictions. To elements essentially the same widely different meanings are often given and, conversely, the same idea or object

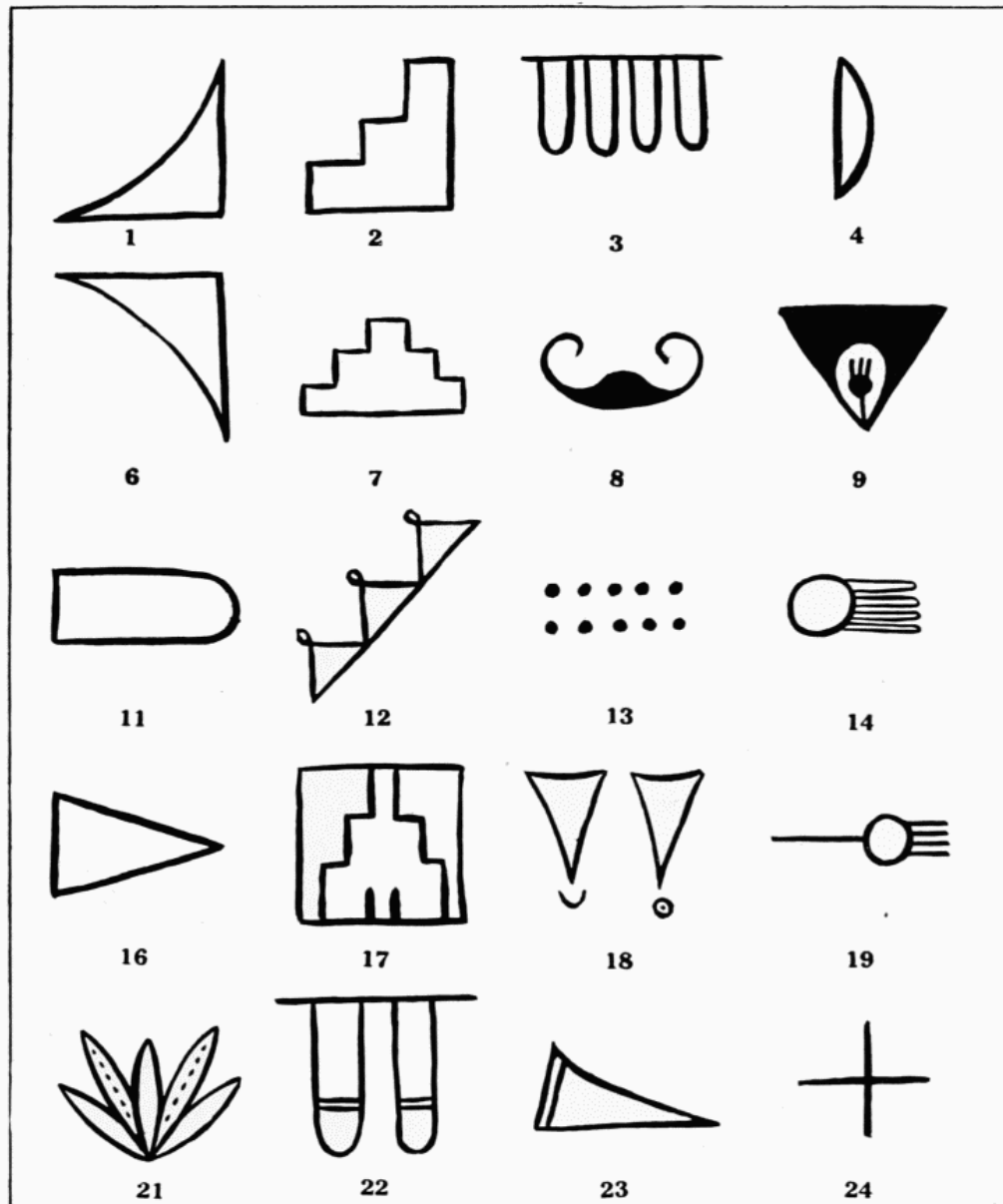


FIG. 11. Elements of design commonly used by the potters of San Ildefonso.

is often represented by entirely different elements. This is also true of more complex figures.

The elements illustrated in figure 11, give examples of the type of idea or object said to be represented, and incidentally illustrate some of the contradictions spoken of above. Numbers 1 and 6 both refer to hills. On the other hand, No. 7 is a mountain, although it bears little resemblance to the "hills". It resembles more closely No. 2 which one informant described as a pueblo and another as a kiva. The same regular zigzag appears in No. 12, which represents a tablita, or dance mask, with small feathers tied to the points of the zigzags. No. 17, in which the zigzag also appears, represents kiva-steps.

Nos. 11 and 16 represent feathers, and No. 21 is a bunch of feathers on the end of a dance-pole. But No. 22 represents the poles of the kiva-ladder, and No. 3 is rain, which is falling a long way off. No. 10 represents fringed woolen armlets, and yet there is as close a resemblance between Nos. 3 and 10, as there is between Nos. 14 and 19, both of which represent the tassels on a man's dance-belt. No. 8 is called rain-water, but a single spiral, as in No. 25, is a buffalo horn. The dots of No. 13 represent rain-drops in dust, but in No. 21 the dots are markings on feathers, and in No. 15 they represent a procession of bugs on the rib of a leaf. No. 18 represents water dripping through a hole in the roof and making a small cup-like depression in the floor, represented by a semicircle, or a whole circle with a dot in the centre of it; but No. 23, which is also a triangle, is called a leaf, as is also the case with No. 15. And yet No. 4 is also a leaf, although if placed horizontally it more closely resembles the clouds represented in No. 5. But No. 9 is also a cloud, this time a big black cloud with an open space in it through which a small cloud may be seen. Nos. 20 and 24 represent respectively the sun and a star.

In the light of the small amount of material obtained on the subject of symbolism, all that can be said is that the evidence is purely negative. If the elements do represent definite ideas and objects, which seems to the writer improbable, the meanings are so deeply hidden that only an intensive specialized study will result in an acceptable solution.

MEANING OF ELEMENTS

(M.—Maria. A.—Antonita)

- 1—"hillside", M.
- 2—"pueblo", M.; "kiva", A.
- 3—"rain falling far away", A.
- 4—"leaf", M.
- 5—"clouds", M.; A.
- 6—"hill", M.
- 7—"mountain", M.
- 8—"rainwater", A.
- 9—"black cloud with open space through which one can see a small cloud", A.
- 10—"fringed woolen dance armlets", M.
- 11—"feather", M.
- 12—"tableta (wooden headdress) with feathers on points", M.
- 13—"rain-drops on dust", A.
- 14—"tassel on man's dance-belt", M.
- 15—"leaf with bugs on it", M.
- 16—"feather", M.
- 17—"kiva-steps, the two small points fireplaces", A.

- 18—"water dripping through holes in roof and making holes in dirt floor", M.
- 19—"tassel on man's dance-belt", M.
- 20—"sun", A.
- 21—"feathers on dance-pole, spots are markings on feathers", M.
- 22—"ladder-poles", A.
- 23—"leaf", M.
- 24—"star", A.
- 25—"buffalo horn", A.

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1915. Flora of New Mexico. *Contributions from the United States National Herbarium*, Vol. 19. Washington 1915.

FOOTNOTES:

[1] A brief history of Pecos, and a description of the work so far done at the site, are contained in "An Introduction to the Study of Southwestern Archaeology, with a Preliminary Account of the Excavations at Pecos", by A. V. Kidder; No. 1 of the present series.

[2] Kidder, 1924.

[3] Kidder, M. A. and A. V., 1917, p. 330.

[4] A searching study of San Ildefonso decoration has been made by K. M. Chapman of Santa Fe. When this is published it will throw much light on the morphology of Pueblo design, and upon the relation of San Ildefonso symbolism to that of other Rio Grande pueblos.

[5] A vocabulary of the Tewa terms for clays, paints, and other ingredients; articles of paraphernalia; types of pottery; processes of pottery making, etc., was recorded by Dr. Guthe, and is available at Andover for consultation.

[6] See Harrington, 1916, Map 19, point 63. Under the nearby point (60; p. 314) he says: "Clay ... is obtained at this place; just where could not be learned."

[7] Stevenson, to whom with Mrs. Stevenson we are indebted for practically our only comparative material, wrote in 1879 of Zuñi: "The clay mostly used by the Zuñians in the manufacture of pottery is a dark, bluish, carbonaceous clayey shale found in layers usually near the tops of the mesas" (1883, p. 329). Later, Mrs. Stevenson described at some length the ceremony of securing pottery clay (1904, p. 374). Among the San Ildefonso Indians of today there is no outward evidence of any such ceremony.

[8] At Zuñi the ingredients are ground and mixed, and the clay is kneaded on a stone slab. (See pl. 28, facing p. 76.)

[9] Stevenson said, in speaking of Zuñi pottery making: "This carbonaceous clay is first mixed with water and kneaded as a baker kneads dough until it reaches the proper consistency" (1883, p. 329). Mrs. Stevenson describes the preparation of clay at Zuñi as follows: "The clay is ground to a powder and mixed with a small quantity of pulverized pottery, fragments of the latter being carefully hoarded for this purpose. The powder thus compounded is mixed with water enough to make a pasty mass, which is kneaded like dough. The more care taken in pulverizing the material and the more time spent in working it, the finer becomes the paste. When the mass reaches such a state of consistency that the fingers can no longer detect the presence of gritty particles, it is still more delicately tested with the tongue, and when found to be satisfactory it is placed in a vessel and covered with a cloth, where it will retain the moisture until wanted for use" (1904, p. 374-5).

[10] Stevenson (1883, p. 329) says: "With this (clay), crushed volcanic lava is sometimes mixed; but the Zuñians more frequently pulverize fragments of broken pottery, which have been preserved for this purpose. This seems to prevent explosion, cracking, or fracture by rendering the paste sufficiently porous to allow the heat to pass through without injurious effect." And again: "The clays used by the Santa Clara Indians are of a brick-red color, containing an admixture of very fine sand, which, no doubt, prevents cracking in burning, and hence dispenses with the necessity of using lava or pottery fragments, as is the custom of the Indians of the western pueblos" (1883, p. 331). Binns, in "The Potter's Craft", describing the preparation of clay, says, "If the clay prove very sandy it must be washed, ... but all the sand should not be removed. Clay with no sand in it will be apt to crack in drying and burning. Clay with too much sand will be short and brittle in working. If the clay is very smooth and shows no sand, a little fine builder's sand may be added with advantage, a definite proportion being used and recorded" (1910, p. 40).

[11] Stevenson, in his catalogue of Zuñi collections, enumerated "*Tierra amarilla*, or yellow micaceous clay, of which the Rio Grande Indians make many varieties of vessels" (1883, p. 368).

[12] Harrington, in speaking of a clay-pit near the village of Las Truchas, writes: "It is said that at this place the best red pottery clay known to the Tewa is obtained. It is pebbly, but makes very strong dishes, and it is used especially for ollas. It is said that Tewa of various pueblos visit this place frequently and carry away the clay. The clay deposit is a mile or two southeast of Truchas town" (1916, p. 340). This may be the claypit referred to by the informant, although the clay, while pebbly, is by no means red.

[13] Stevenson, writing of the Zuñi, said of their white slip: "A paint or solution is then made, either of a fine white calcareous earth, consisting mainly of carbonate of lime, or of a milk-white indurated clay, almost wholly insoluble in acids, and apparently derived from decomposed feldspar with a small proportion of mica" (1883, p. 329). Mrs. Stevenson, in a later publication, said: "A white clay is dissolved in water and then made

into cones which are dried in the sun. When required for use these cones are rubbed to powder on a stone, again mixed with water, and applied in a liquid state to the object" (1904, p. 375).

[14] Harrington mentions this bed as "a deposit of bright red paint situated about two miles east of Santa Fe, the informants think north of Santa Fe Creek, in high land a few hundred yards from that creek. This paint was used for body painting. It is said that Jicarilla Apache still go to the deposit to get this paint and sometimes sell it to the Tewa" (1916, p. 354). Stevenson is speaking of the black ware of Santa Clara and other Rio Grande pueblos, mentions "a solution of very fine ochre-colored clay" (1883, p. 331), which was probably this same substance. Mrs. Stevenson also mentions "red ochre" (1904, p. 375).

[15] Stevenson, writing of the Zuñis in 1881, said: "The material used to produce the red or brown colors is a yellowish impure clay, colored from oxide of iron; indeed it is mainly clay, but contains some sand and a very small amount of carbonate of lime"; it "is generally found in a hard, stony condition, and is ground in a small stone mortar ... and mixed with water so as to form a thin solution" (1883, p. 330). Mrs. Stevenson added later: "Ferruginous clays which on heating burn to yellow, red, or brown are employed for decorating" (1904, p. 375).

[16] *Peritoma serrulatum* (Pursh); synonyms: *Cleome serrulata*, and *Cleome integrifolia*; (Wootton and Standley, 1915, p. 290). Stevenson wrote: "It is said that among the Cochiti, Santa Clara, and some other pueblos a vegetable matter is employed to produce some of their decorative designs; this, however, I was unable to verify, though some of the Indians assured me of the fact, and furnished me a bunch of the plant, which Dr. Vasey, of the Agricultural Department, found to be *Cleome integrifolia*, a plant common throughout the Western Territories" (1883, p. 331). Again, "That of the decorated class is ornamented with the juice of *Cleome integrifolia*, which is fixed to the ware in the process of burning" (loc. cit., p. 430). Harrington writes: "This is a very important plant to the Tewa, inasmuch as black paint for pottery decoration is made from it. Large quantities of young plants are collected, usually in July. The plants are boiled well in water; the woody parts are then removed and the decoction is again allowed to boil until it becomes thick and attains a black color. This thick fluid is poured on a board to dry and soon becomes hardened. It may be kept in hard cakes for an indefinite period. When needed these are soaked in hot water until of the consistency needed for paint. Guaco is also used as a food. The hardened cakes are soaked in hot water, and then fried in grease. The finely ground plants are mixed with water and the liquid is drunk as a remedy for stomach disorders; or sometimes fresh plants wrapped in a cloth are applied to the abdomen" (1916, a, pp. 58, 59). (It should be noted that the accounts obtained by the writer from the informants of San Ildefonso do not agree in detail with the one given by Harrington.) Mrs. Stevenson wrote of Zuñi: "Water from boiled *Cleome serrulata* (Mexican name waco) is mixed with black pigment (a manganiferous clay containing organic matter) in decorating pottery" (1904, p. 375); and in more detail: "The entire plant, minus the root, is boiled for a considerable time, and the water in which it is cooked is allowed to evaporate. The firm paste secured from precipitation is used in conjunction with a black mineral paint for decorating pottery" (1915, p. 92).

[17] Stevenson, in 1879, saw the Rio Grande Pueblo potters use "coarsely broken dried manure obtained from the corrals", and added that for the polished black ware "another application of fuel, finely pulverized, is made" (1883, p. 331).

[18] Stevenson, in 1879, observed at Zuñi that "the bottoms of old water jars and bowls form stands for the articles while being worked by the potter. The bowls are filled with sand when objects of a globular form are to be made" (1883, p. 329). At Pojuaque (a Tewa pueblo near San Ildefonso, now nearly extinct) he collected two "pottery moulds for bottoms of vessels" (loc. cit. p. 439). Mrs. Stevenson also noticed these olla bottoms at Zuñi (1904, p. 374).

[19] Harrington writes: "Various shaped section of gourd are used by women to smooth and shape pottery in the making" (1916, a, p. 102). Stevenson, in 1879, noted at Zuñi "a small trowel fashioned from a piece of gourd or fragment of pottery, the only tool employed in the manufacture of pottery" (1883, p. 329). Mrs. Stevenson also speaks of these trowels (1904, p. 374).

[20] Stevenson, in speaking of the making of pottery at Santa Clara, mentions "the process of polishing—with smooth, fine-grained stones", (1883, p. 331). In his catalogue of stone objects collected at Zuñi in 1881, he differentiates between "small stones used in polishing pottery", "small stones used in polishing unburned vessels", and the "small polisher for glazing and smoothing pottery", giving three different Indian names for the three kinds. He does not, however, explain the difference (1884, pp. 525, 526). He also catalogues "fifteen rubbing or smoothing stones for pottery" obtained at Walpi (1884, p. 587). Mrs. Stevenson remarked: "Polishing stones are used to finish the surface" of vessels at Zuñi (1904, p. 375).

[21] Stevenson, in 1879 at Zuñi, noted that the paint was applied to pottery “with brushes made of the leaves of the yucca. These brushes are made of flat pieces of the leaf, which are stripped off and bruised at one end, and are of different sizes adapted to the coarse or fine lines the artist may wish to draw” (1883, p. 330). Mrs. Stevenson wrote, again of Zuñi: “*Yucca glauca* Nutt. Soapweed.... The brushes employed for decorating pottery are made from the leaves of this plant, which, for this purpose, are cut the proper lengths and fringed at one end” (1915, p. 82).

[22] Mrs. Stevenson says the Zuñis used a “rabbit skin mop” (1904, p. 375).

[23] Mrs. Stevenson wrote of Zuñi: “The pieces to be fired are placed upon stones to raise them a few inches from the ground”, implying the absence of a grate (1904, p. 376).

[24] I use the words “very skillfully” advisedly. Repeated attempts on my part to imitate the process resulted in rolls of exceedingly variable diameter. At Zuñi the roll is made by the apparently simpler and more efficient method of rolling the clay back and forth on a flat stone slab with one hand (see [pl. 29, a](#)).

[25] Zuñi potters apply the roll to the outside instead of the inside of the growing vessel wall; otherwise the handling of the roll is exactly as at San Ildefonso (see [pl. 29, b](#)).

[26] Throughout this report, the puki is spoken of as moving “clockwise” or “counterclockwise” and the sectors worked in are referred to the numerals on the face of the clock, as, “the six-to-eight-o’clock sector”. The point of view taken is that of the potter with “six o’clock” referring to that part of the bowl nearest her.

[27] The “rim” of an unfinished vessel means the upper edge of the last roll added to it.

[28] See, however, [pl. 29, b](#); the Zuñi potter in this case has not obliterated the preceding rolls.

[29] Stevenson, 1883, p. 331.

[30] M. C. Stevenson, 1904, p. 375.

[31] Binns, 1910, pp. 69-71.

[32] The process is identical at Zuñi, see [pl. 30, a](#).

[33] From the earliest periods Southwestern pottery has been made in more or less the manner just described, that is, by the addition of successive rings of clay. During prehistoric times, however, there was developed a ware usually known as “corrugated”, in the building of which the roll was so lengthened as to become a spiral coil. The roll of paste forming the coil was very thin, averaging a little over one-quarter of an inch in diameter; and it was applied to the outside of the temporary rim, as at Zuñi (see [pl. 29, b](#)), instead of to the inside, as at San Ildefonso. Furthermore, the junctions between the coils were not obliterated on the exterior of the vessel, thus producing a ridged or corrugated effect. Corrugated pottery was principally used for cooking pots (see Holmes, 1886, p. 273).

[34] One informant stated that these bowls, both with and without the terraces, were formerly, and are still, used to hold sacred meal.

[35] The two-mouthed vase is rarely made at San Ildefonso. It is a type more commonly produced by the potters of Santa Clara.

[36] Potsherds from pre-Spanish ruins show that handles have been applied to vessels in the ways described above for many centuries. The lug-type of handle, which is frequently found in ancient pottery, is no longer made at San Ildefonso. The probabilities are, however, that these lugs were welded to the exterior of the vessel, as in the first method described, rather than riveted on as in the second method.

[37] The potter placed the responsibility for this crack upon a kitten which had jumped into the olla during the night. In the animal’s attempt to climb out, the vessel suffered some rather severe bumps. The story of the kitten’s adventure was told before the crack was noticed. Later it was used to explain the presence of the imperfection.

[38] The expression “get warm” comes from the fact that while the vessel is damp the evaporating moisture causes it to feel cool; when thoroughly dry it feels warm.

[39] The red slips are not used today in the designs, although some of the older vessels contain red elements which appear to have been made of the dark-red slip.

[40] Stevenson says of the Zuñi: “This solution (of a fine white calcareous earth) is applied to the surface of the vessel and allowed to dry; it is then ready for the decorations” (1883, pp. 329, 330). Mrs. Stevenson, writing at a later date, gives more details: “A white clay is dissolved in water and then made into cones which are dried in the sun. When required for use these cones are rubbed to powder on a stone, again mixed with

water, and applied in a liquid state to the object with a rabbit-skin mop. Polishing-stones are used to finish the surface” (1904, p. 375).

[41] At Santa Clara and San Juan some polished ware is polished only from the rim to the shoulder and the lower half is apparently unslipped.

[42] Stevenson in speaking of the polished black pottery of the Rio Grande pueblos says: “A solution of very fine ochre-colored clay is applied to the outside and inside near the top, or to such parts of the surface as are to be polished. While this solution thus applied is still moist, the process of polishing begins by rubbing the part thus washed with smooth, fine-grained stones until quite dry and glossy. The parts thus rubbed still retain the original red color of the clay. The vessels are again placed in the sun and allowed to become thoroughly dry, when they are ready for baking” (1883, p. 331). Mrs. Stevenson also gives an account somewhat similar: “In many of the pueblos the pottery is undecorated, the surface being finished in plain red or black. The ware is made of a yellowish clay, in the manner heretofore described, and the vessels are placed in the sun, where they remain for hours. They are then washed with a solution of red ochre, and while wet the process of polishing begins, the woman with untiring energy going over the surface again and again with her polishing-stone, every little while passing a wet cloth over the vessel to keep the surface moist. When the polishing is completed, the vessel is again placed in the sun for a short time before receiving its final baking in the oven” (1904, p. 375.)

[43] This second potter, however, did use lard in polishing vessels to which the red slip had been applied.

[44] The substitution of the chamois for a cloth was inaugurated by the potter using it.

[45] The use of this paint for producing designs upon polished black ware was discovered by Maria Martinez in June, 1921. On inquiring why other potters did not copy the process, it was learned that because of certain taboo-like restrictions the secret of making this kind of ware would not be disclosed until a year after its discovery. Although the matter could not be further investigated because of the unwillingness of the Indians to discuss such things, it would appear that we have in this case a sort of primitive patent-right. At the present time (1925) practically all San Ildefonso potters make this ware, which has proved extraordinarily popular; Maria, however, still produces by far the finest pieces (see [pl. 8, a](#)).

[46] Occasionally such precautions are disregarded, apparently through laziness. One potter did her painting in the same room in which some meat was being cooked. The combined odors of the cooking meat and the guaco attracted all the flies in the neighborhood, but fortunately screens kept some of them out of the room.

[47] Stevenson gives no details of the painting at Zuñi; “When the pigment is properly reduced, and mixed with water so as to form a thin solution, it is applied with brushes made of the leaves of the yucca. These brushes are made of flat pieces of the leaf, which are stripped off and bruised at one end, and are of different sizes adapted to the coarse or fine lines the artist may wish to draw. In this manner all the decorations on the pottery are produced” (1883, p. 330). Mrs. Stevenson is just as brief: “After a thorough drying of this foundation, the slip, the designs are painted with brushes made of yucca needles, the pigments having been ground in stone mortars and made into a paste with water to which a syrup of yucca fruit is added” (1904, p. 375). The stone mortar for pigment grinding may be seen in [pl. 31, a](#).

[48] On pp. 78 to 84 will be found notes on the actual painting of several typical designs.

[49] One woman tried her best to teach me how to tell these shades apart, but it proved utterly impossible for me to distinguish the darker from the lighter.

[50] This variant in polished black ware is a new departure. It was discovered, probably accidentally, by one of the San Ildefonso potters early in the summer of 1921. The process by which the red blotch is made has not as yet been brought completely under control.

[51] It is interesting to note that in making the preliminary pencil draft, she uses entirely different strokes from those made by the brush.

[52] The subject of symbolism is touched upon in this report only with the greatest diffidence. No two students of Southwestern ceramics seem to entertain the same theory, in all details, upon this subject. The statements made here are given for what they may be worth. The time devoted to the work did not permit of a careful and exhaustive study of the subject.

[53] One potter, after glancing at a small vessel which was undoubtedly old, informed the writer that it had once belonged to the “summer people”, a social-religious division of the community.

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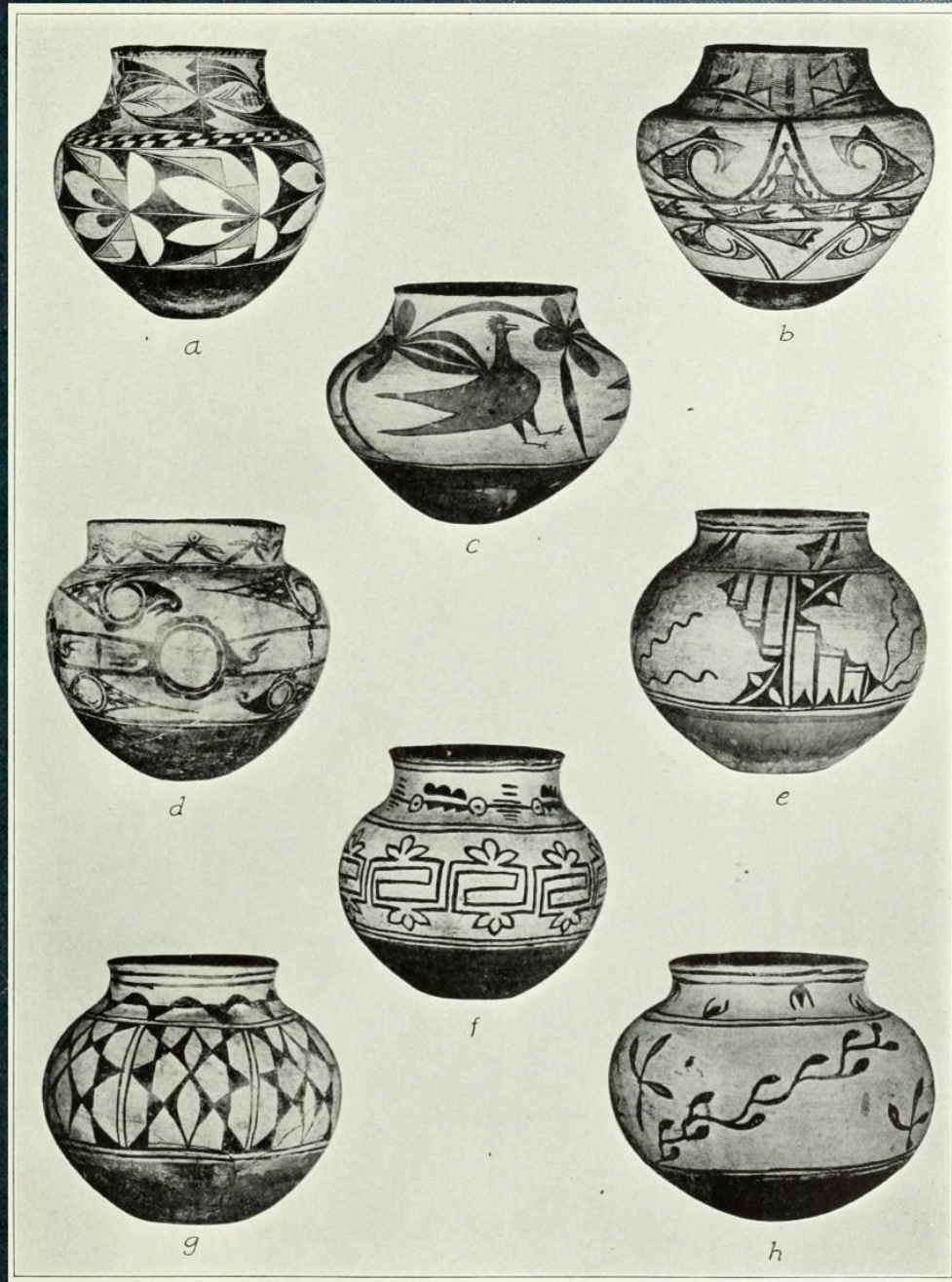
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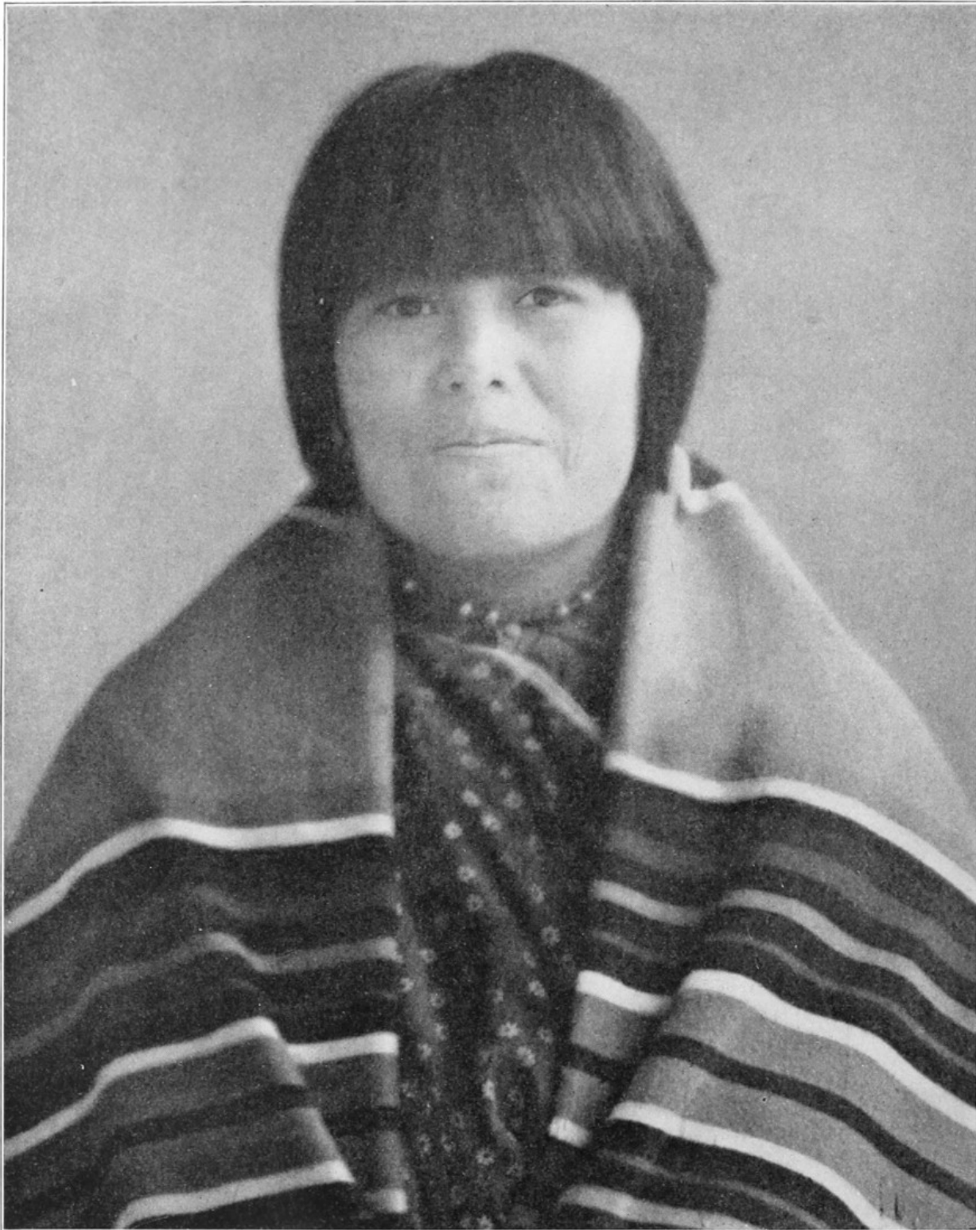
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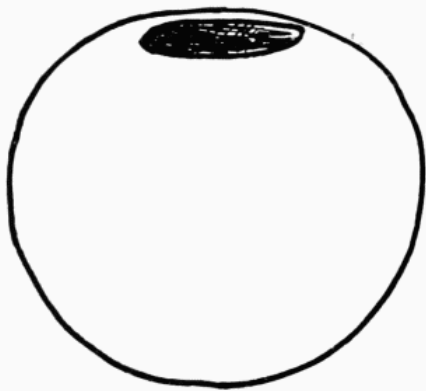


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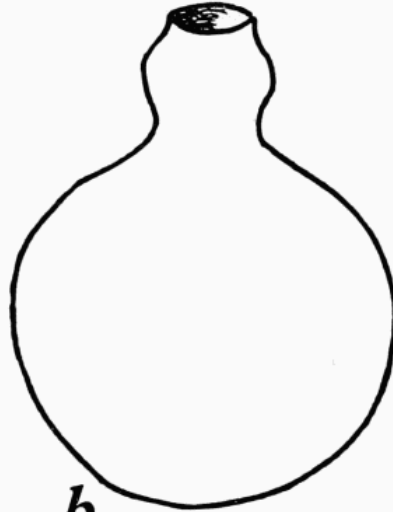


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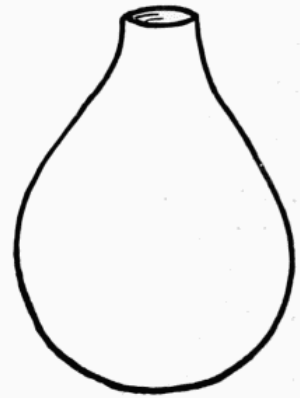
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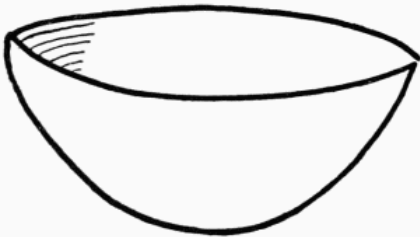
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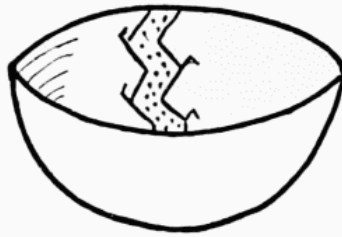
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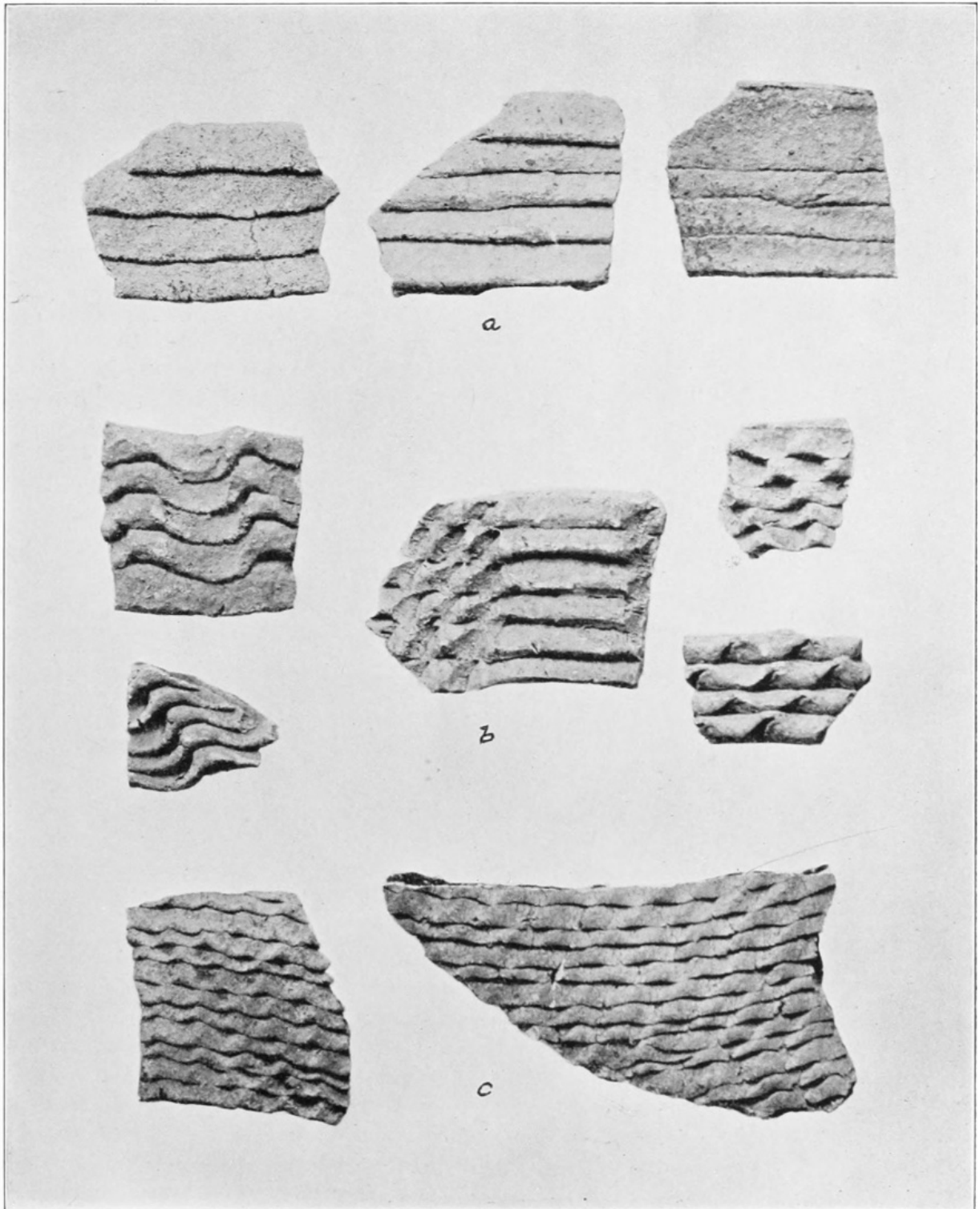


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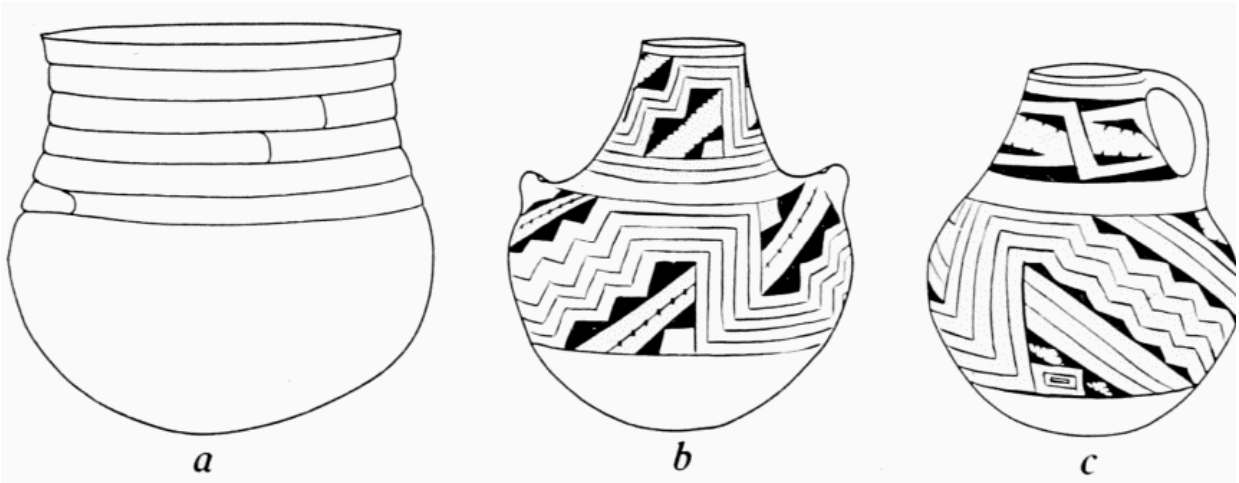


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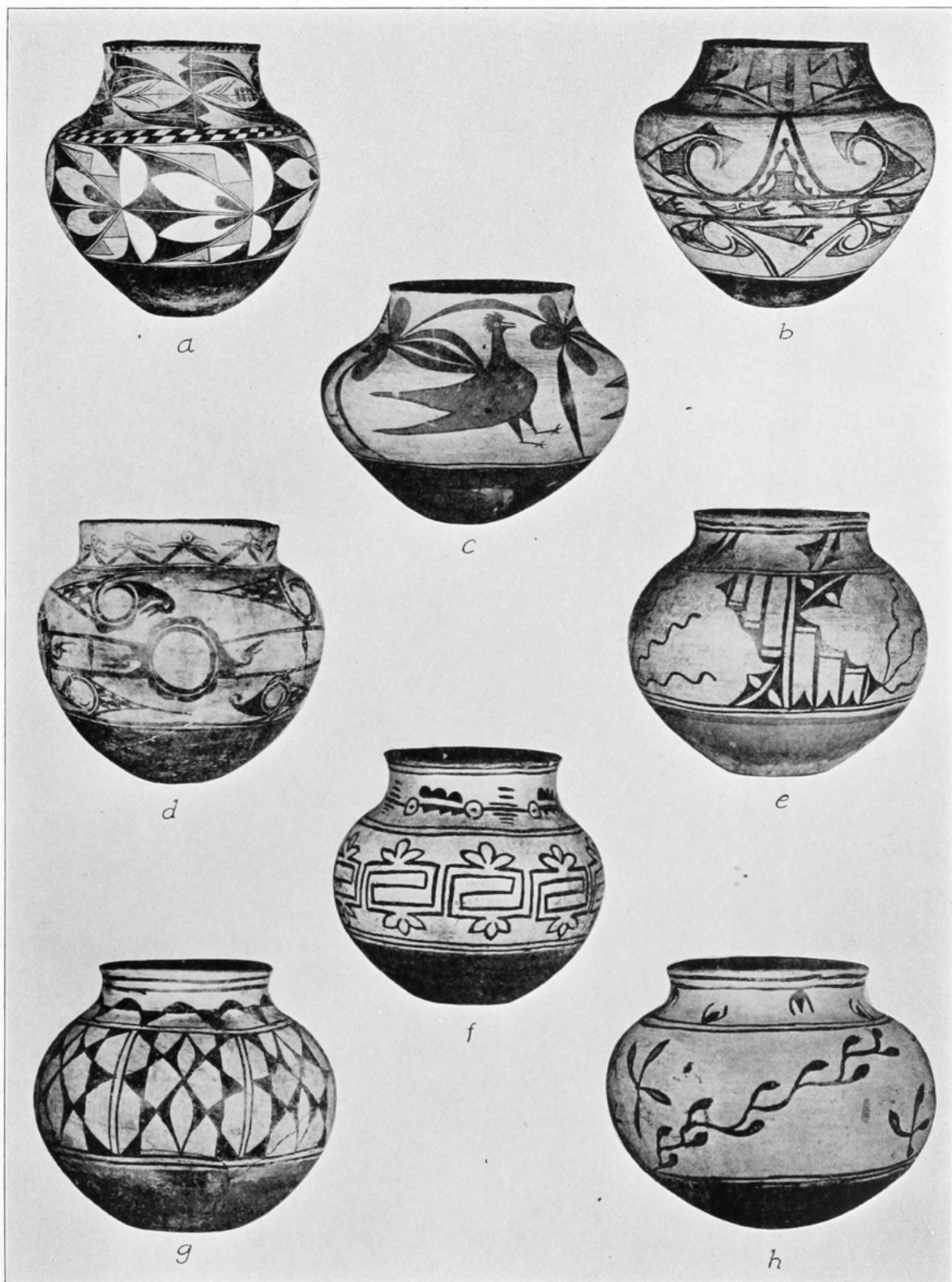
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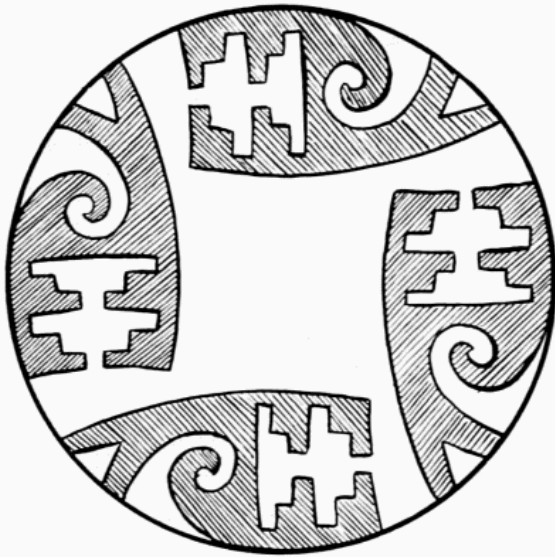
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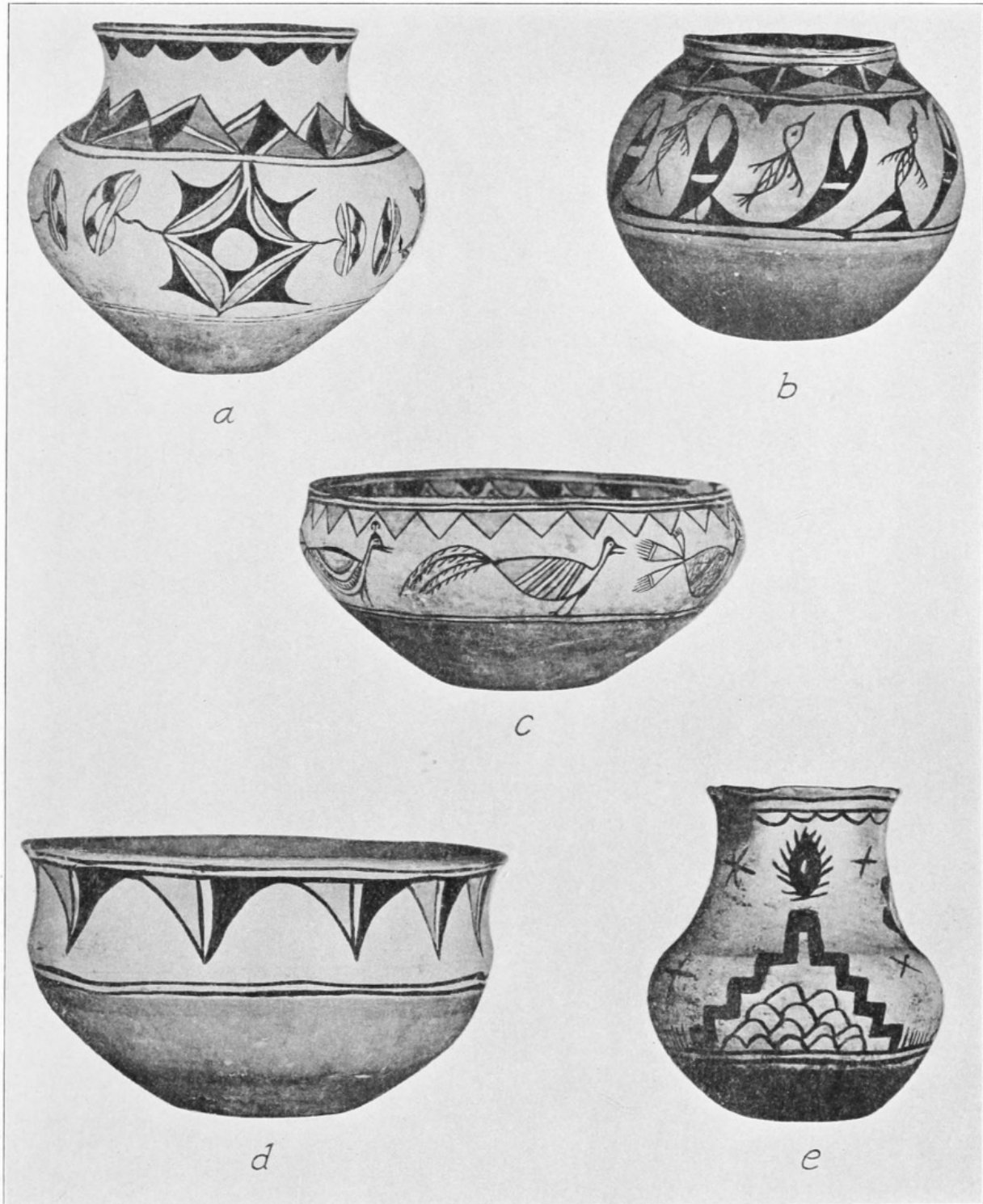


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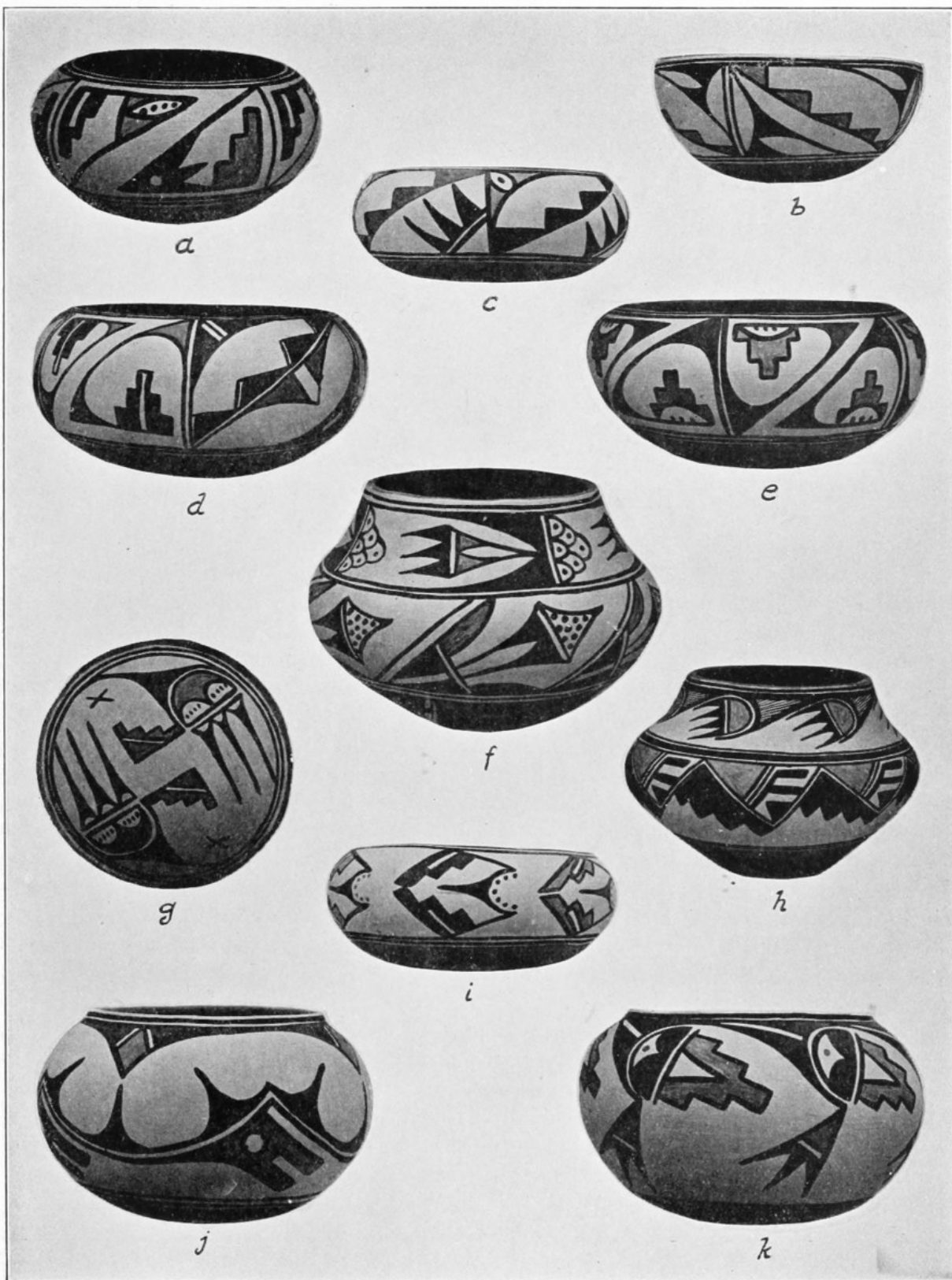


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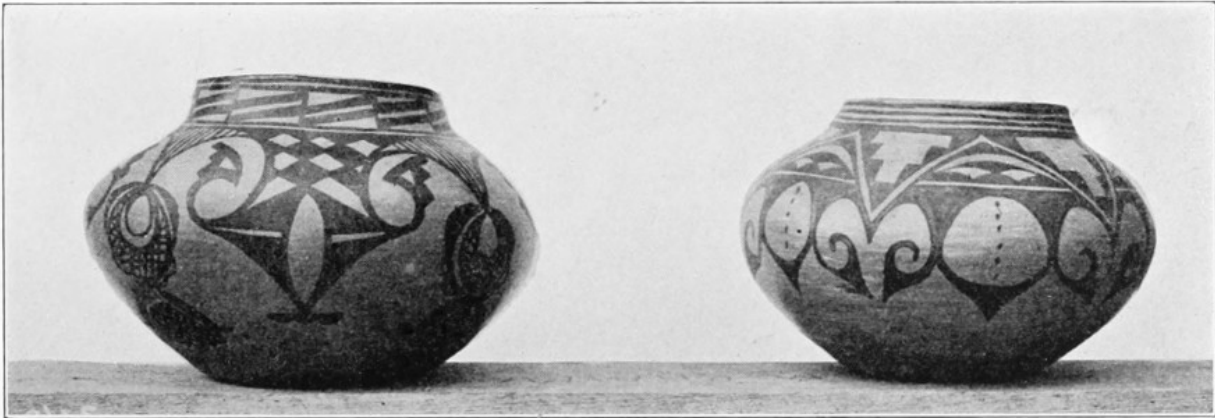


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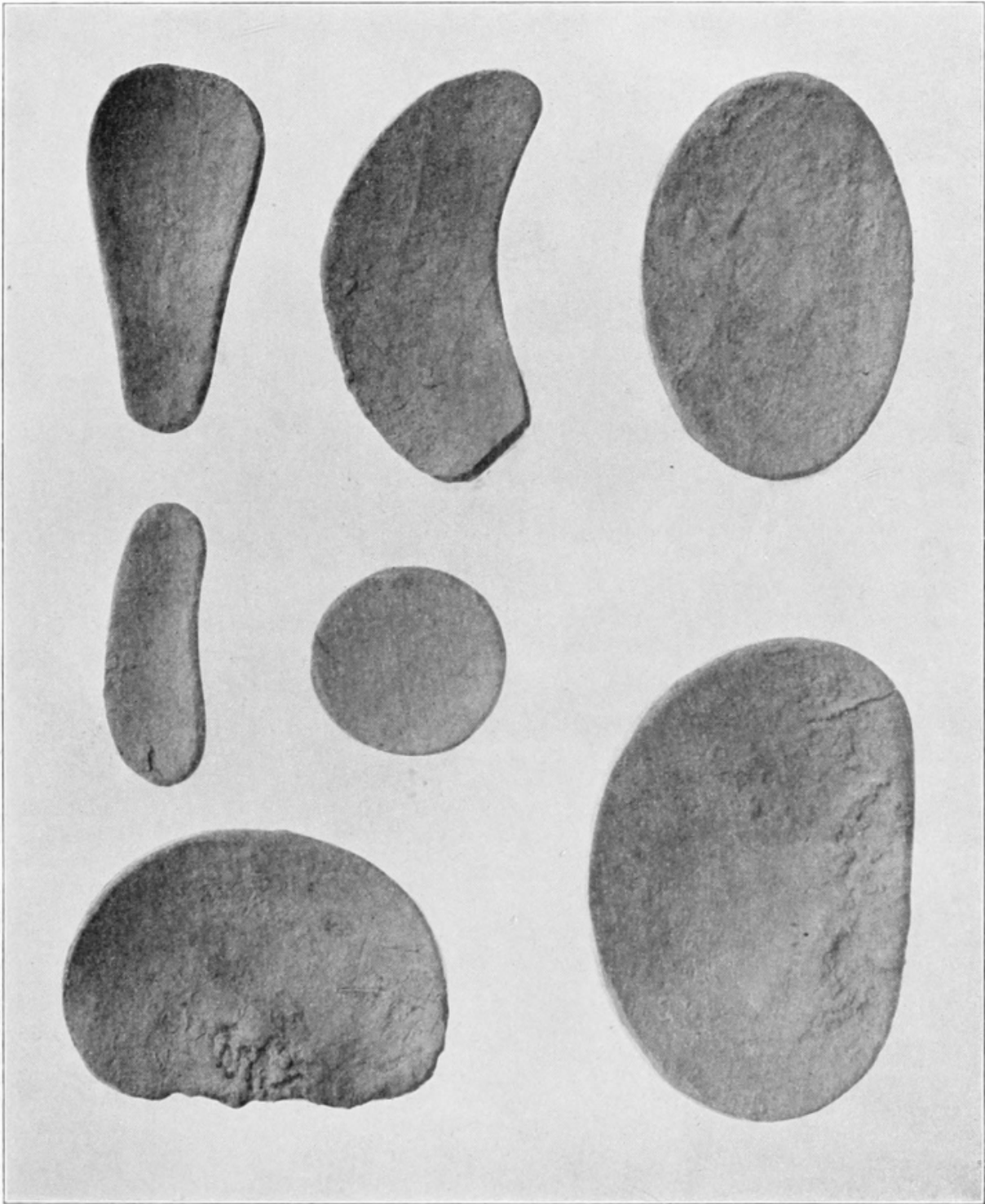


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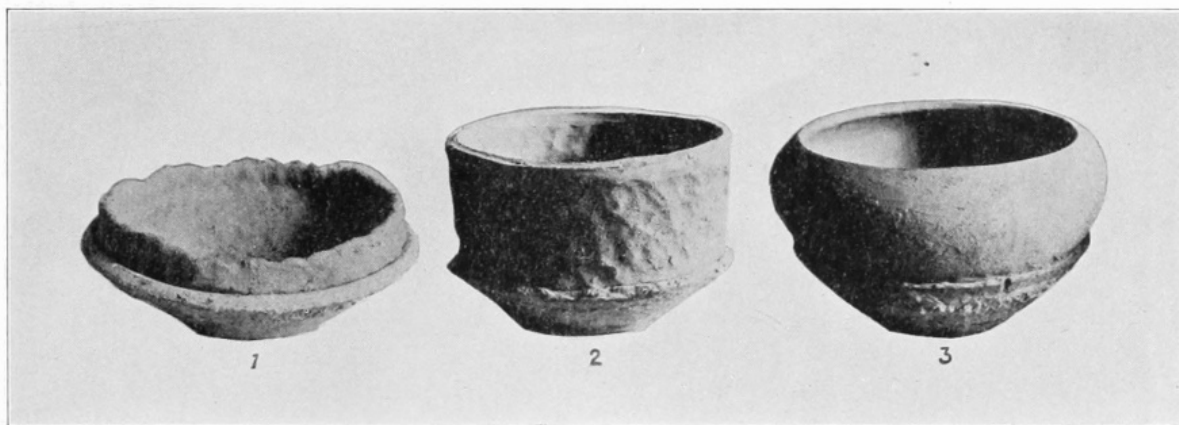
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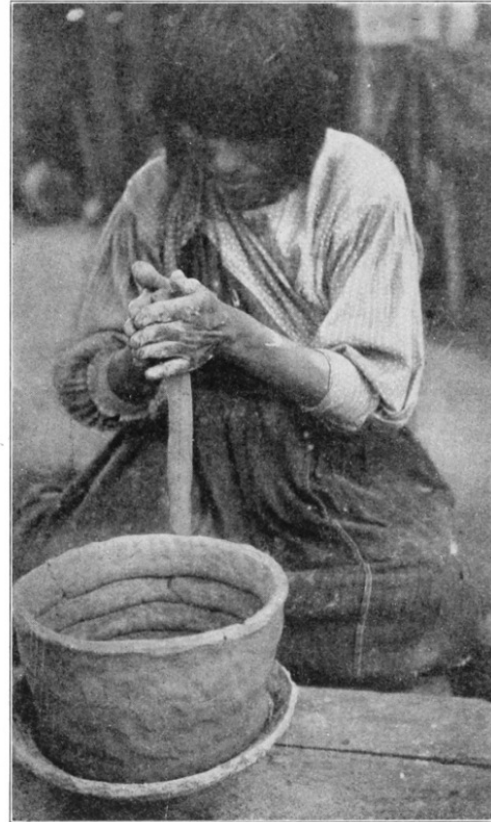


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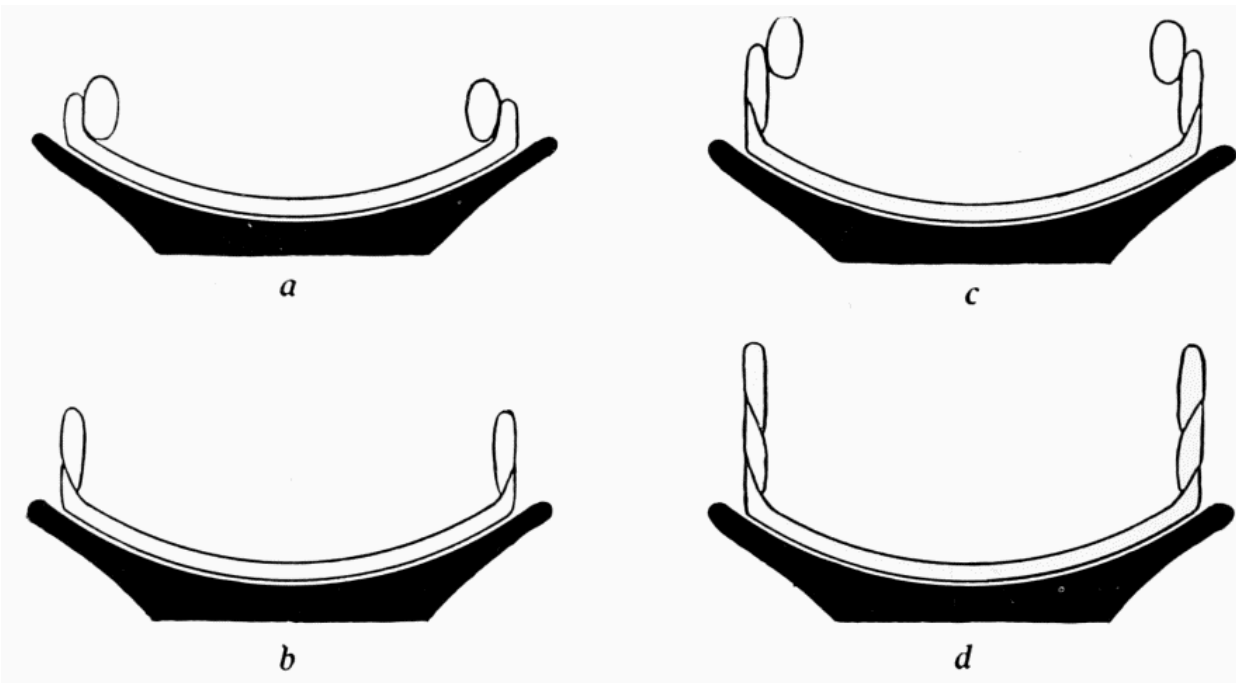


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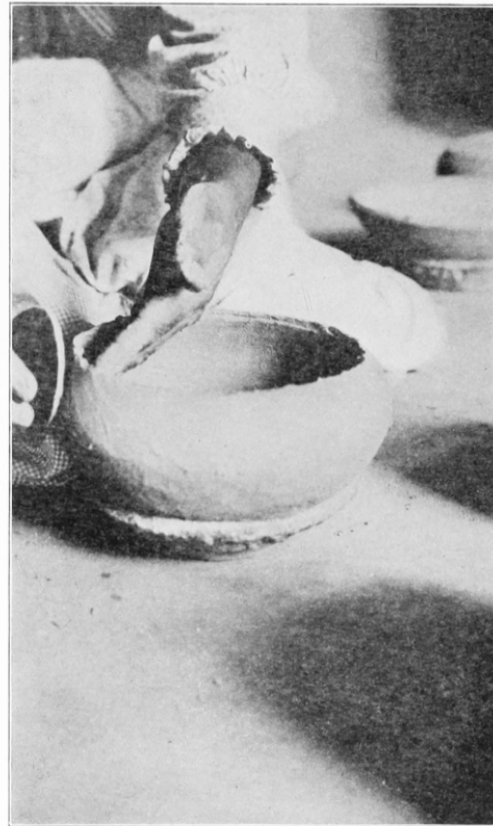
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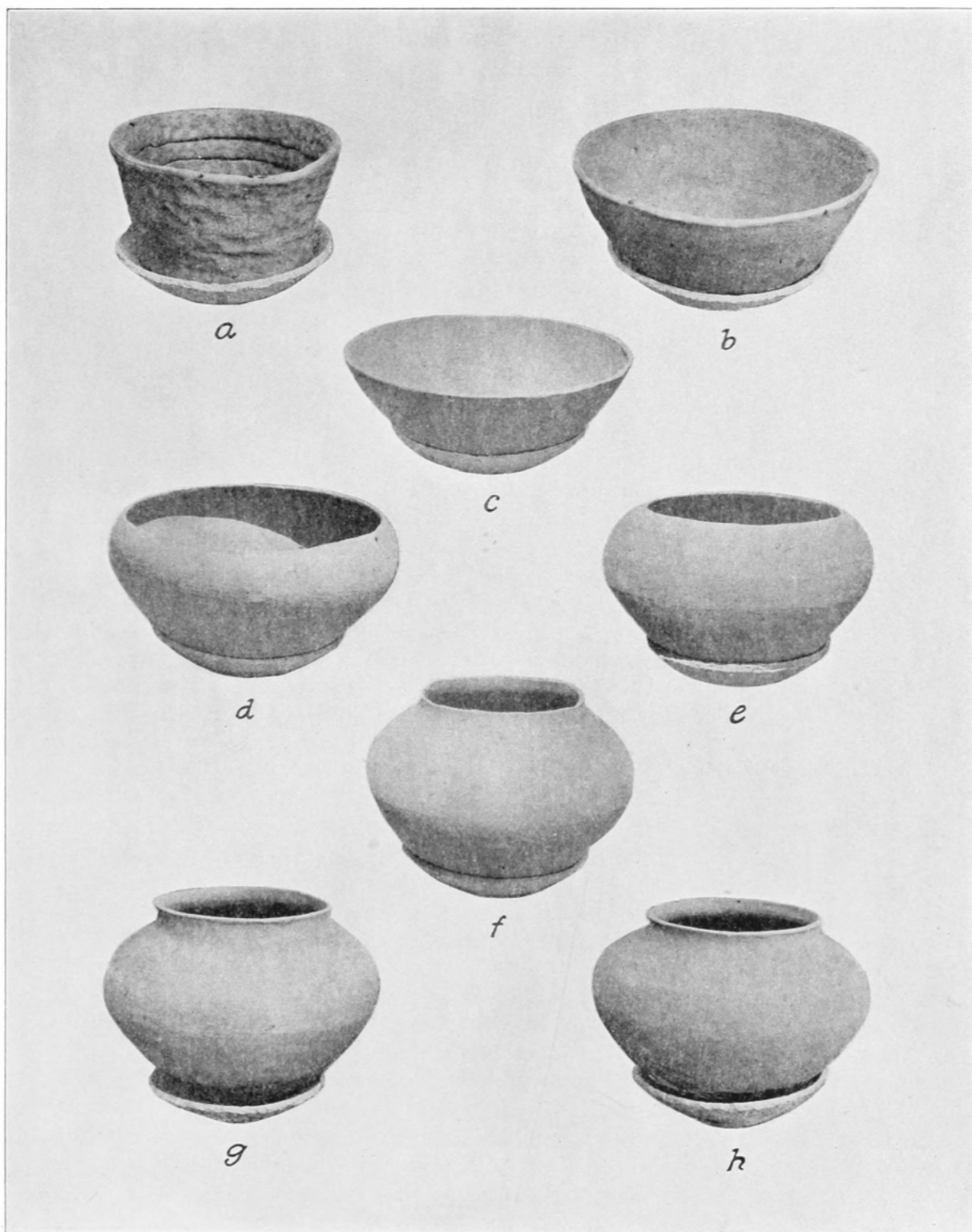


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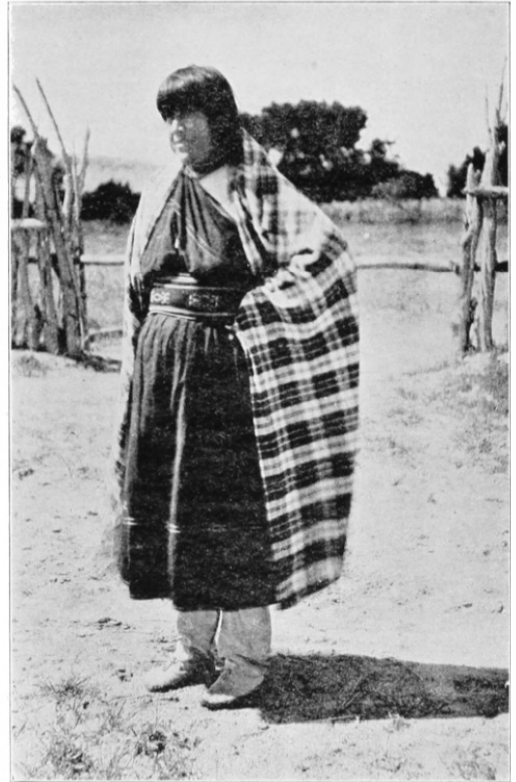
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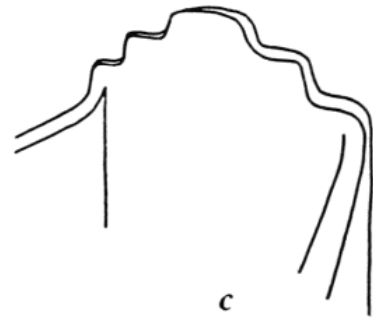
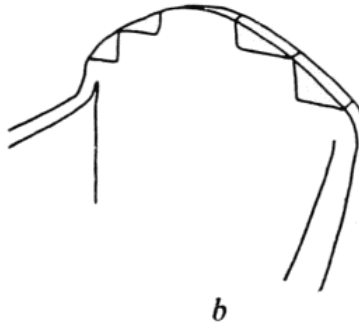
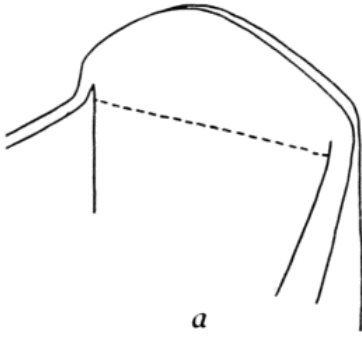


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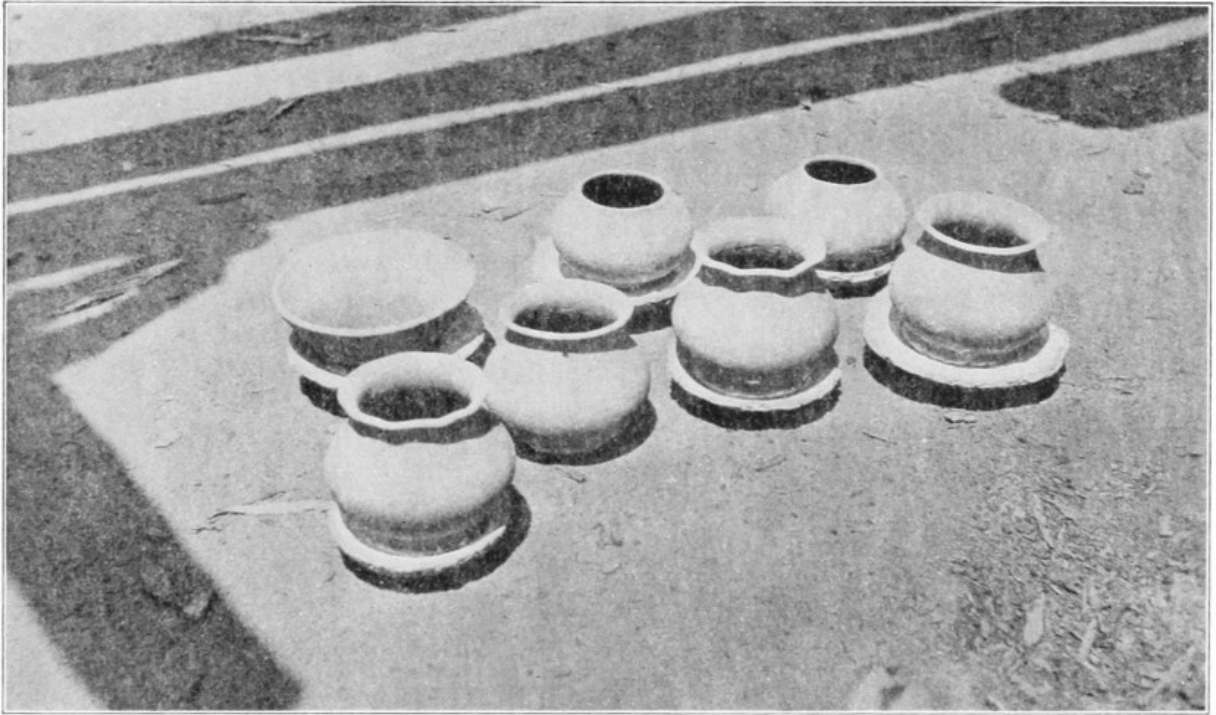


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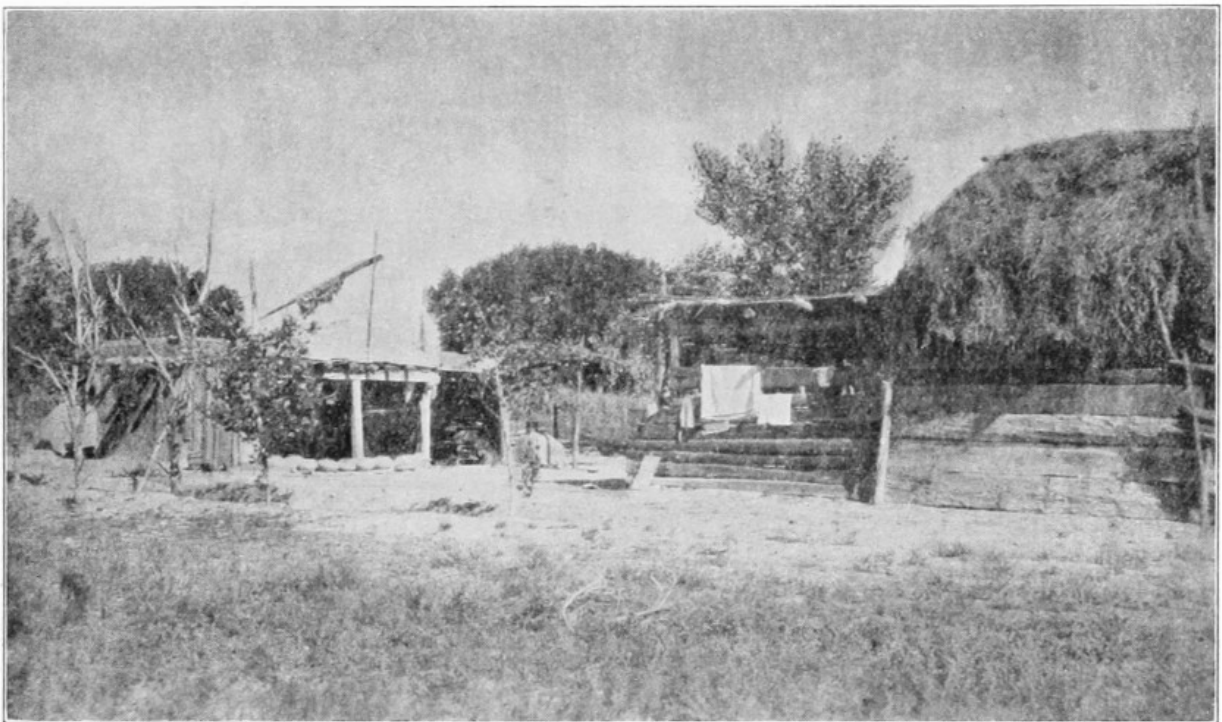
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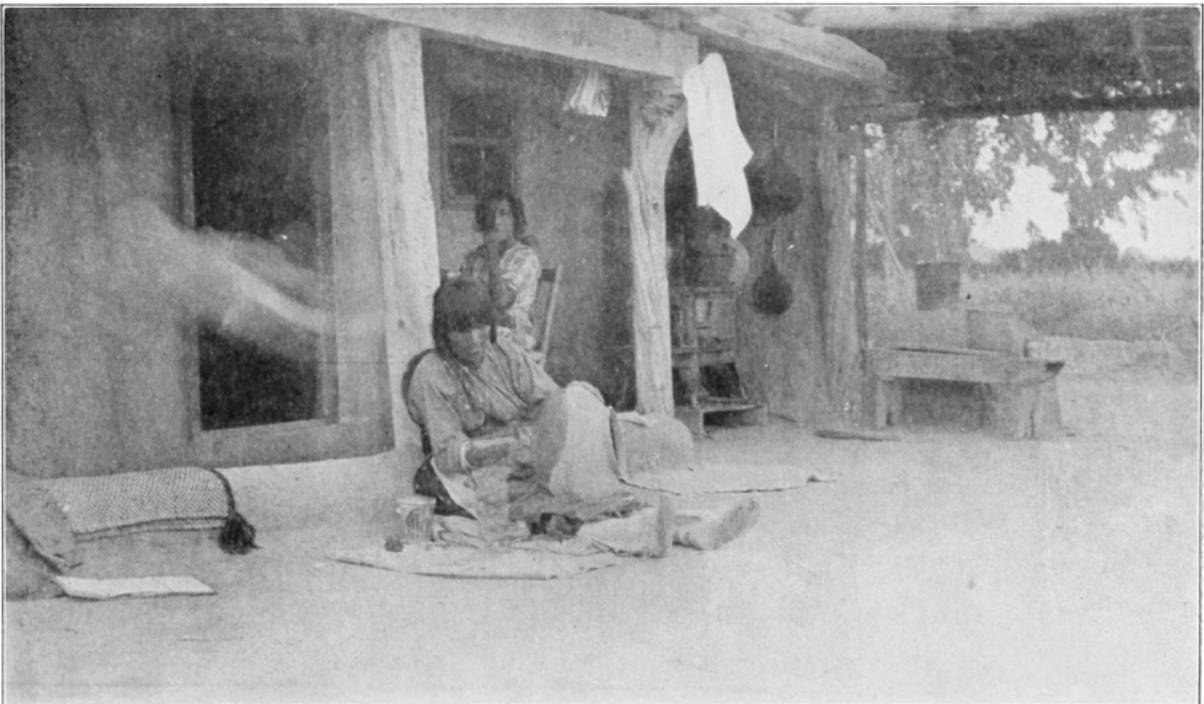


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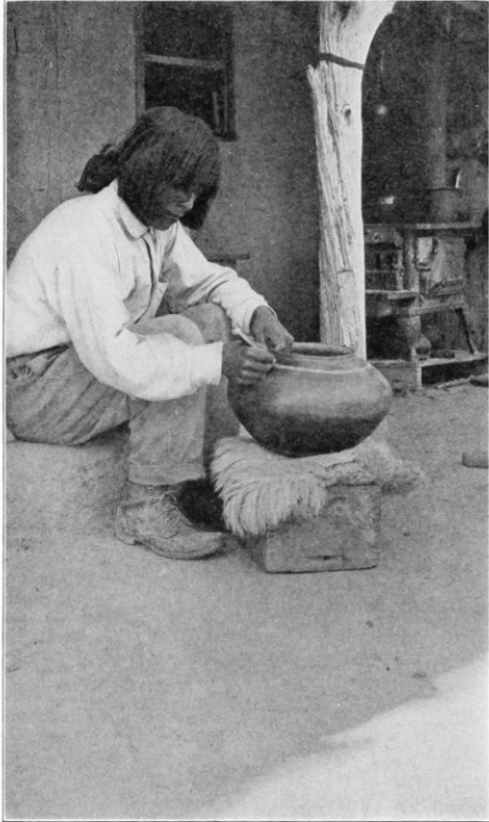


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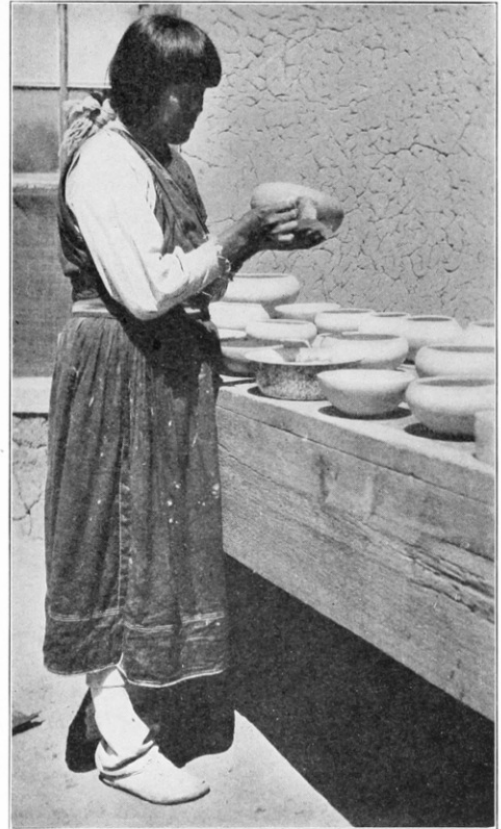


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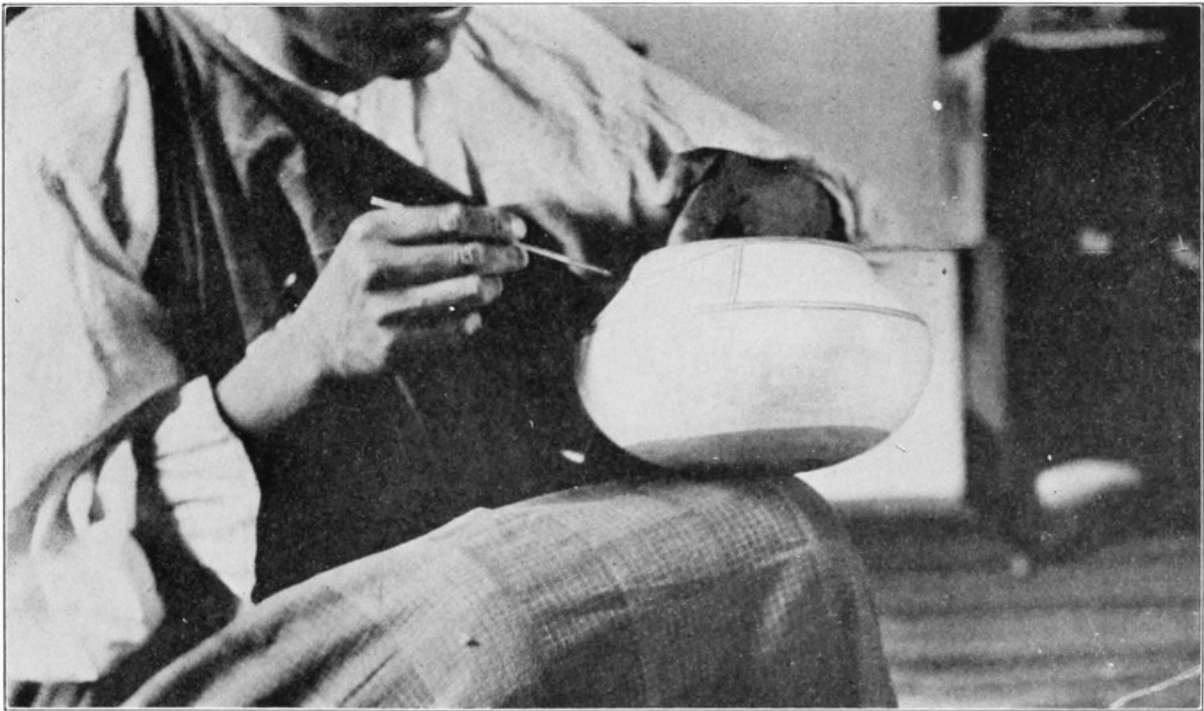


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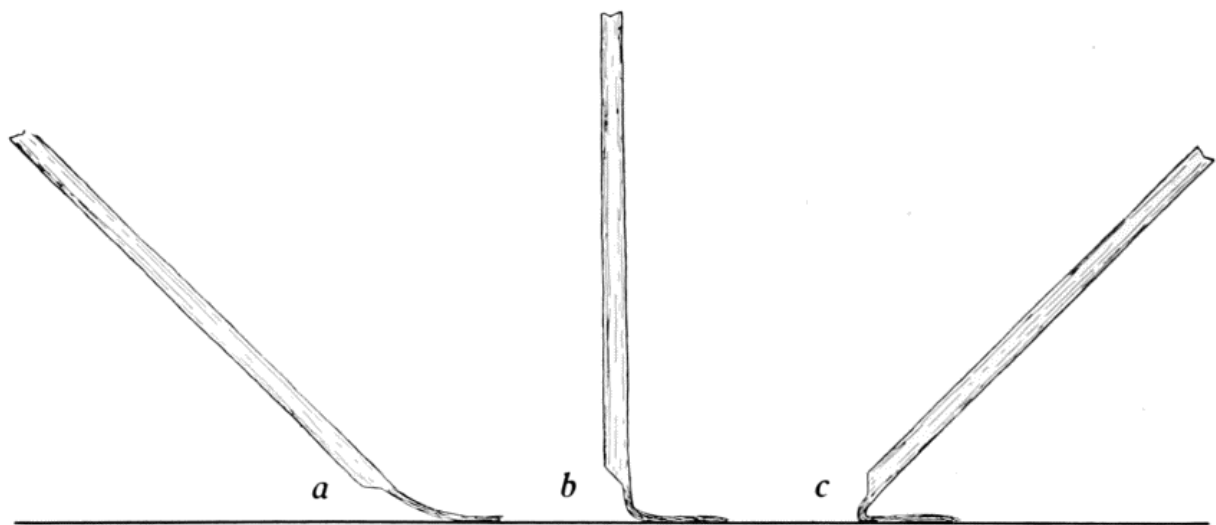


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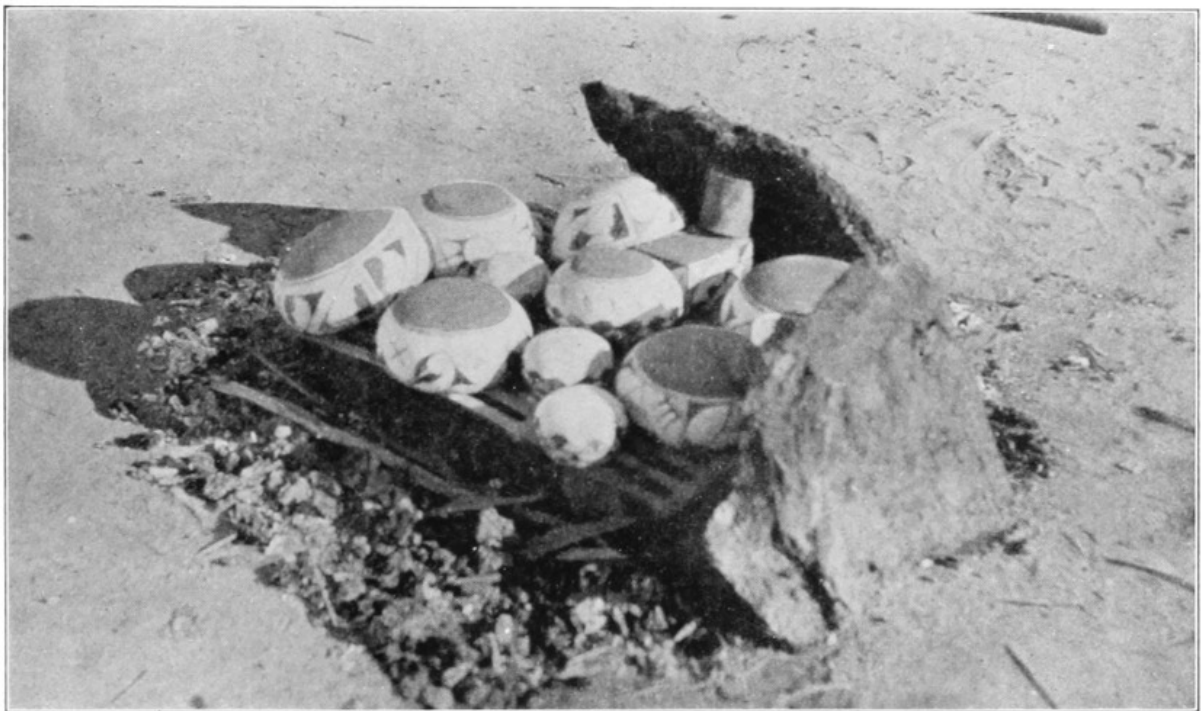


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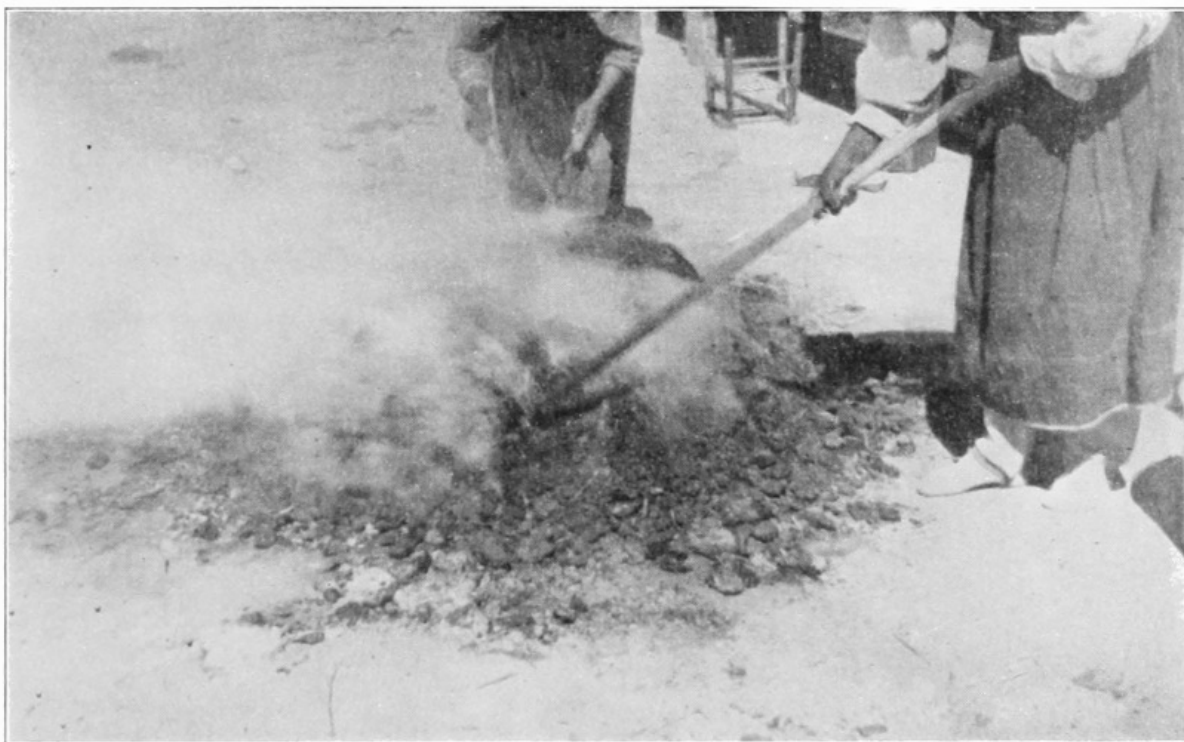


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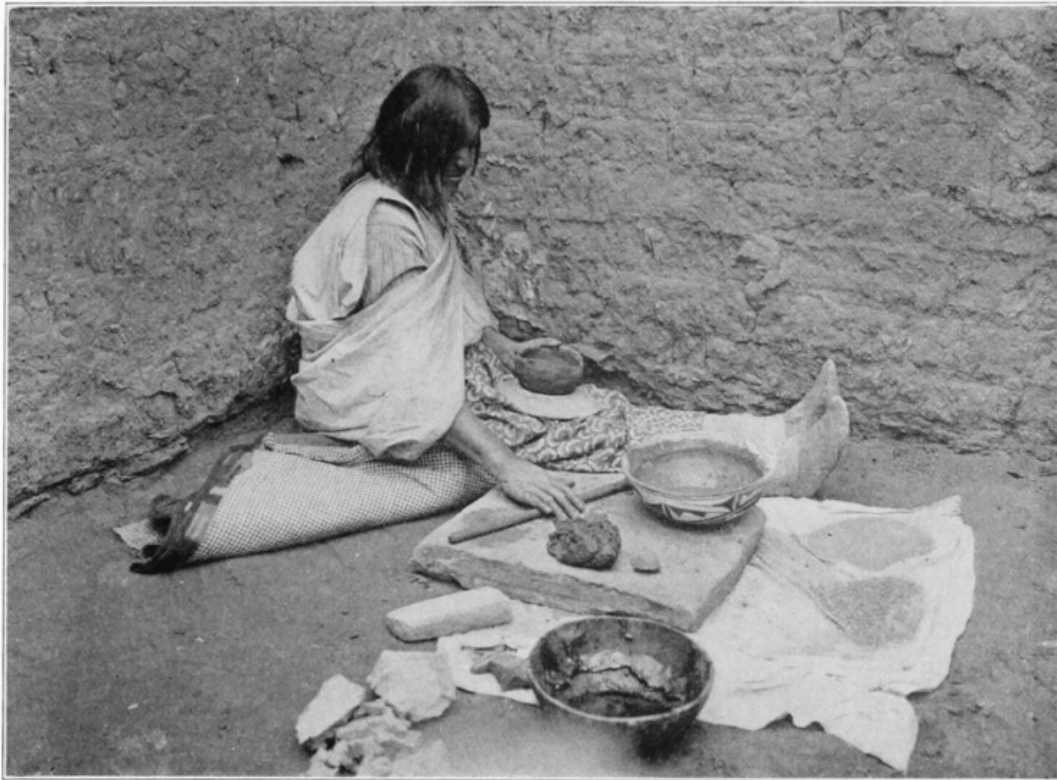
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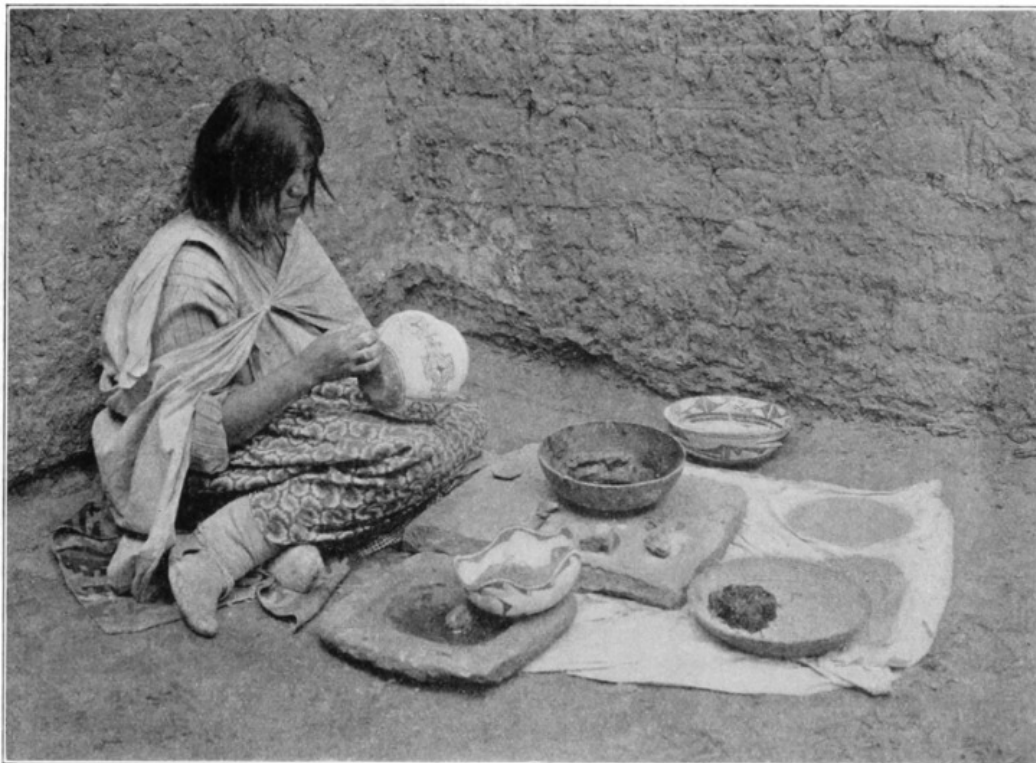
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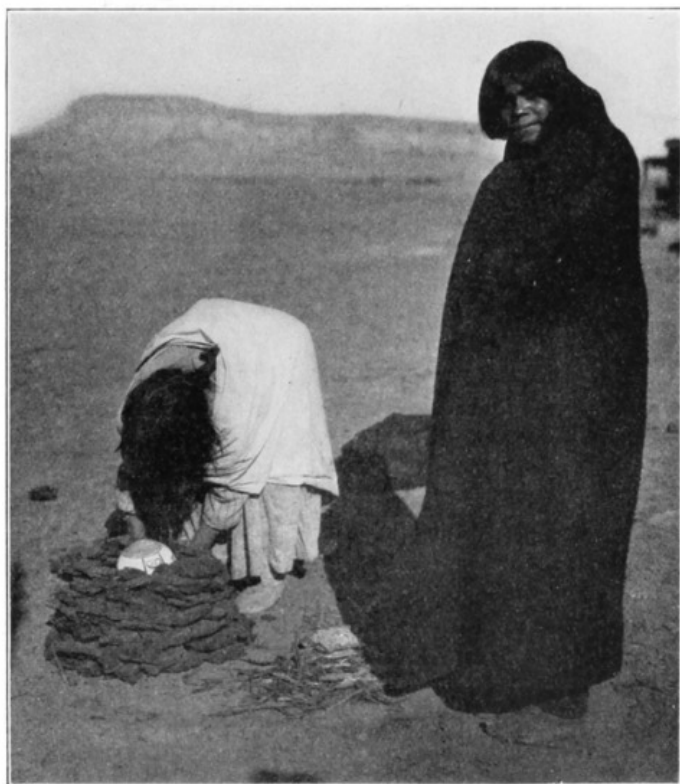
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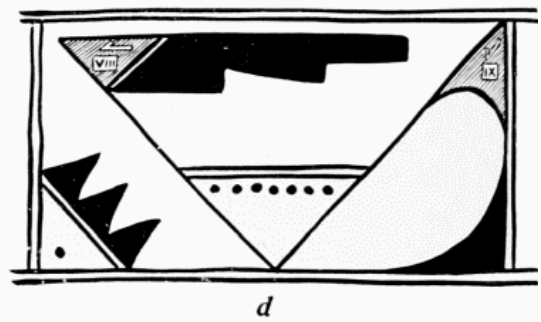
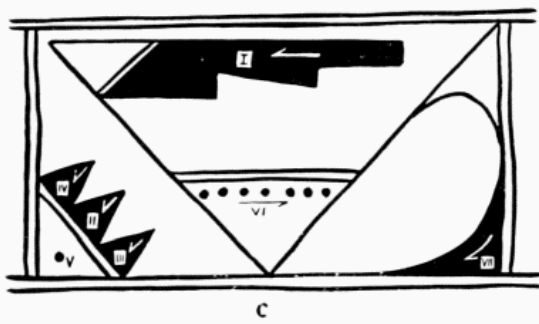
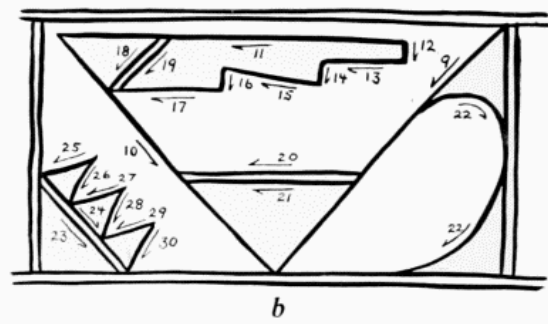
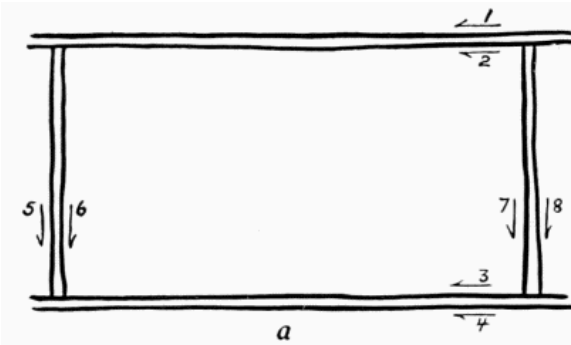


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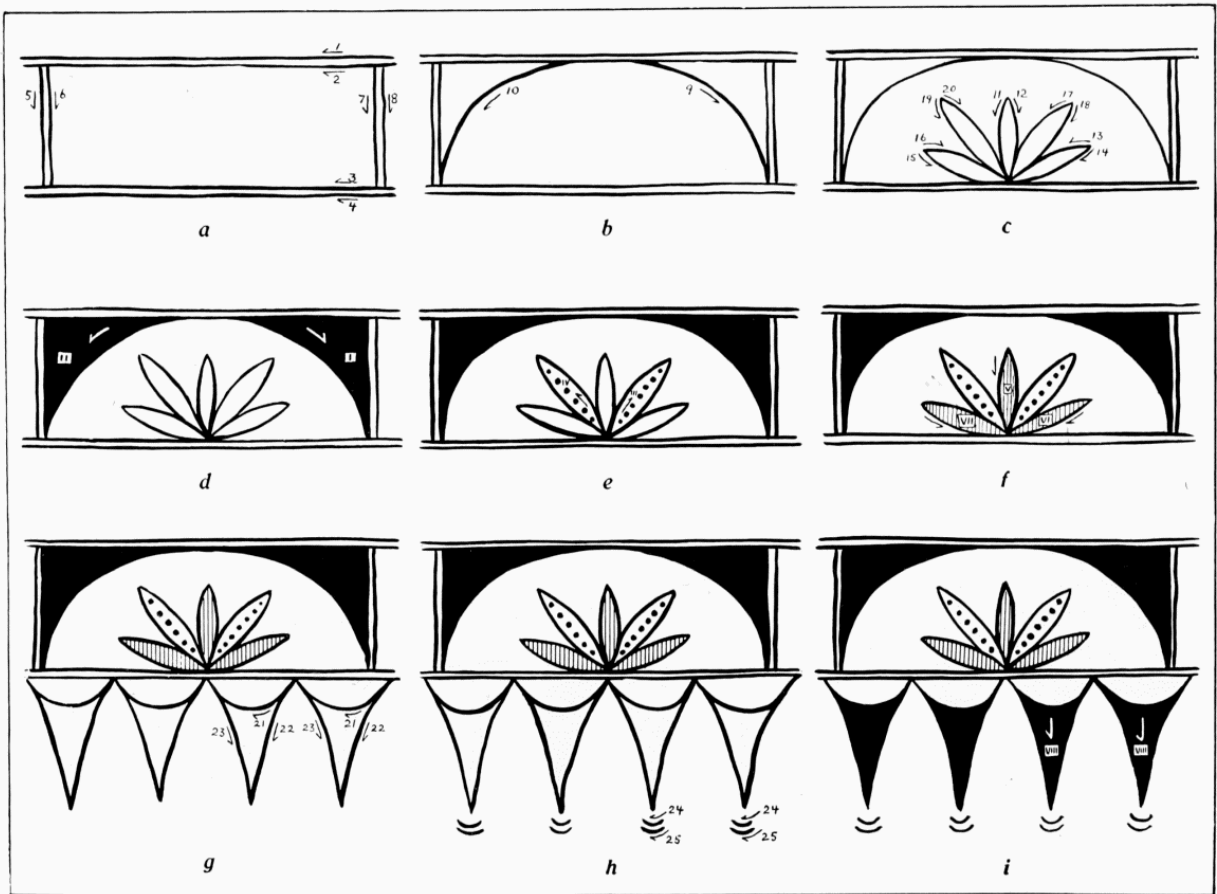


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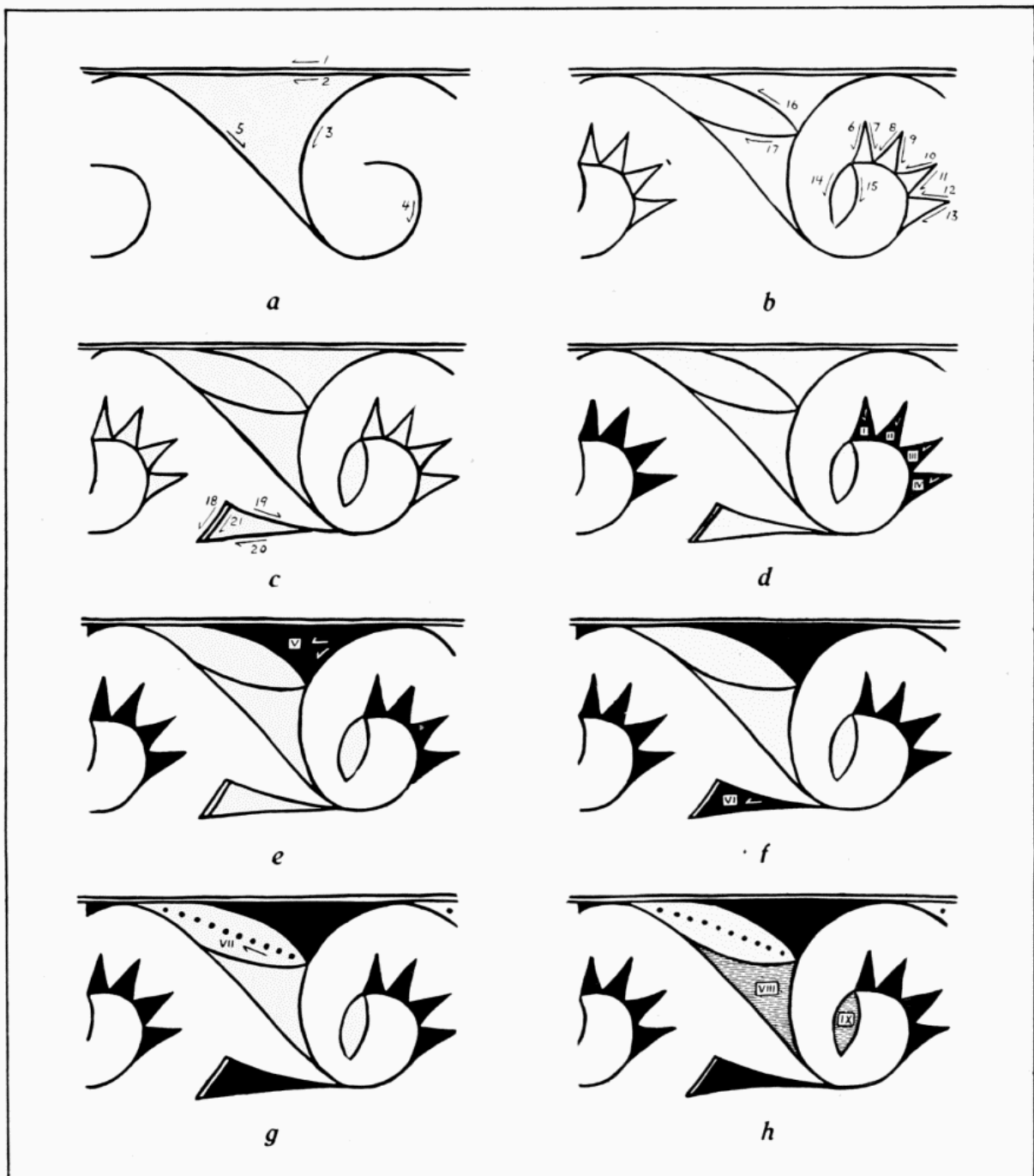
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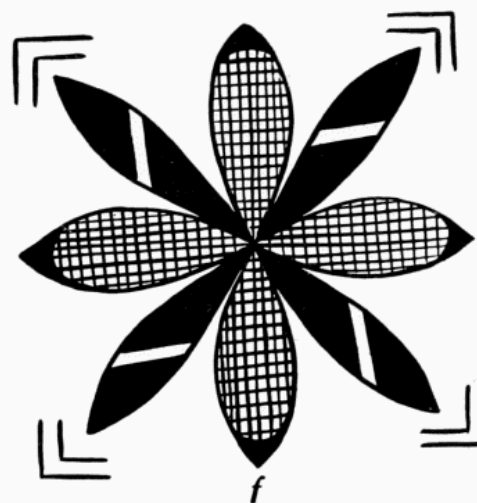
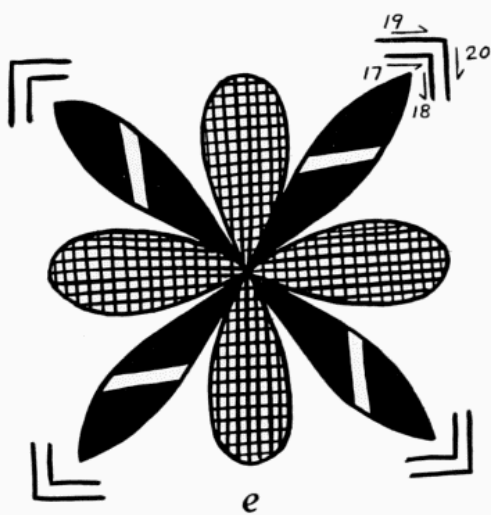
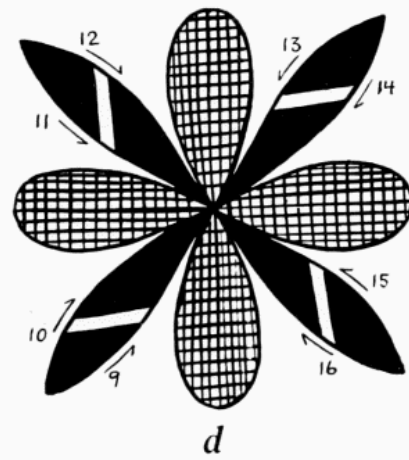
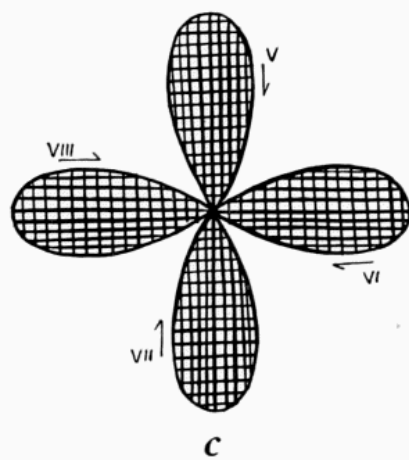
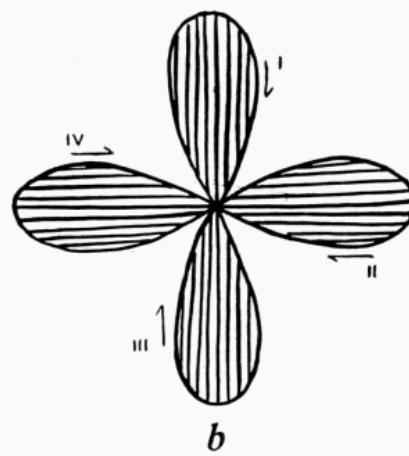
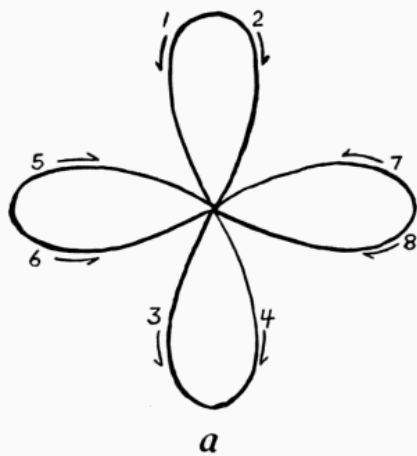
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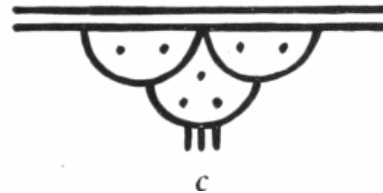
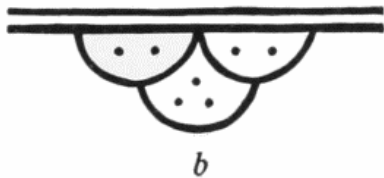
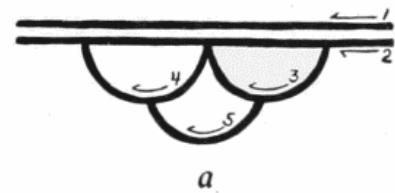
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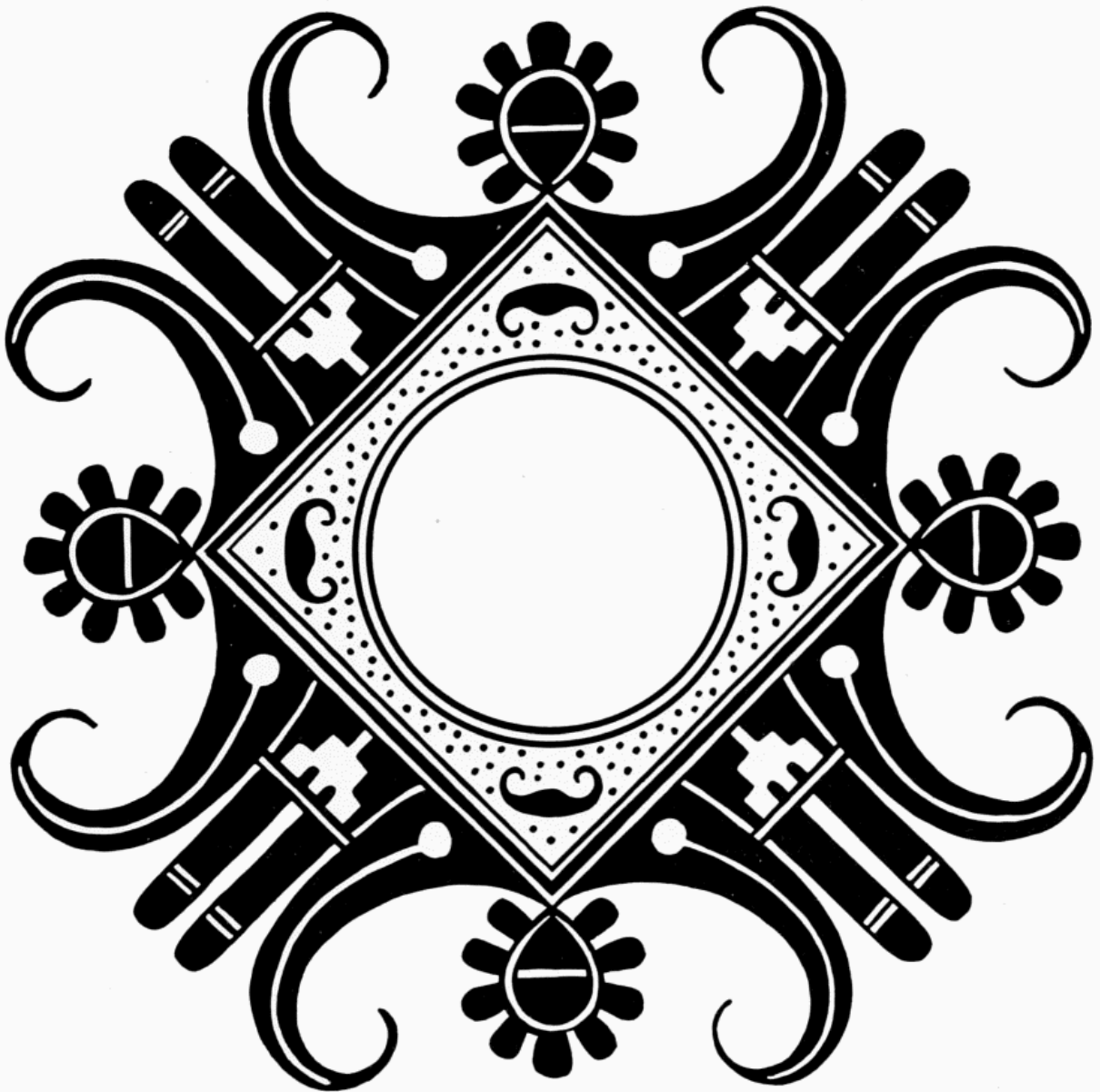
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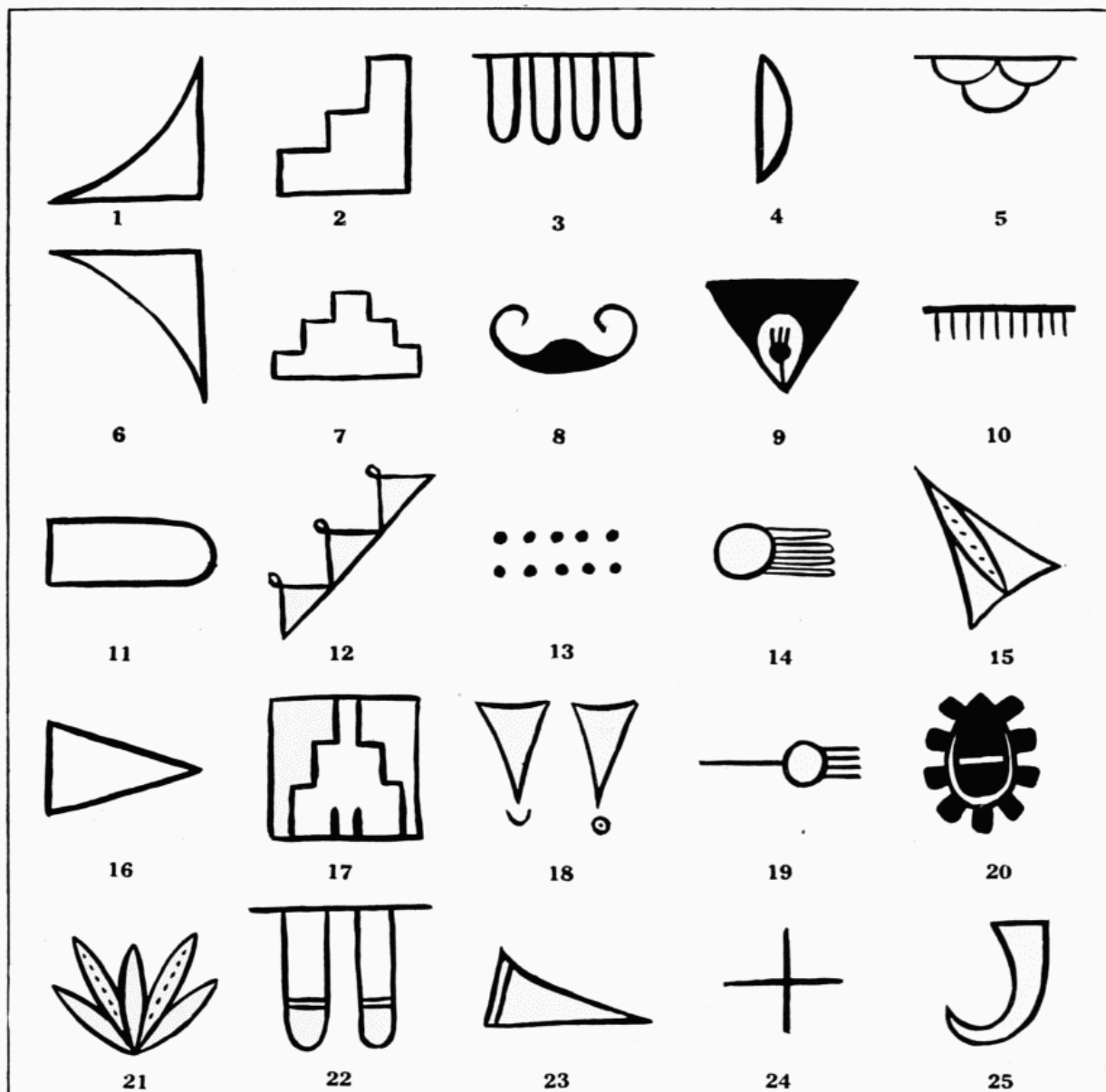
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